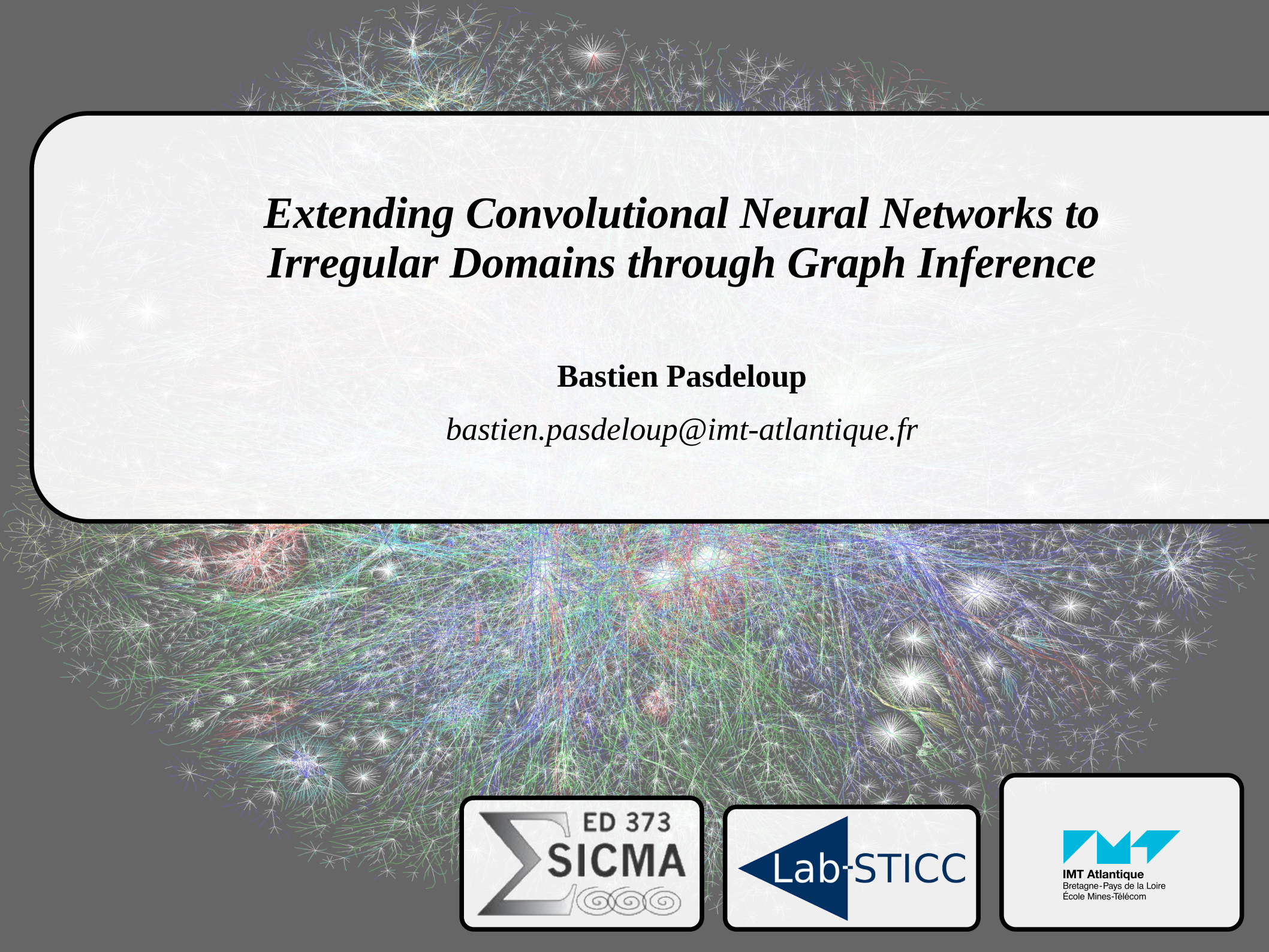
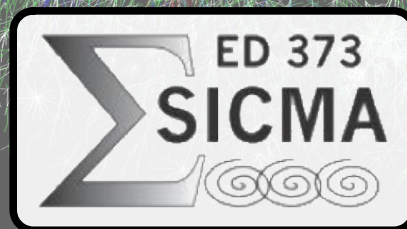




Extending Convolutional Neural Networks to Irregular Domains through Graph Inference

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Who am I?



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► A few words on graph signal processing

- * What is it?
- * The main ideas
- * Some problems and applications

► Graph inference from signals

- * Why it may be necessary
- * The case of stationary signals

► Translations on irregular graphs

- * How we define them
- * How we identify them
- * Examples and applications

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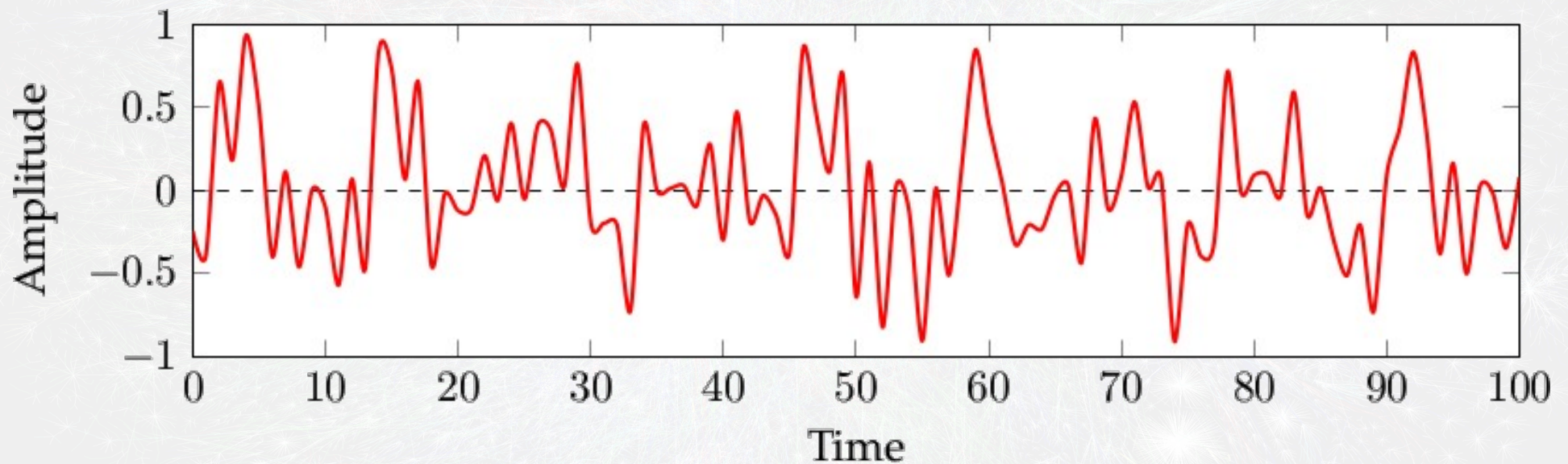
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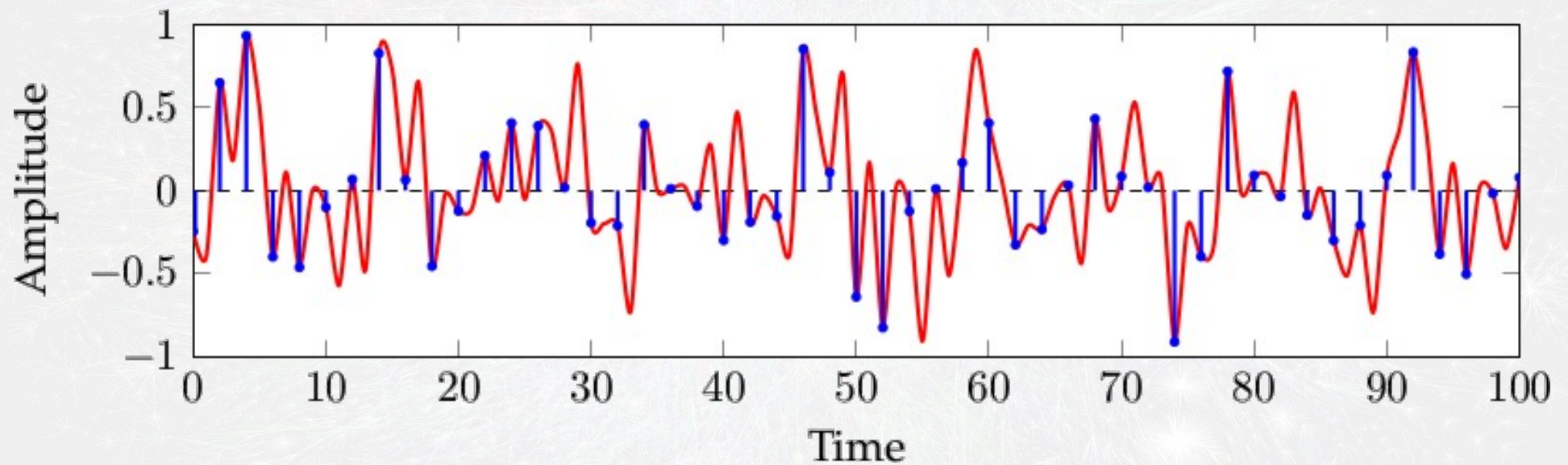
A few words on graph signal processing

What is it? – Discrete Fourier analysis



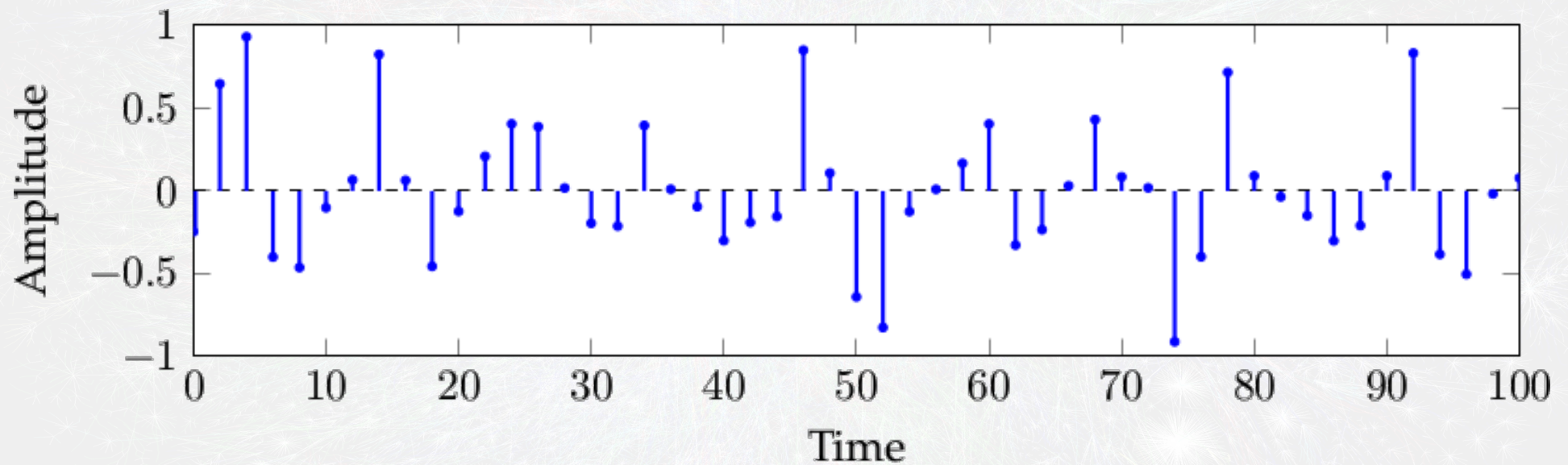
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What is it? – Discrete Fourier analysis



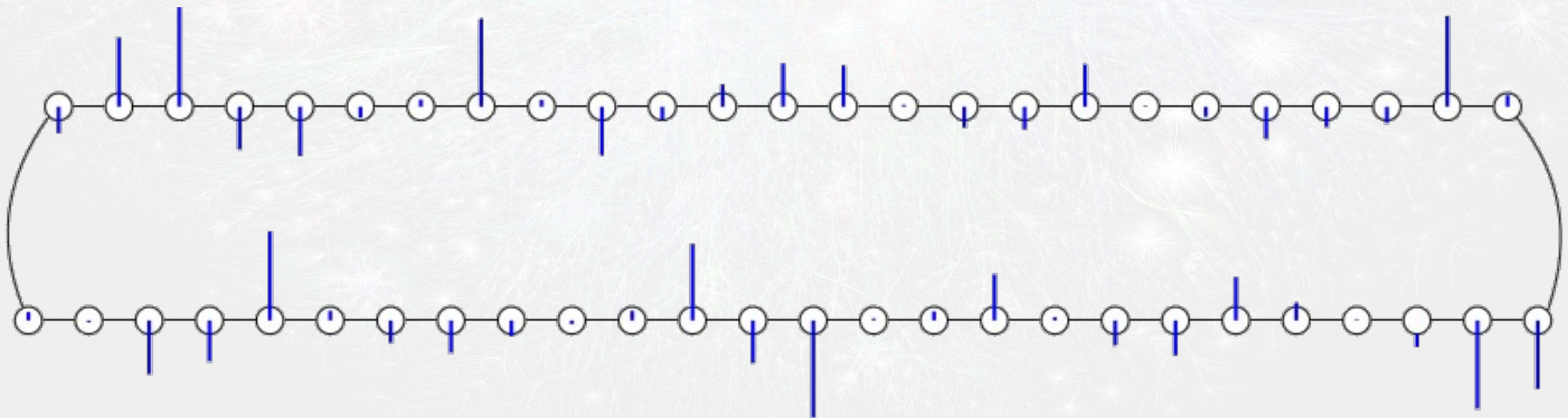
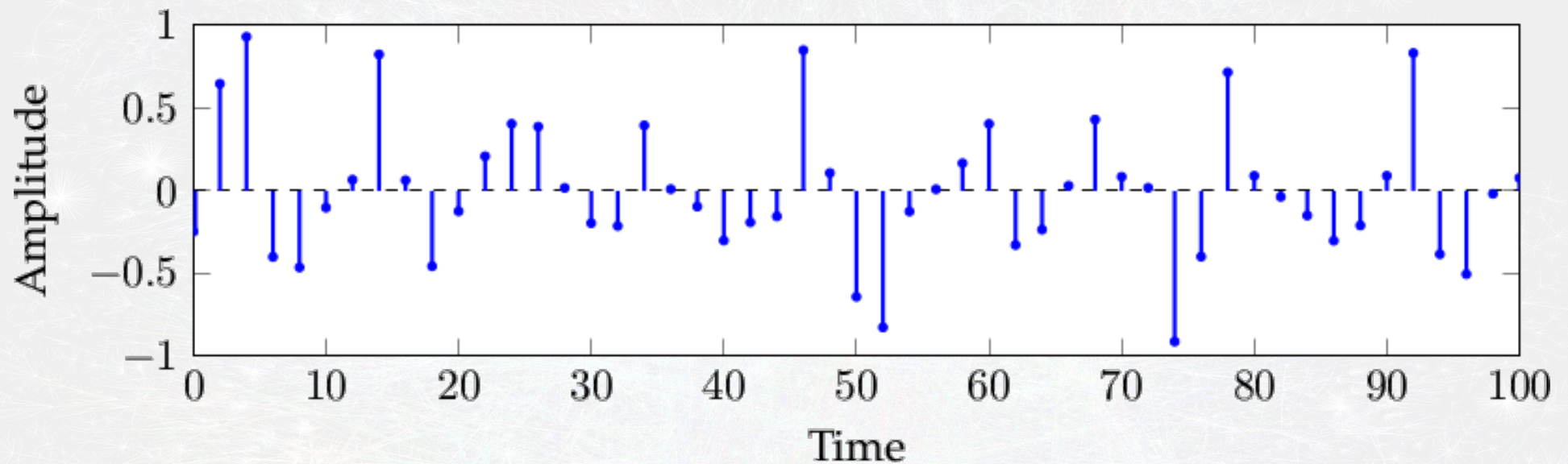
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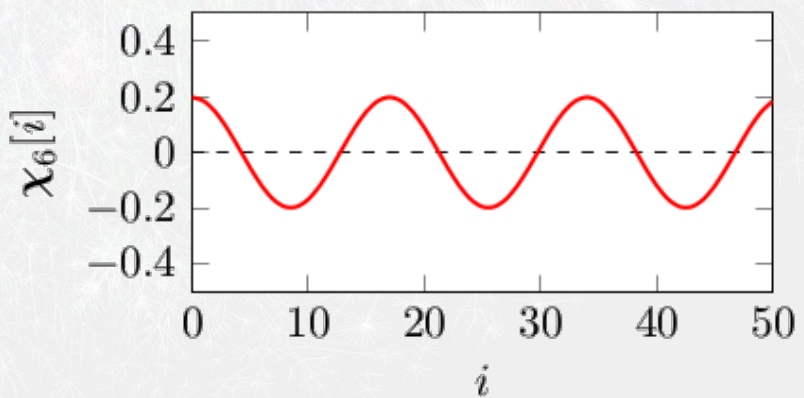
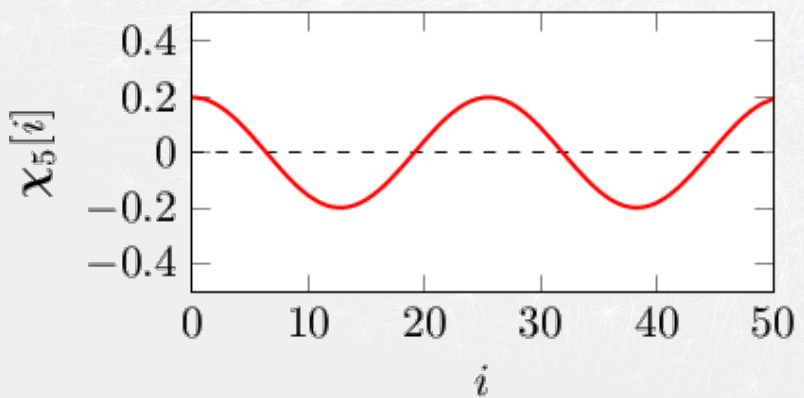
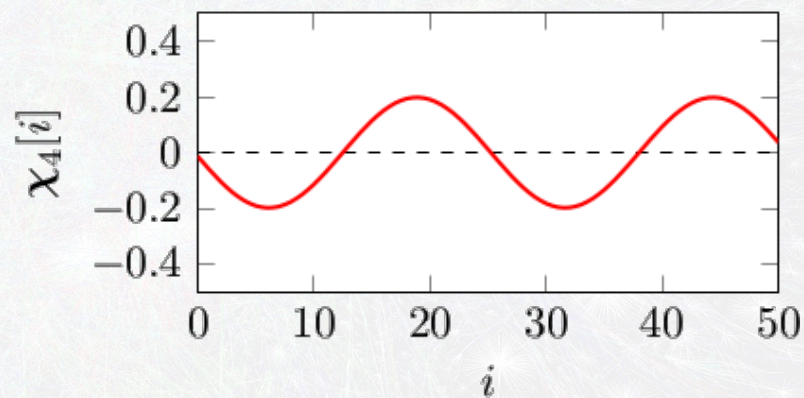
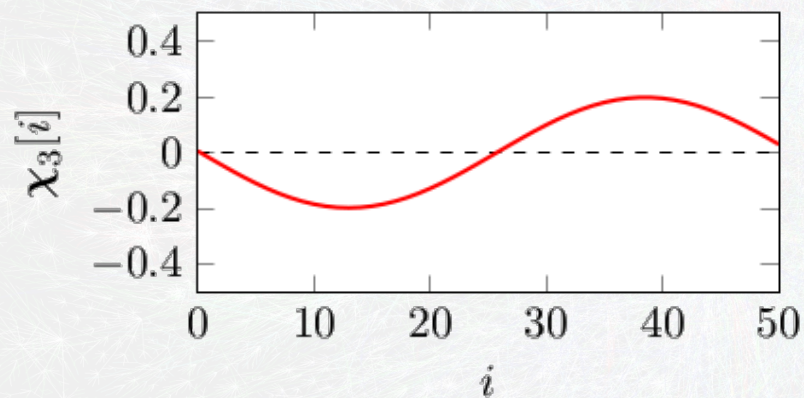
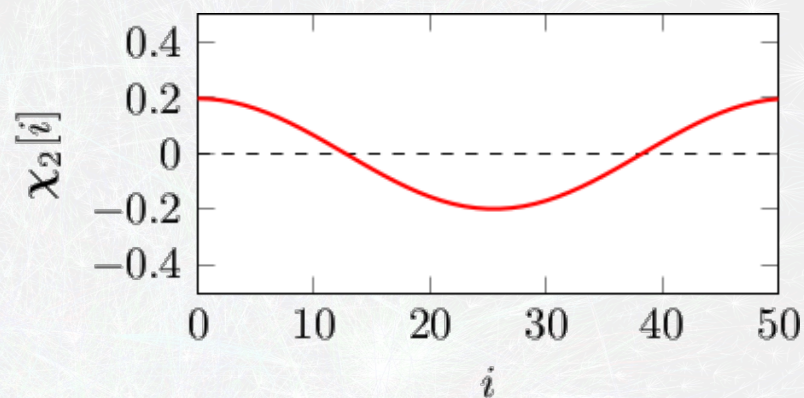
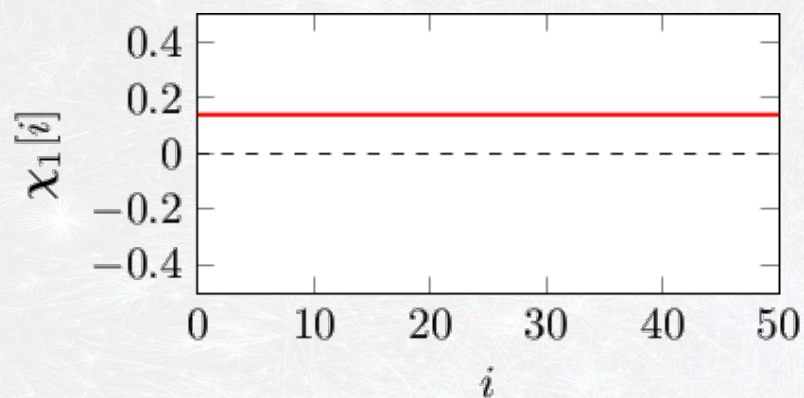
A few words on graph signal processing

What is it? – Modelisation of periodic time as a graph



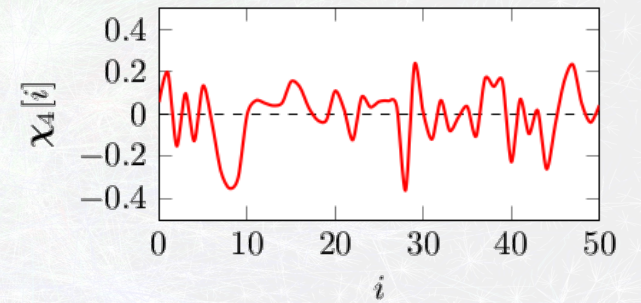
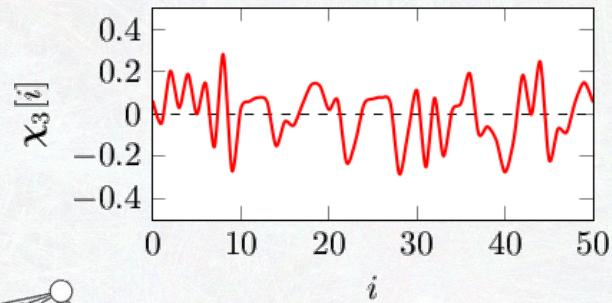
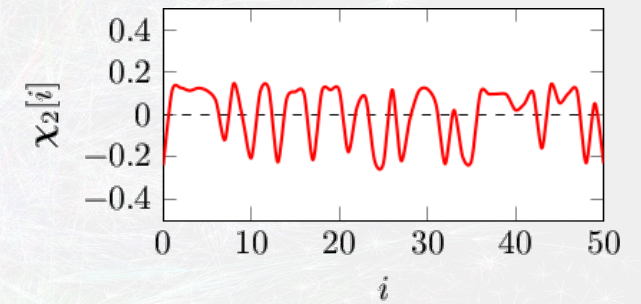
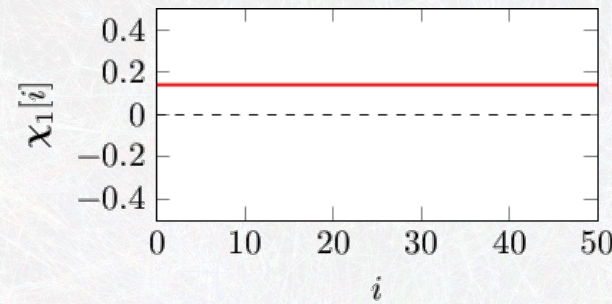
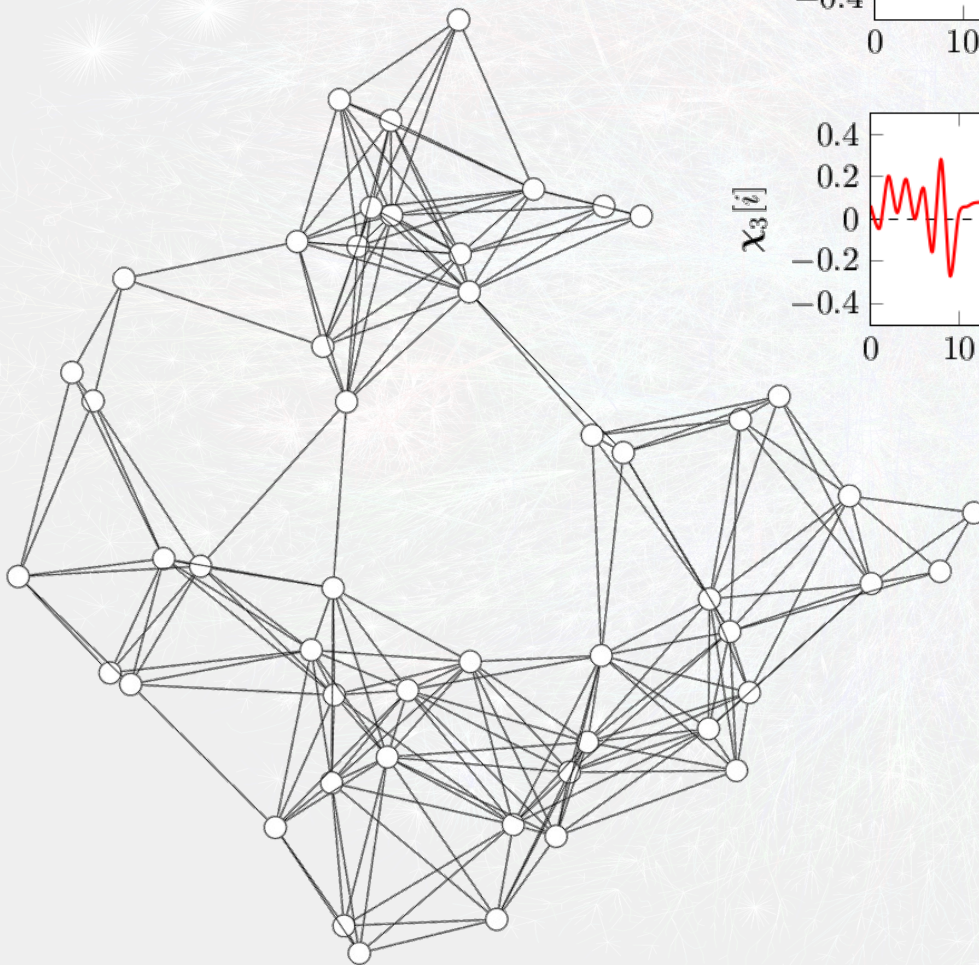
A few words on graph signal processing

What is it? – The first eigenvectors of the Laplacian matrix of the ring graph



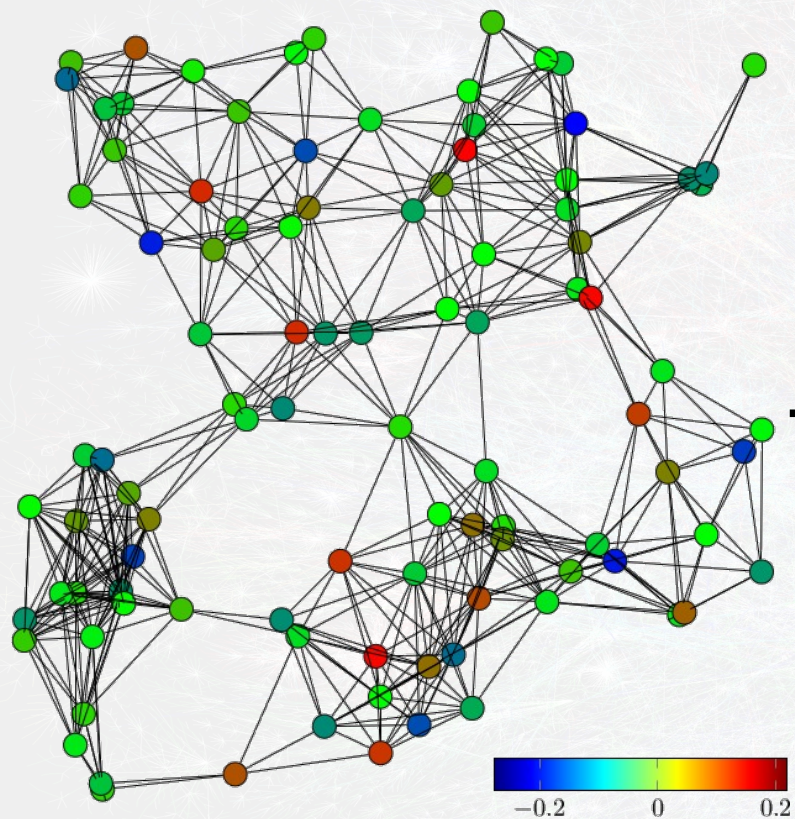
A few words on graph signal processing

What is it? – Analogy with the previous result

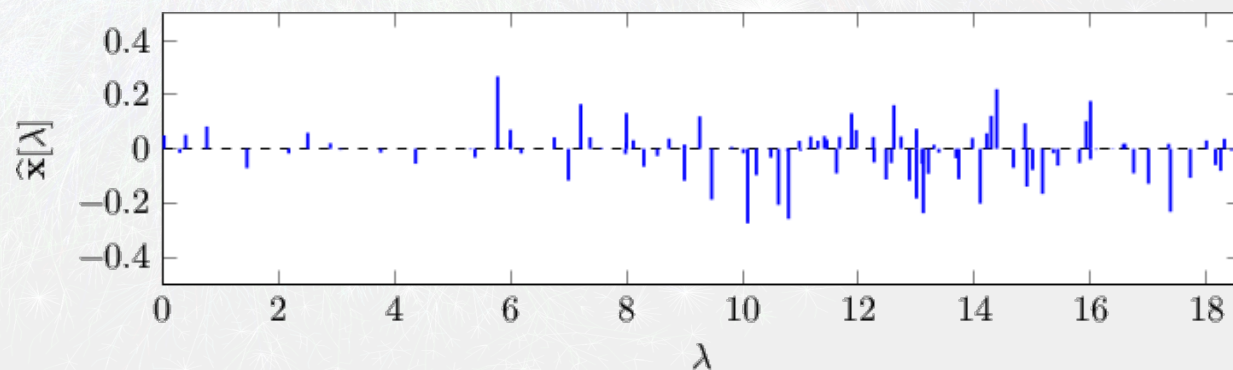


A few words on graph signal processing

The main ideas – The graph Fourier transform

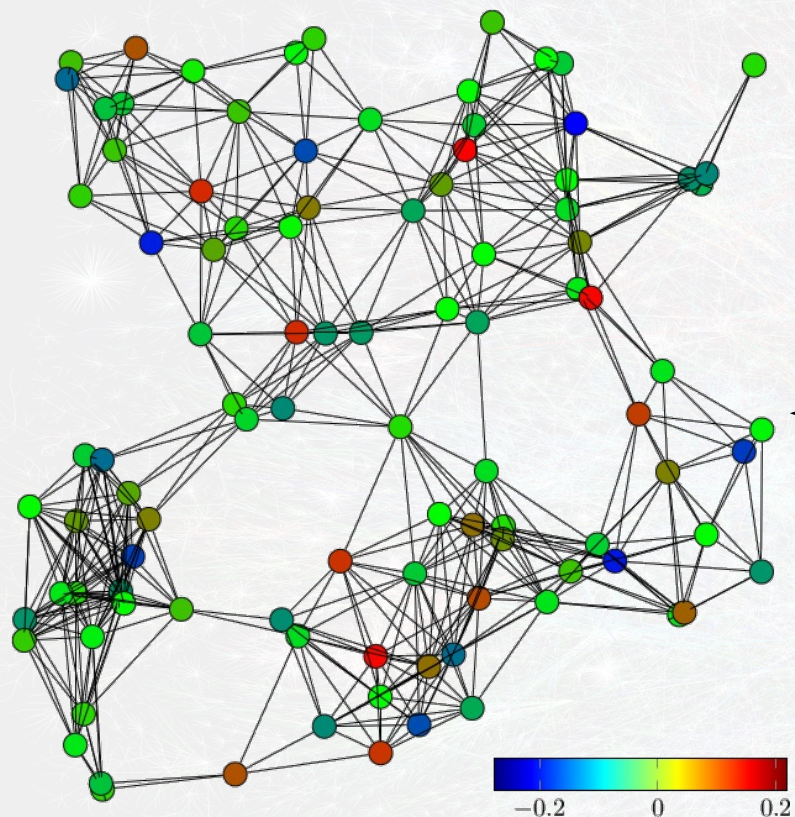


$$GFT(\mathbf{x}) = \hat{\mathbf{x}} = \mathcal{X}^\top \mathbf{x}$$

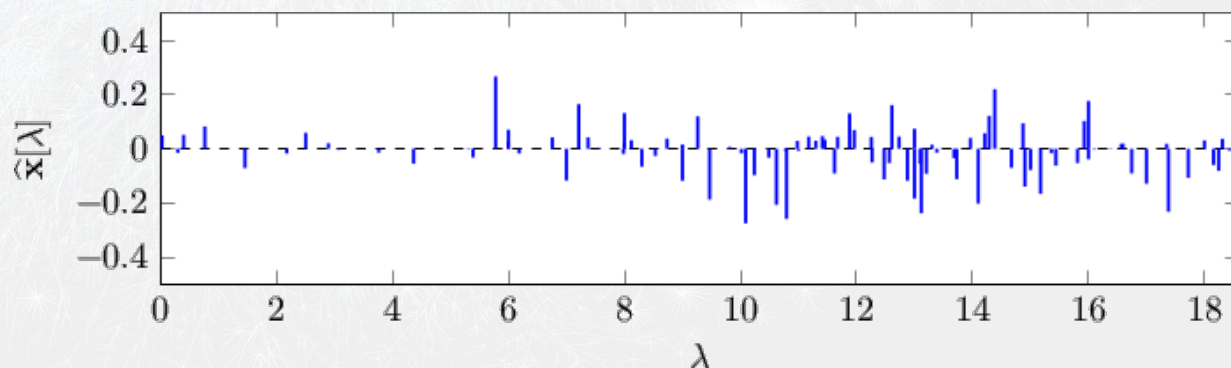


A few words on graph signal processing

The main ideas – The graph Fourier transform

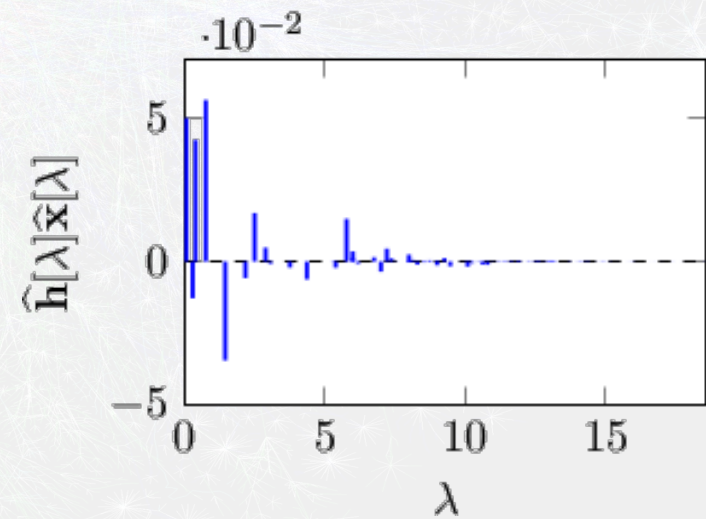
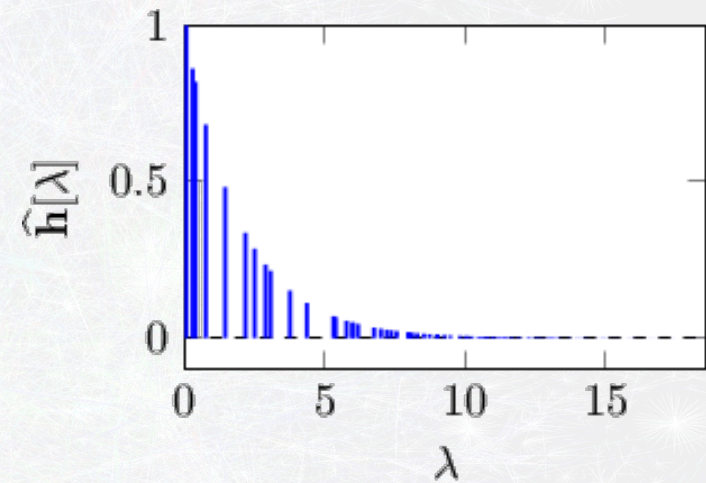
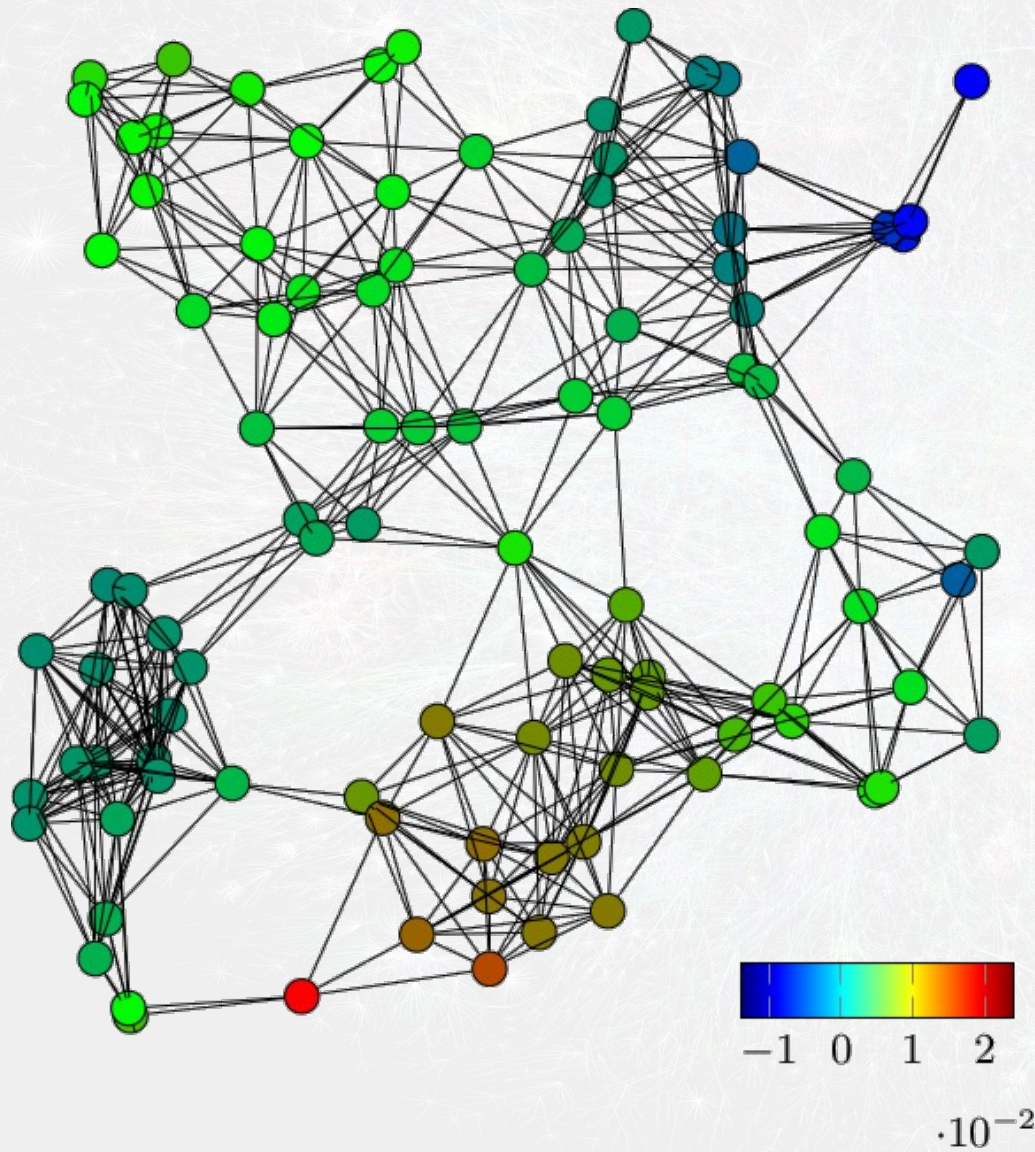


$$GFT^{-1}(\hat{\mathbf{x}}) = \mathbf{x} = \mathcal{X}\hat{\mathbf{x}}$$



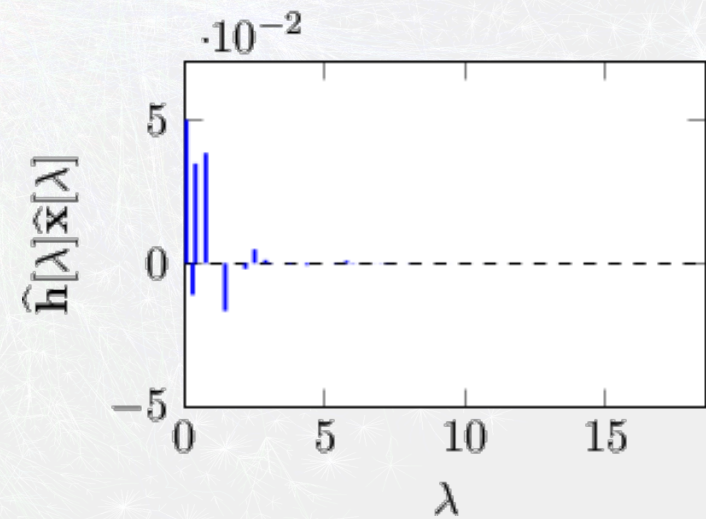
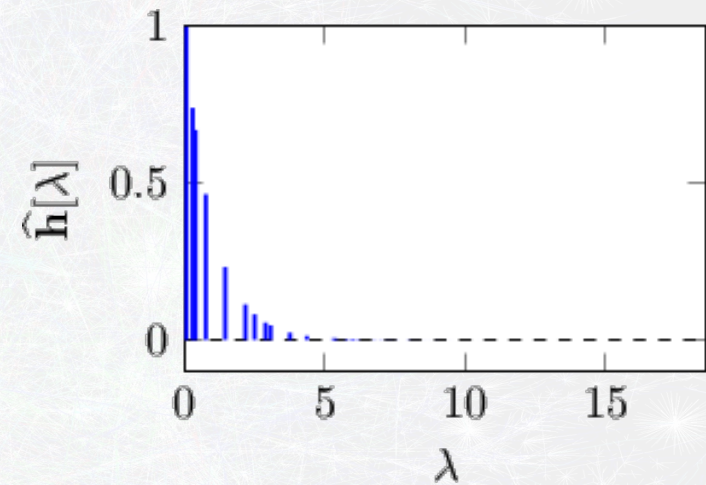
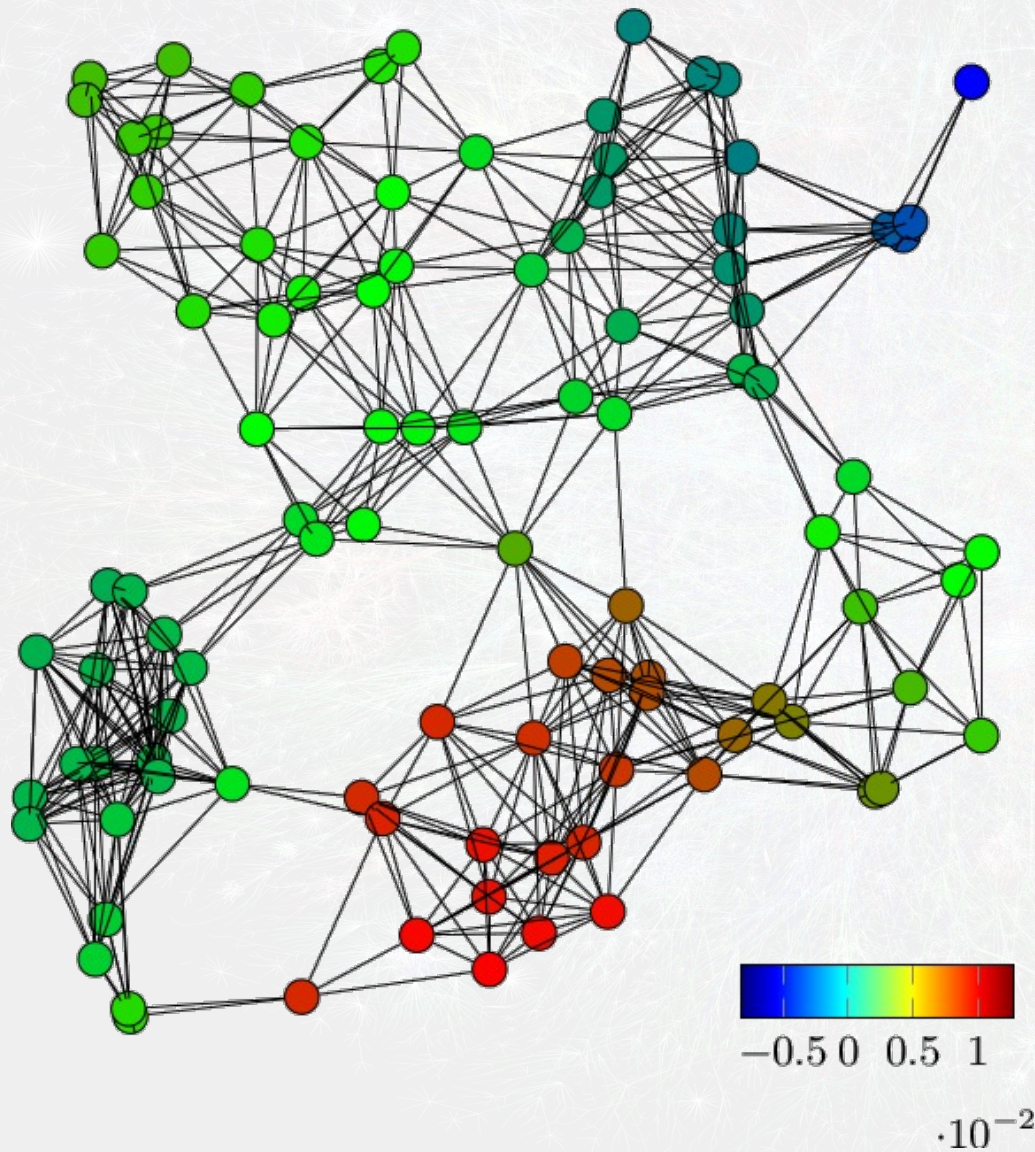
A few words on graph signal processing

The main ideas – Filtering signals



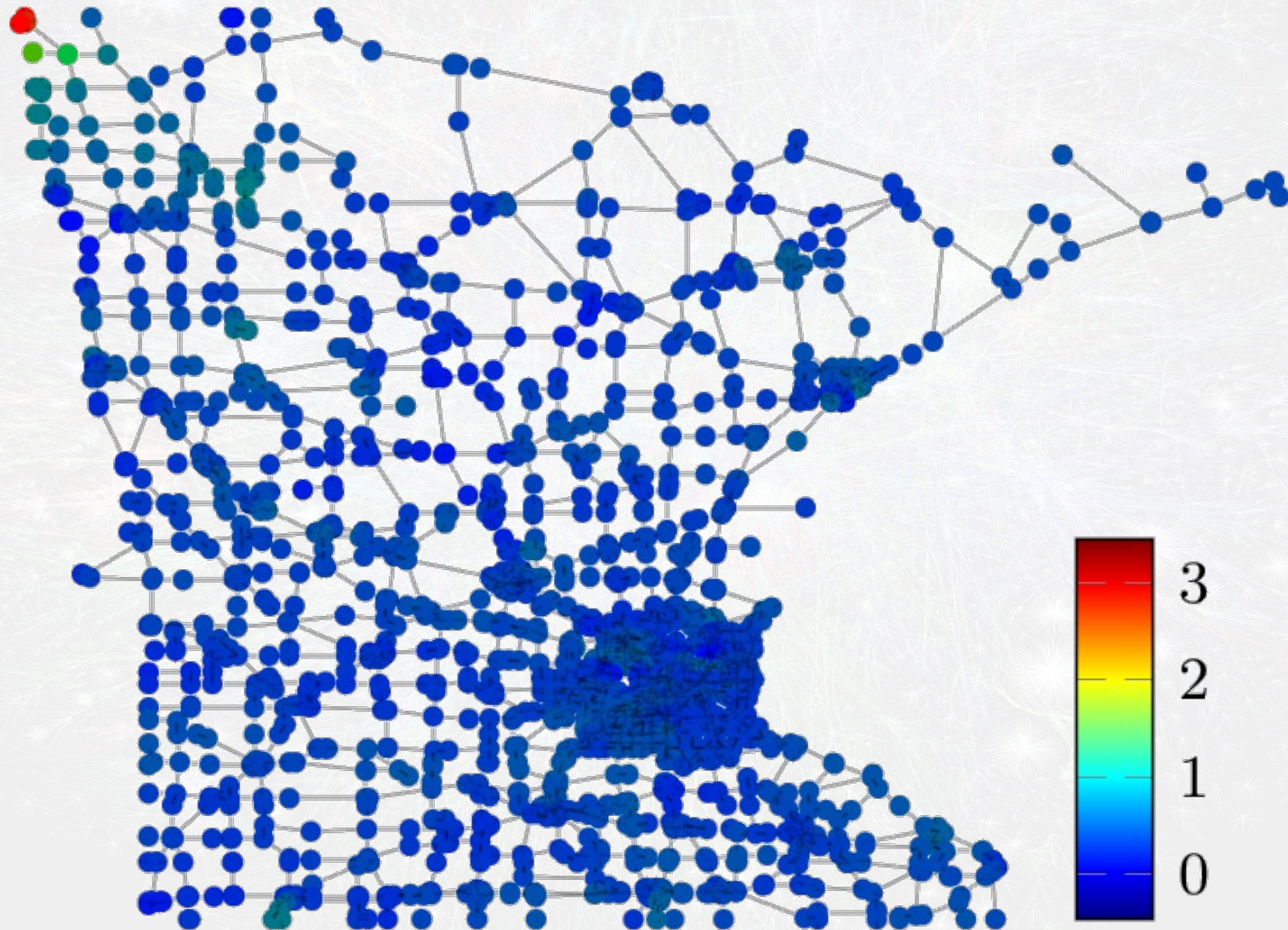
A few words on graph signal processing

The main ideas – Filtering signals



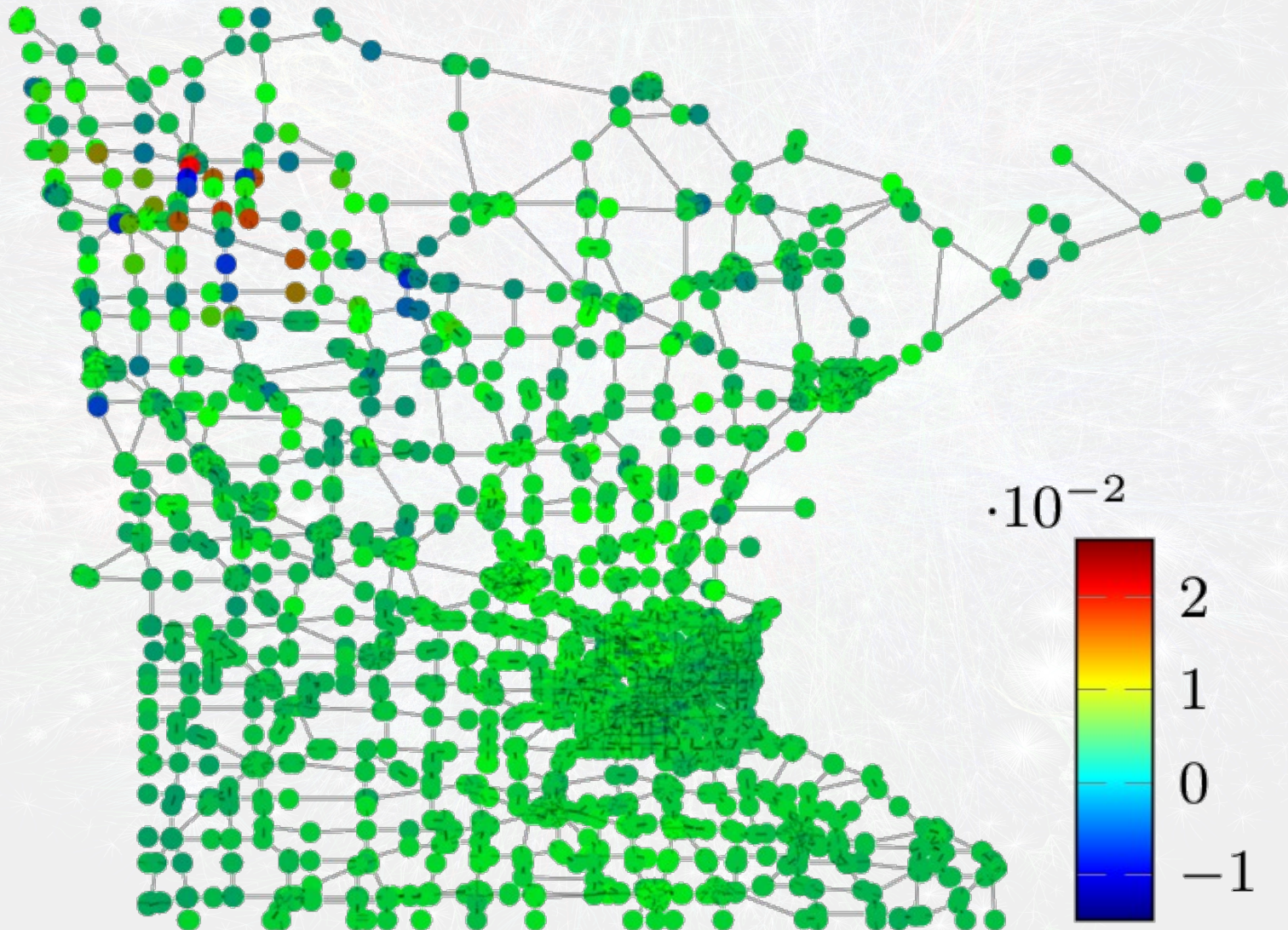
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The main ideas – Translating signals



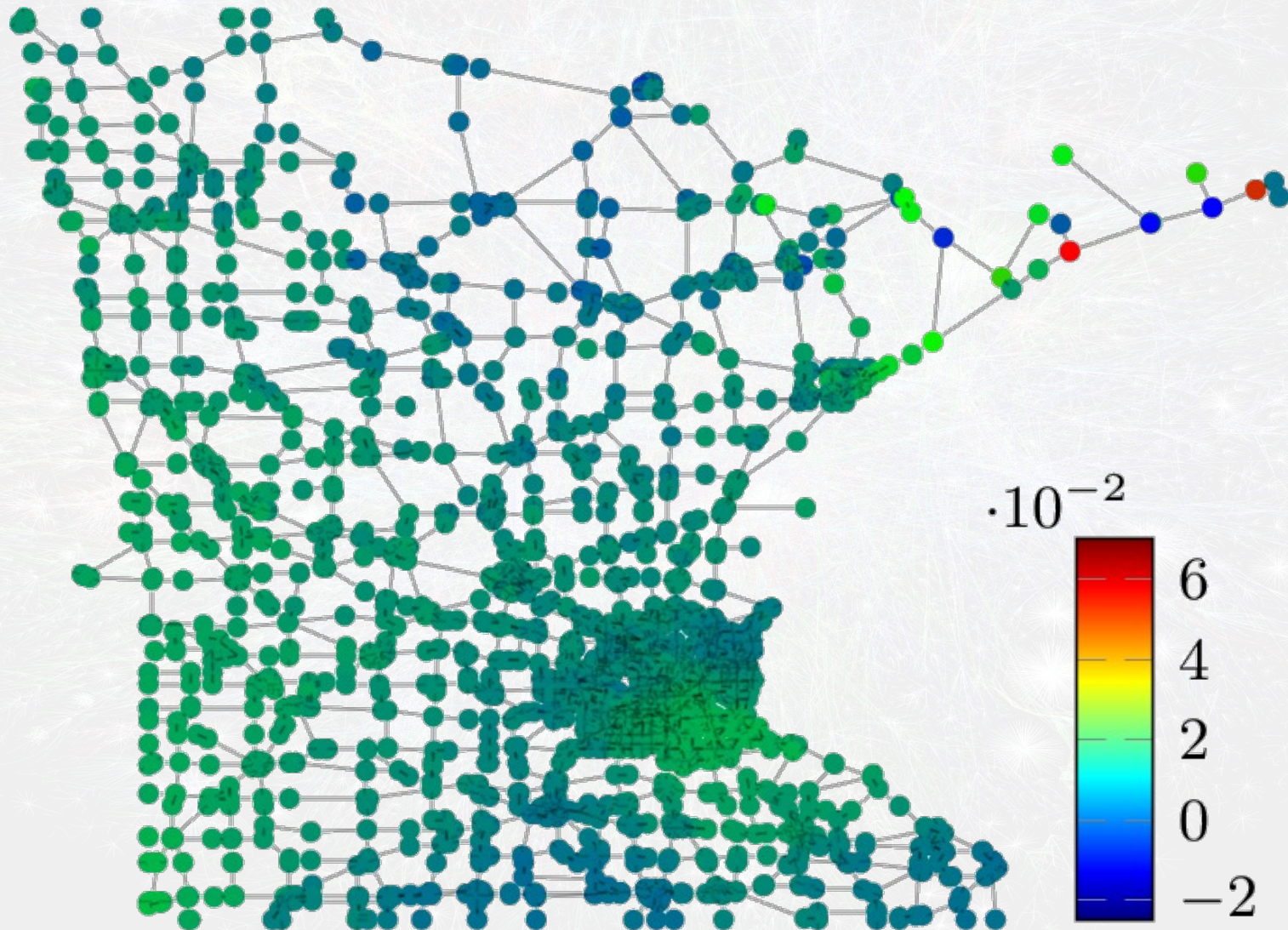
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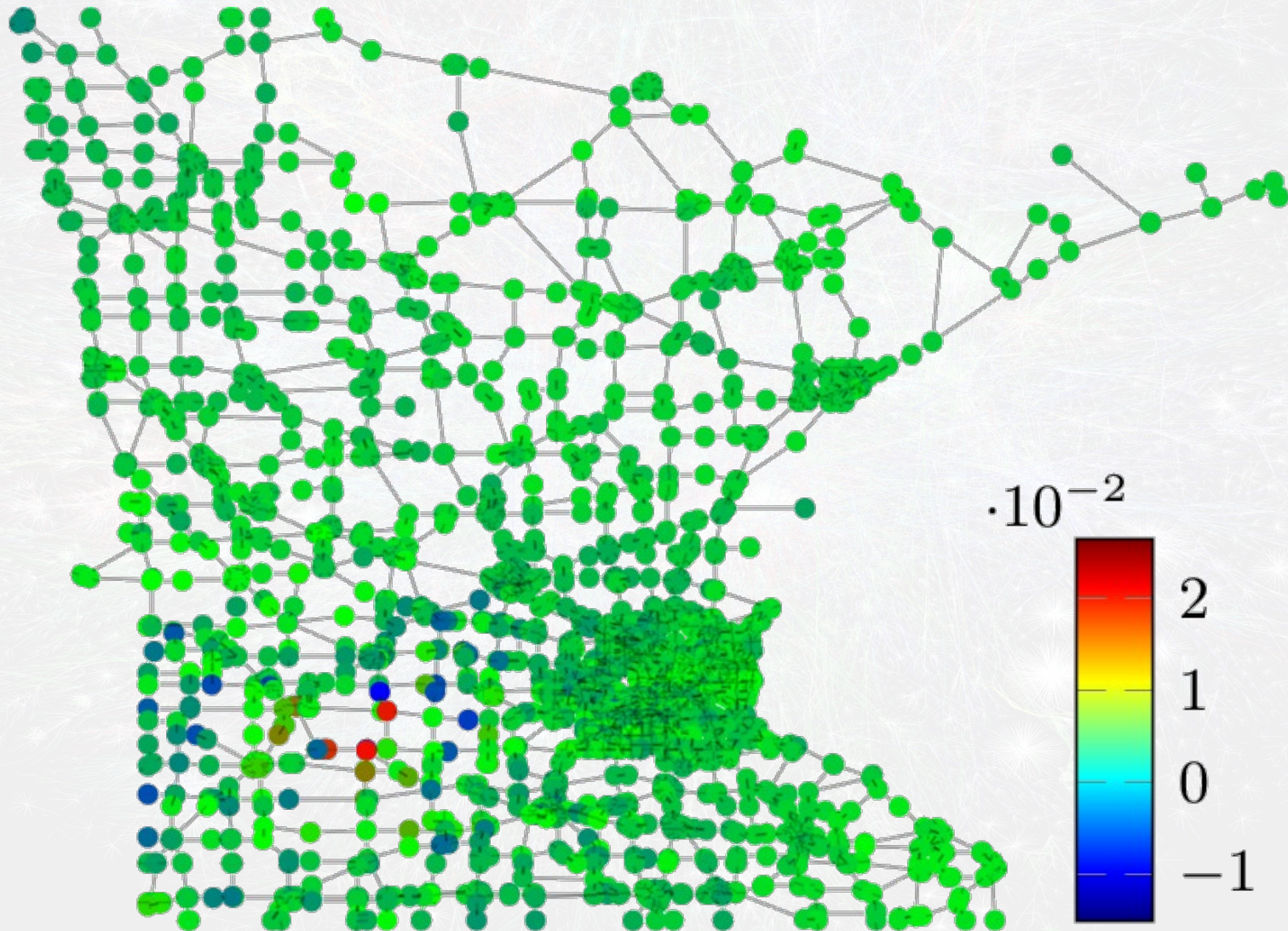
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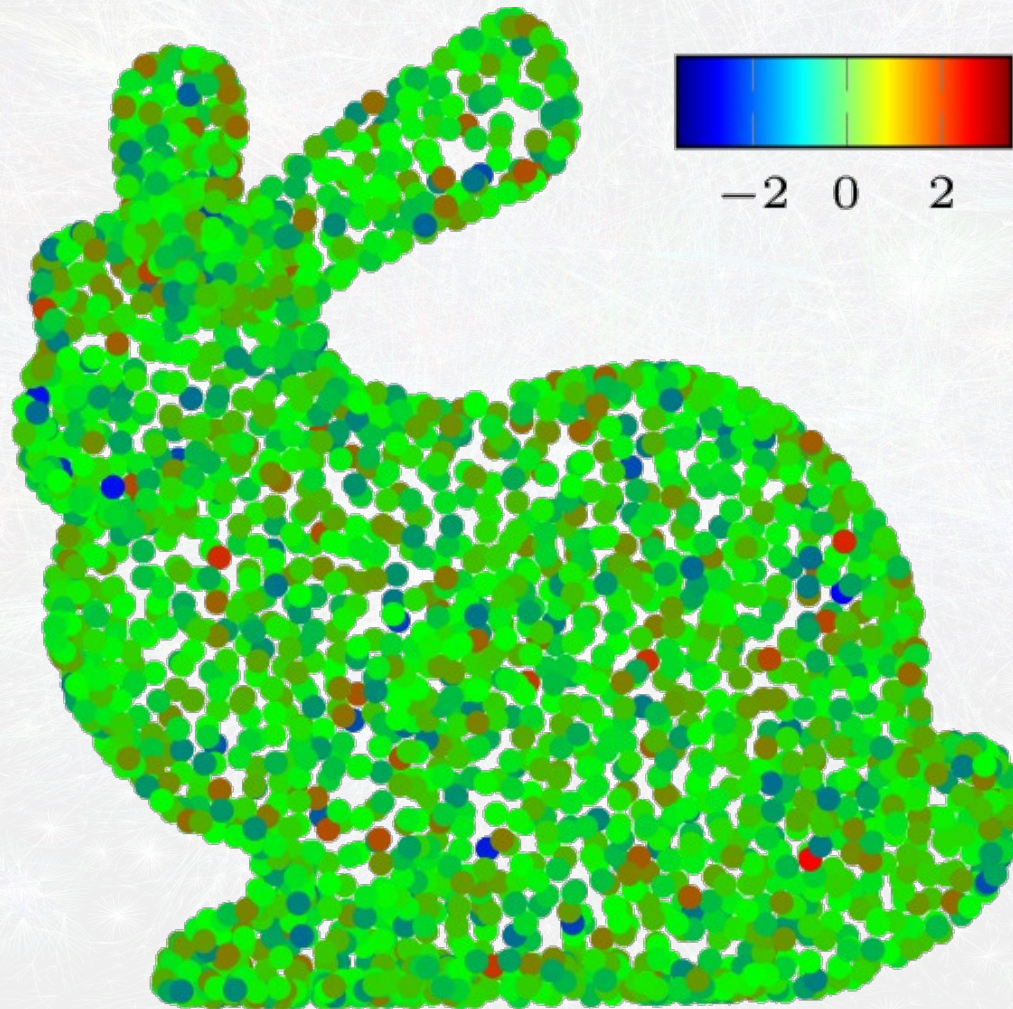
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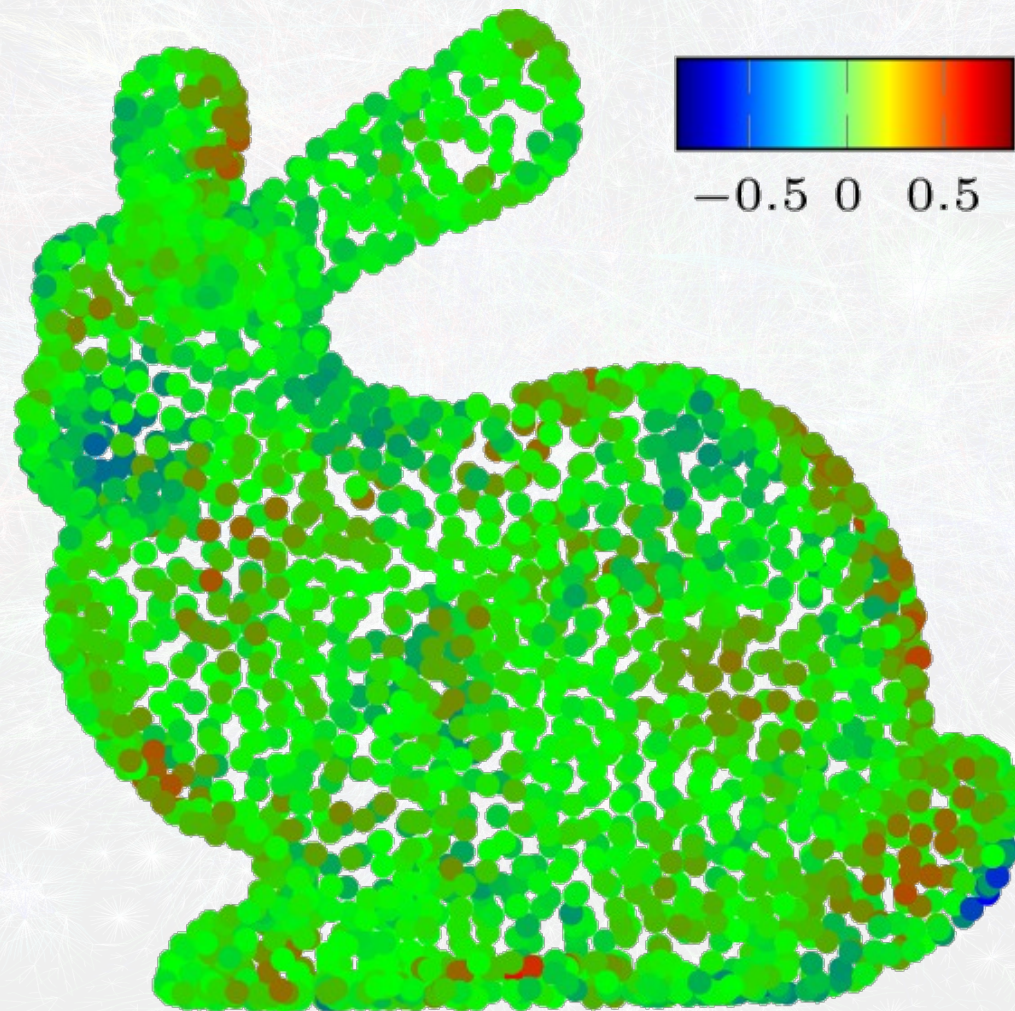
A few words on graph signal processing

The main ideas – Diffusing signals



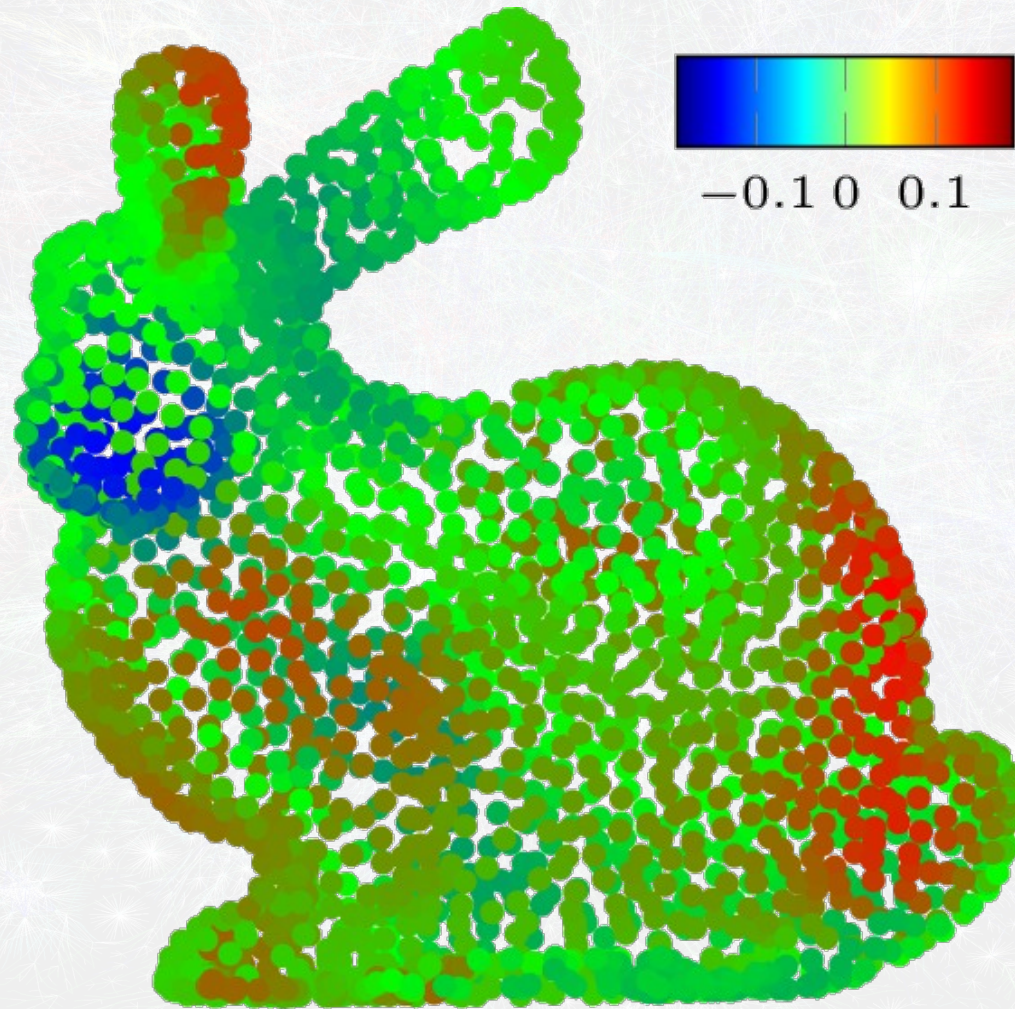
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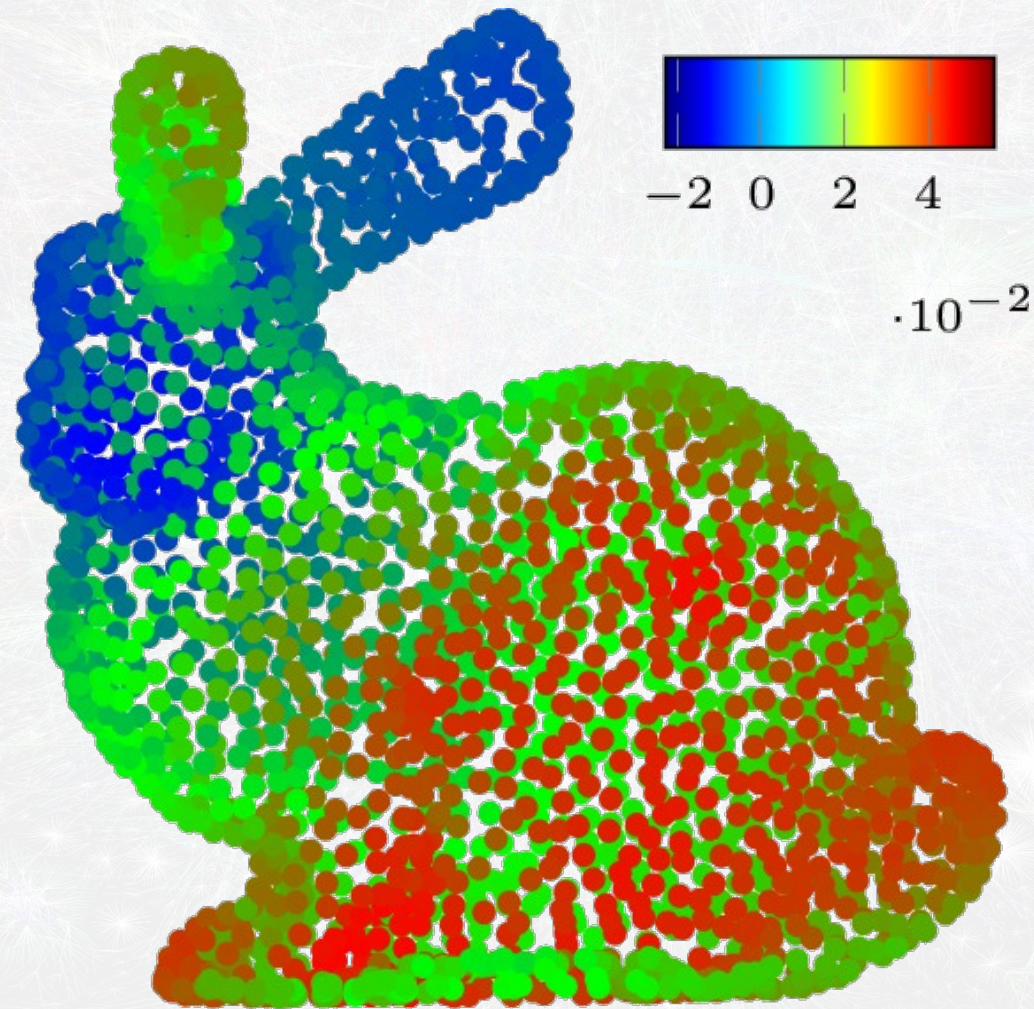
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A few words on graph signal processing

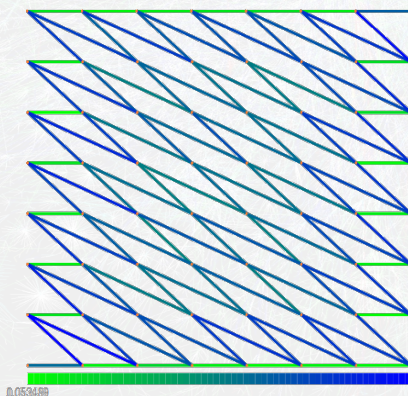
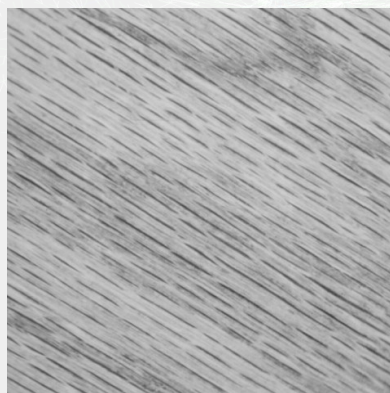
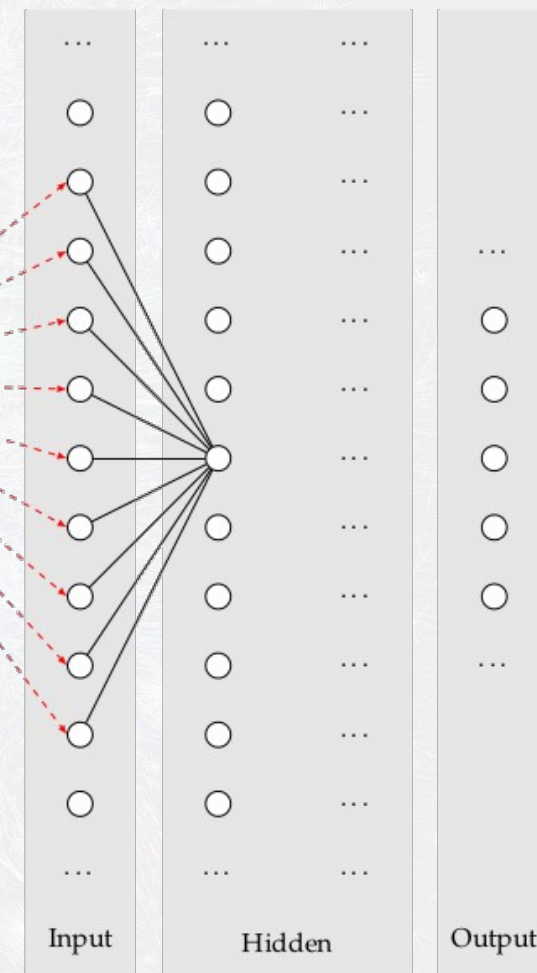
The main ideas – Diffusing signals



A few words on graph signal processing

The main ideas – Some problems and applications

- ▶ Denoising
- ▶ Downsampling
- ▶ Compression of signals
- ▶ Patterns recognition
- ▶ Machine learning
- ▶ Understanding the brain
- ▶ ...



► A few words on graph signal processing

- * What is it?
- * The main ideas
- * Some problems and applications

► Graph inference from signals

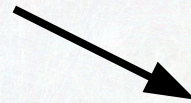
- * Why it may be necessary
- * The case of stationary signals

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- * How we define them
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- * Examples and applications

Graph inference from signals

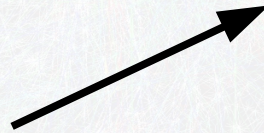
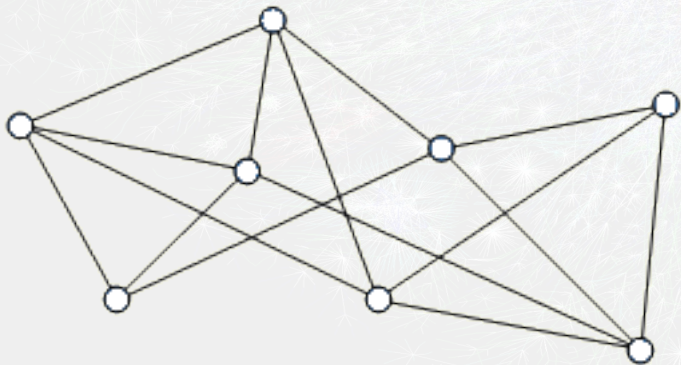
Why it may be necessary? – The problem



GSP
tools

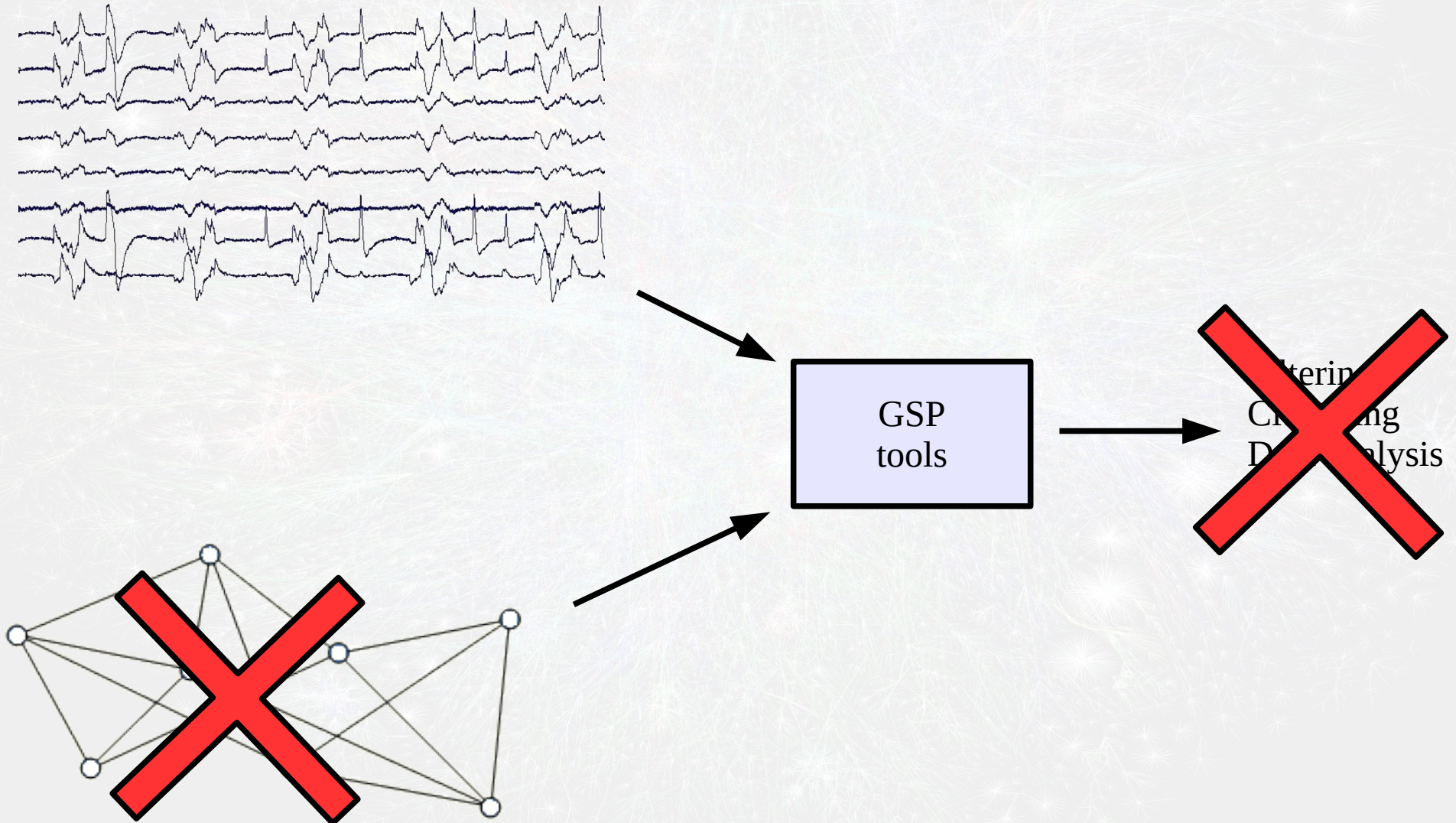


Filtering
Clustering
Data analysis
...



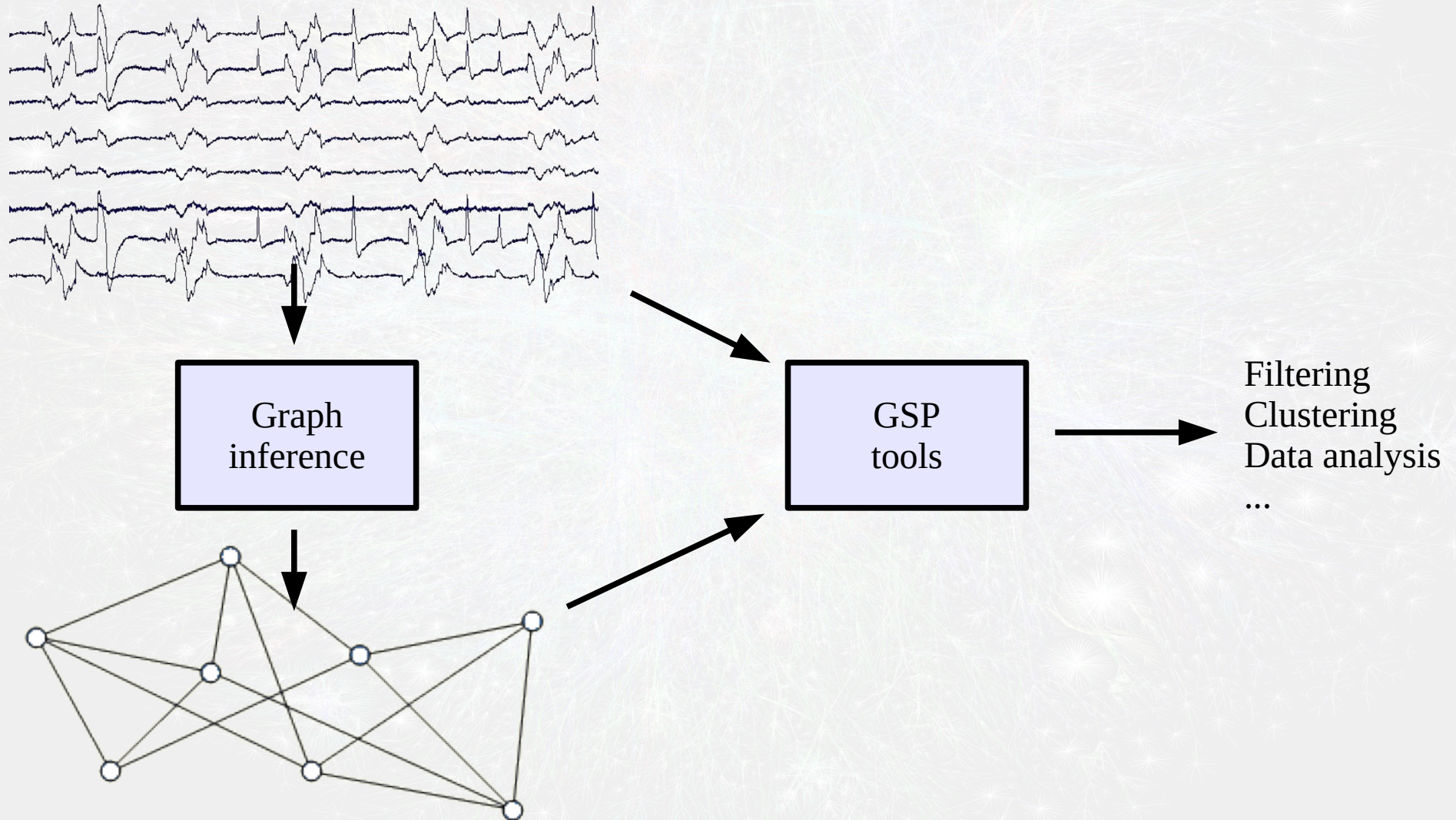
Graph inference from signals

Why it may be necessary? – The problem



Graph inference from signals

Why it may be necessary? – A possible solution

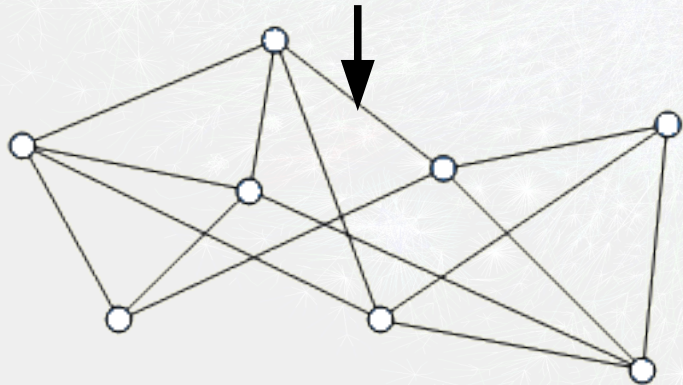


Graph inference from signals

Why it may be necessary? – An ill-posed problem



Graph
inference



Signals:

Need for a model

Graph:

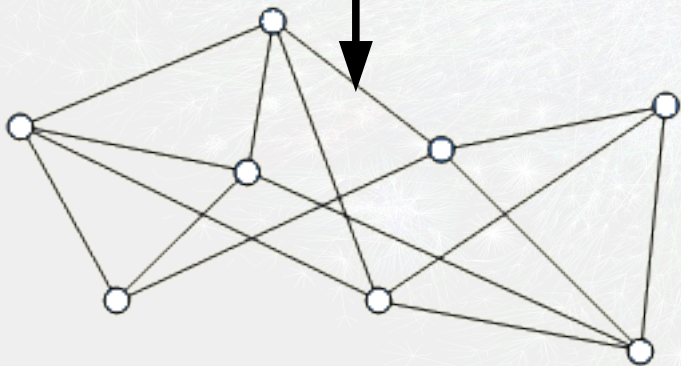
Need for assumptions

Graph inference from signals

Why it may be necessary? – An ill-posed problem



Graph
inference



Signals:

Need for a model

- ▶ Stationary processes

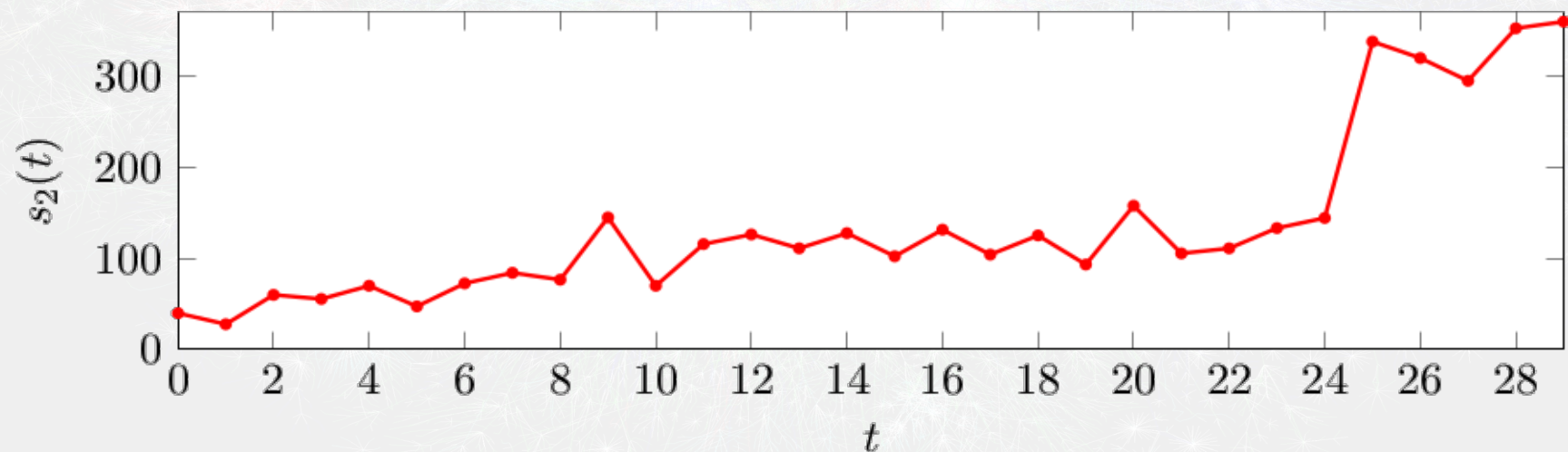
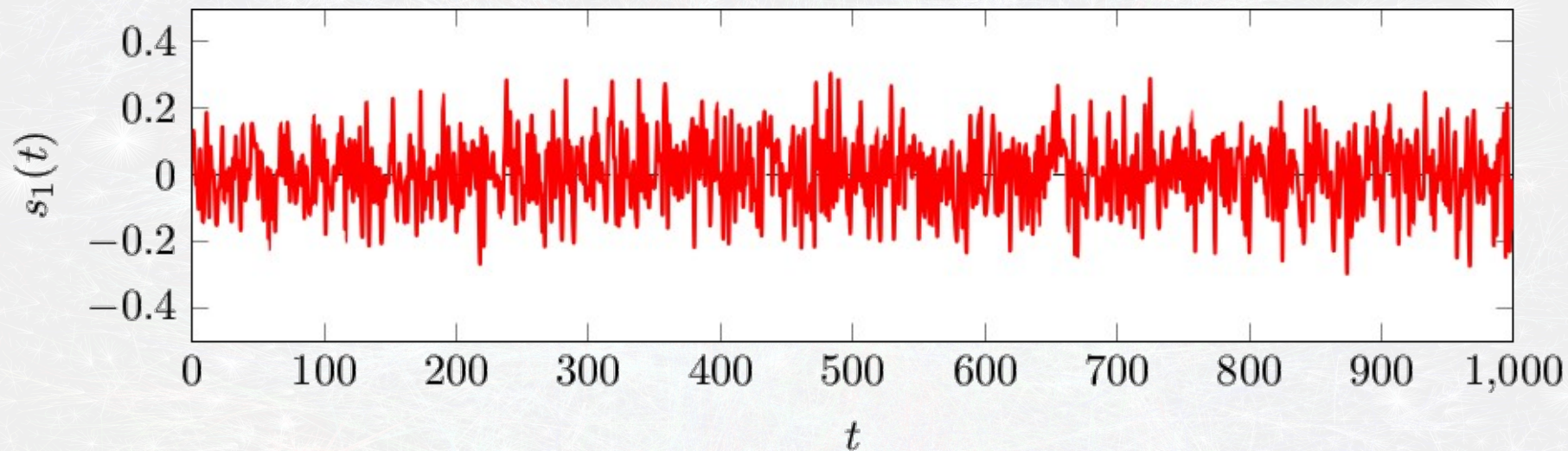
Graph:

Need for assumptions

- ▶ Valid for stationary processes
- ▶ Desired features

Graph inference from signals

The case of stationary signals – Stationarity in classical signal processing



Graph inference from signals

The case of stationary signals – Stationarity in graph signal processing

Definition:

A stochastic graph signal $\mathbf{x}$ defined on the vertices of a graph $\mathcal{G}$ is called *graph wide-sense (or second order) stationary (GWSS)*, if and only if it satisfies the following properties:

- * Its first moment is constant over the vertex set: $m_{\mathbf{x}}[i] = \mathbb{E}\{\mathbf{x}[i]\} = c \in \mathbb{R}$
- * Its covariance is invariant with respect to the localization operator:
$$\Sigma_{\mathbf{x}}[i, n] = \mathbb{E}\{(\mathbf{x}[i] - m_{\mathbf{x}})(\mathbf{x}[n] - m_{\mathbf{x}})\} = \mathcal{T}_i \gamma_{\mathbf{x}}[n]$$

Remark:

White noise is GWSS

Theorem:

When a graph filter g is applied to a GWSS signal, the result remains GWSS

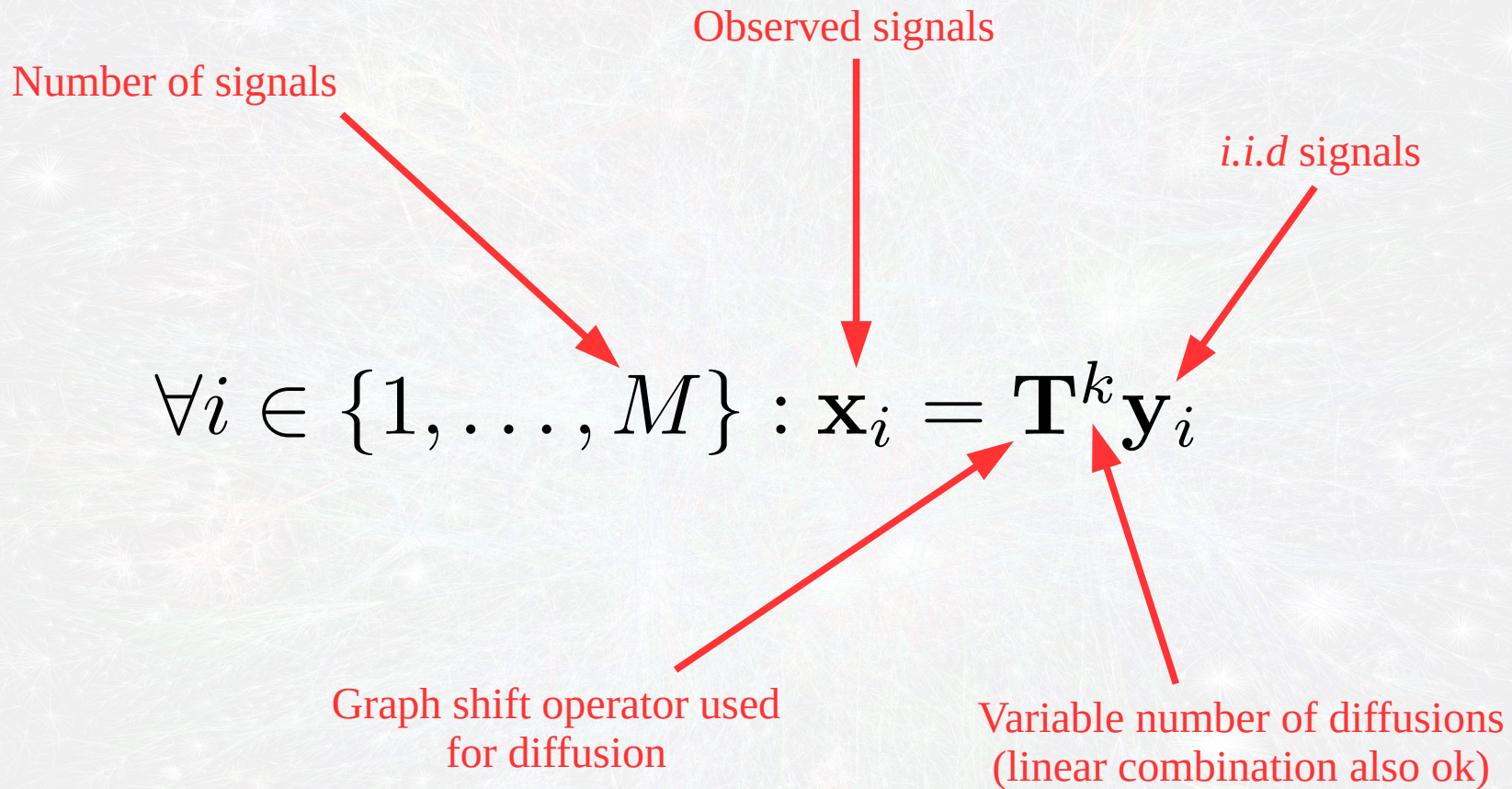
Graph inference from signals

The case of stationary signals – Diffusion as a particular case of stationarity

$$\forall i \in \{1, \dots, M\} : \mathbf{x}_i = \mathbf{T}^k \mathbf{y}_i$$

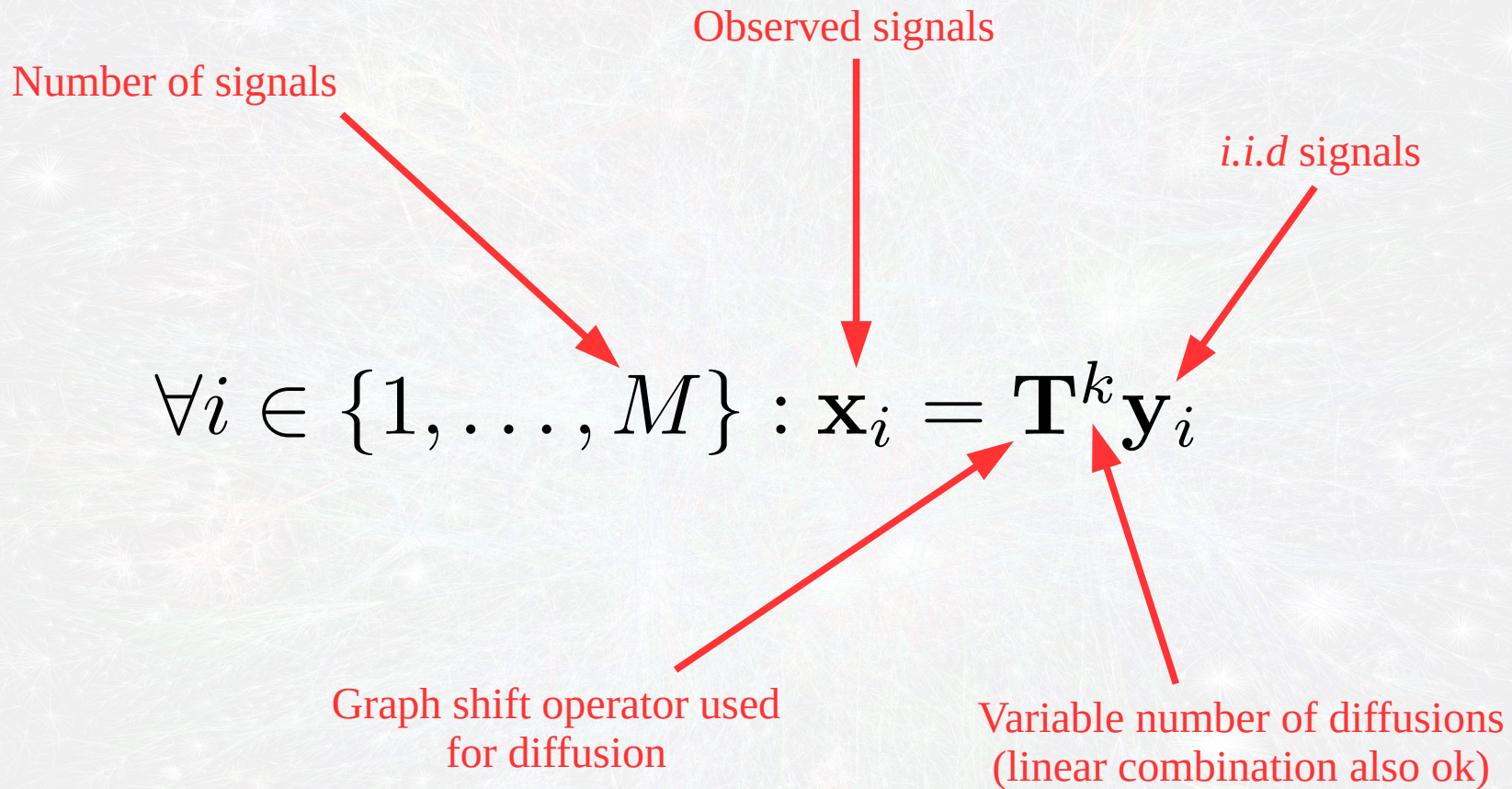
Graph inference from signals

The case of stationary signals – Diffusion as a particular case of stationarity



Graph inference from signals

The case of stationary signals – Diffusion as a particular case of stationarity

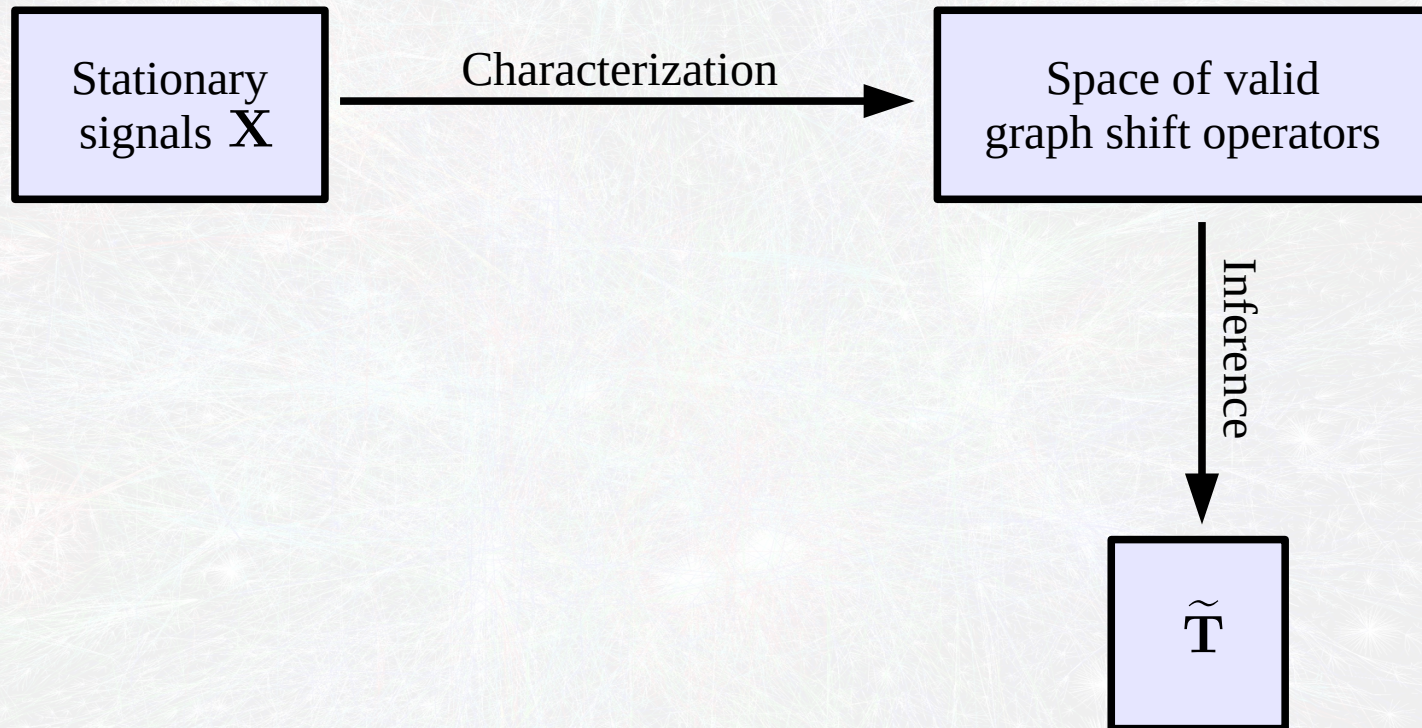


For every value of k :

$$\mathbf{\Sigma} = \mathbb{E}[\mathbf{X}\mathbf{X}^\top] = \mathbb{E}[\mathbf{T}^k \mathbf{Y}\mathbf{Y}^\top \mathbf{T}^{k\top}] = \mathbf{T}^k \mathbb{E}[\mathbf{Y}\mathbf{Y}^\top] \mathbf{T}^{k\top} = \mathbf{T}^{2k}$$

Graph inference from signals

The case of stationary signals – Problem formulation



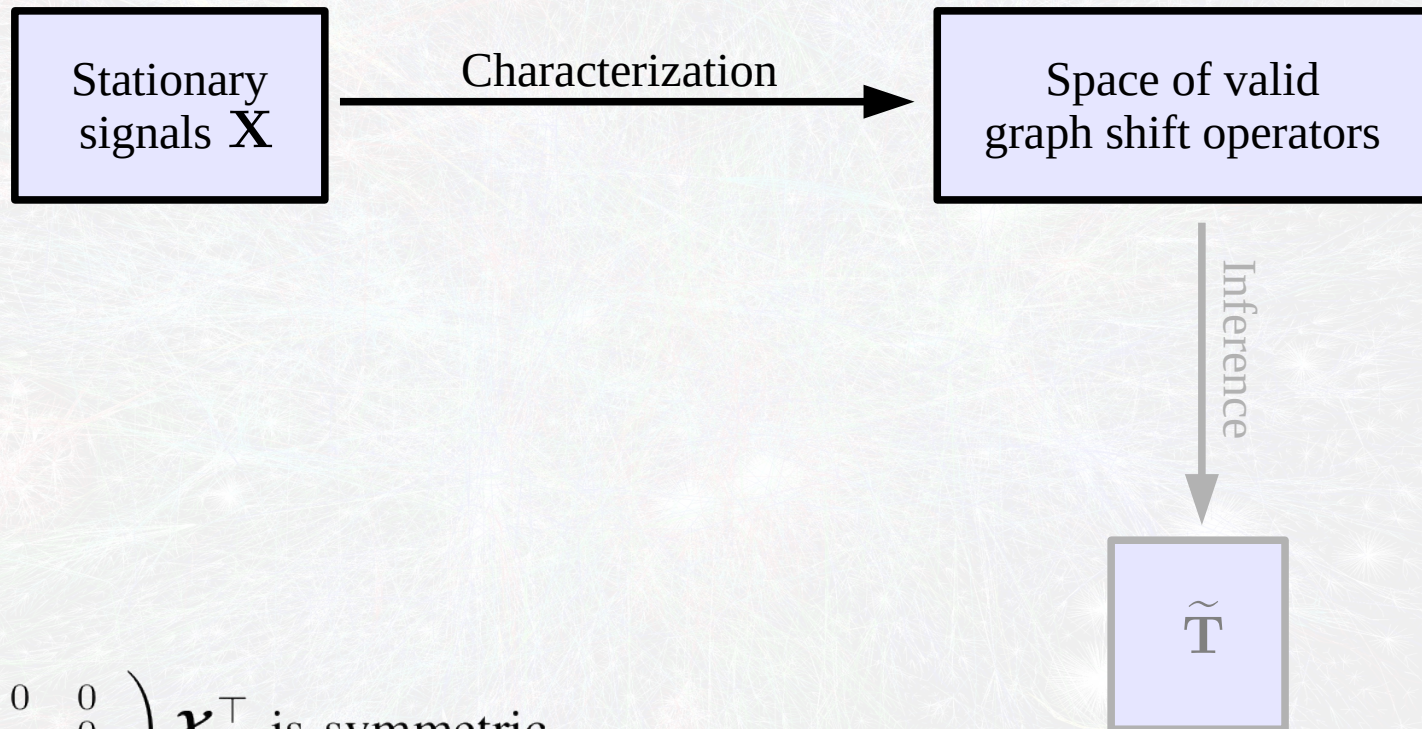
Remarks:

- * Characterization step enforces coherence between the signal model and the topology
- * Inference requires selection criteria

Graph inference from signals

The case of stationary signals – Characterization of the valid graphs

Characterization step enforces coherence between the signal model and the topology



- $\tilde{\mathbf{T}} \triangleq \mathbf{X} \begin{pmatrix} \tilde{\lambda}_1 & 0 & 0 \\ 0 & \dots & 0 \\ 0 & 0 & \tilde{\lambda}_N \end{pmatrix} \mathbf{X}^\top$ is symmetric.
- $\forall i, j \in \{1, \dots, N\}; j \geq i : \tilde{\mathbf{T}}(i, j) \geq 0$.
- Let χ_1 be the constant-sign eigenvector in $\mathbf{X}$: $\tilde{\lambda}_1 = 1$.
- $\forall i \in \{1, \dots, N\} : \tilde{\lambda}_i \in [-1, 1]$.

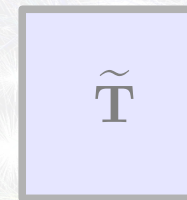
Graph inference from signals

The case of stationary signals – Characterization of the valid graphs

Characterization step enforces coherence between the signal model and the topology



Inference



Eigenvectors of the covariance matrix



- $\tilde{\mathbf{T}} \triangleq \mathbf{X} \begin{pmatrix} \tilde{\lambda}_1 & 0 & 0 \\ 0 & \dots & 0 \\ 0 & 0 & \tilde{\lambda}_N \end{pmatrix} \mathbf{X}^\top$ is symmetric. } Diffusion matrix properties
- $\forall i, j \in \{1, \dots, N\}; j \geq i : \tilde{\mathbf{T}}(i, j) \geq 0.$ }
- Let χ_1 be the constant-sign eigenvector in $\mathcal{X}$: $\tilde{\lambda}_1 = 1.$ } Scale factor
- $\forall i \in \{1, \dots, N\} : \tilde{\lambda}_i \in [-1, 1].$ }

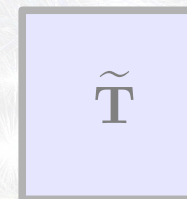
Graph inference from signals

The case of stationary signals – Characterization of the valid graphs

Characterization step enforces coherence between the signal model and the topology



Inference



Eigenvectors of the covariance matrix

Unknown

- $\tilde{\mathbf{T}} \triangleq \mathbf{X} \begin{pmatrix} \tilde{\lambda}_1 & 0 & 0 \\ 0 & \dots & 0 \\ 0 & 0 & \tilde{\lambda}_N \end{pmatrix} \mathbf{X}^\top$ is symmetric. } Diffusion matrix properties
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Graph inference from signals

The case of stationary signals – The constraints yield a convex polytope

$$\forall i, j \in \{1, \dots, N\}; j \geq i : \tilde{\mathbf{T}}(i, j) \geq 0$$

$$\tilde{\mathbf{T}} = \mathbf{x} \begin{pmatrix} \widetilde{\lambda}_1 & 0 & 0 \\ 0 & \dots & 0 \\ 0 & 0 & \widetilde{\lambda}_N \end{pmatrix} \mathbf{x}^\top$$

$$\tilde{\mathbf{T}}(i, j) = \alpha_{ij1} \widetilde{\lambda}_1 + \dots + \alpha_{ijN} \widetilde{\lambda}_N$$

Graph inference from signals

The case of stationary signals – The constraints yield a convex polytope

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Graph inference from signals

The case of stationary signals – Example

3x3 matrix:

$$\mathbf{T}(2, 3) = \underbrace{\mathcal{X}(2, 1)\mathcal{X}(3, 1)}_{\alpha_{231}} \lambda_1 + \underbrace{\mathcal{X}(2, 2)\mathcal{X}(3, 2)}_{\alpha_{232}} \lambda_2 + \underbrace{\mathcal{X}(2, 3)\mathcal{X}(3, 3)}_{\alpha_{233}} \lambda_3$$

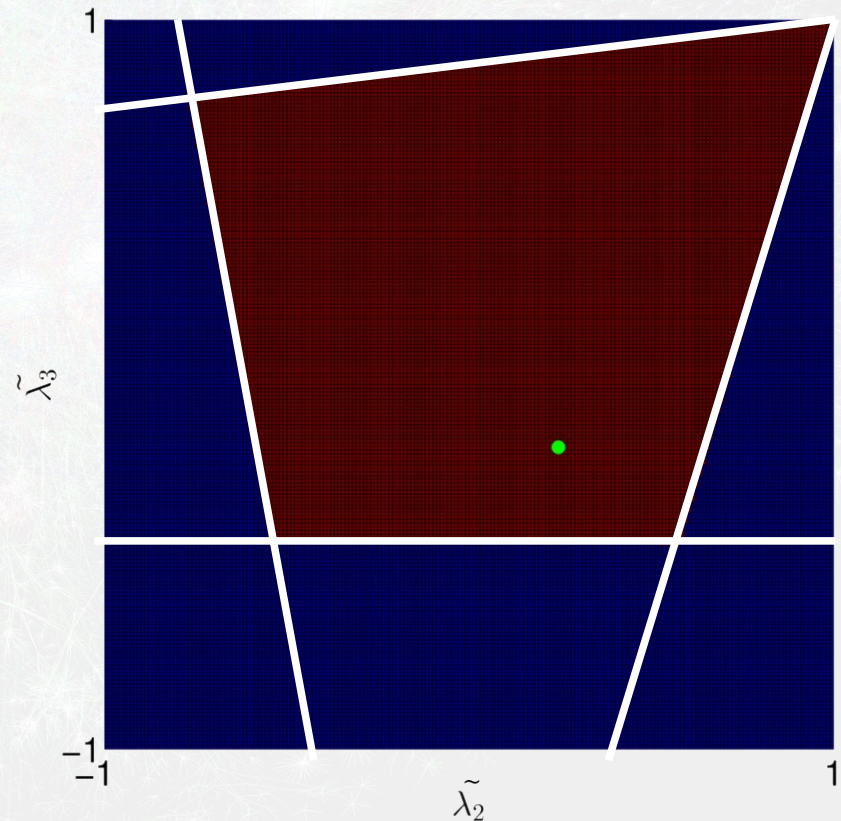
Graph inference from signals

The case of stationary signals – Example

3x3 matrix:

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$$\mathbf{W} = \begin{pmatrix} 0.417 & 0.302 & 0.186 \\ 0.302 & 0.147 & 0.346 \\ 0.186 & 0.346 & 0.397 \end{pmatrix}$$



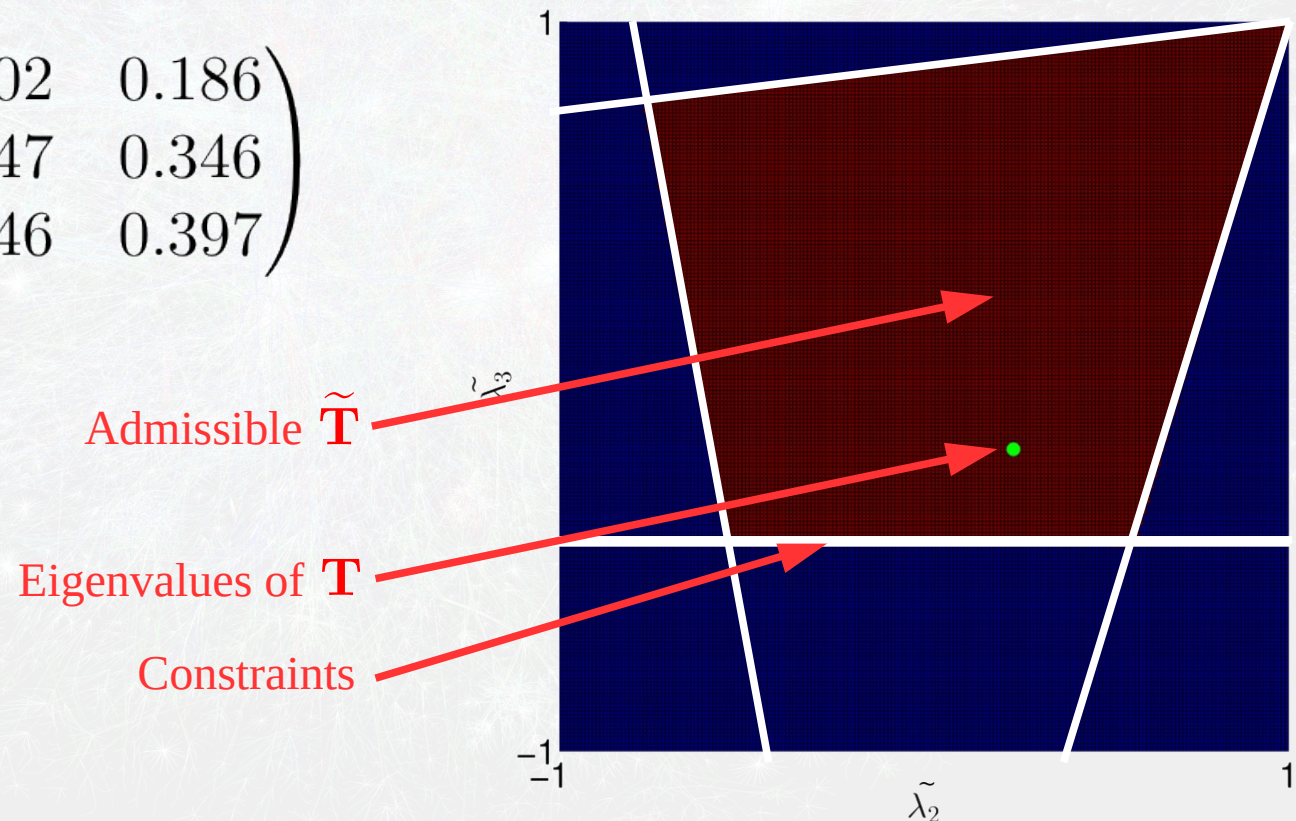
Graph inference from signals

The case of stationary signals – Example

3x3 matrix:

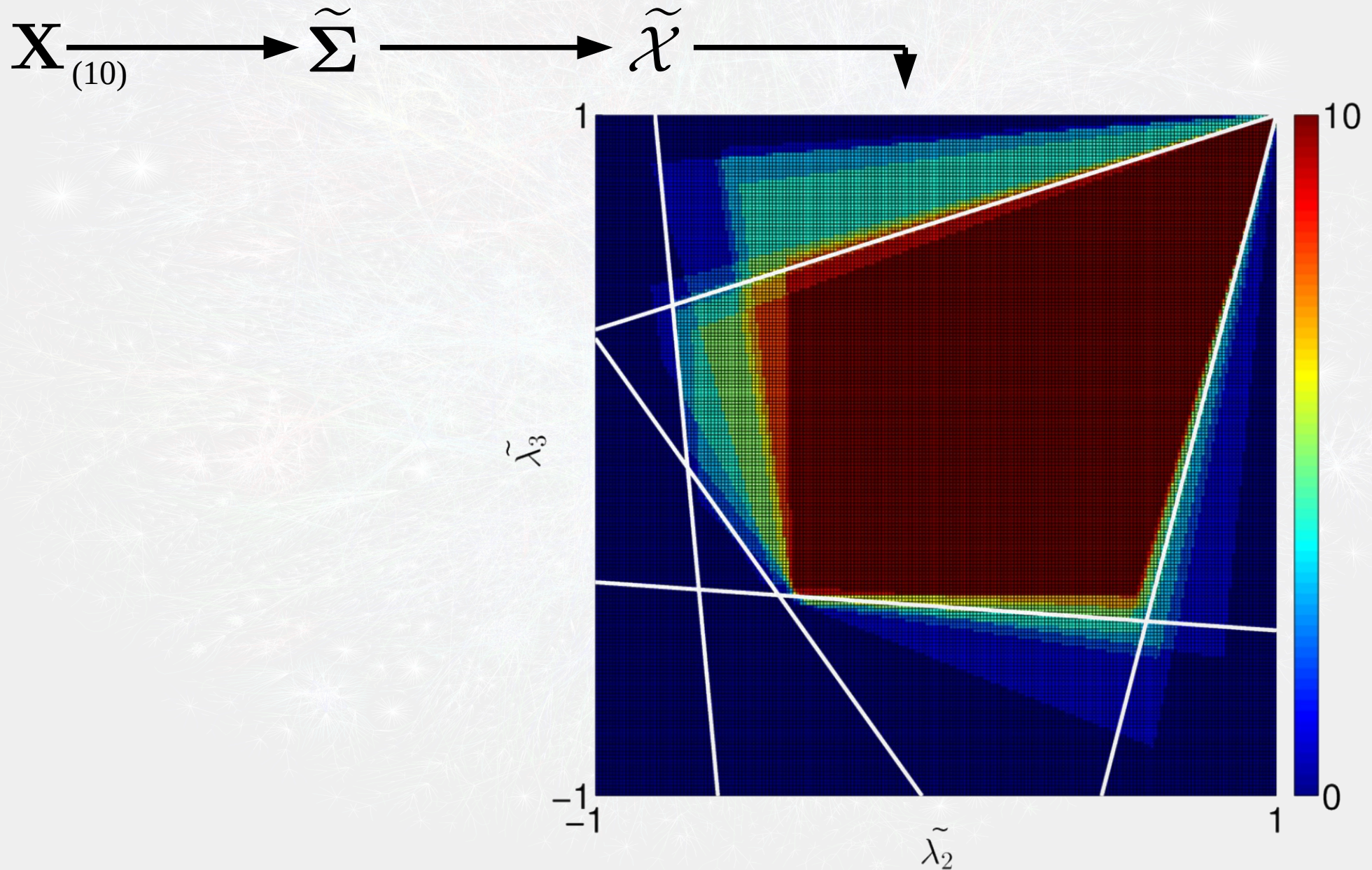
$$\mathbf{T}(2, 3) = \underbrace{\mathcal{X}(2, 1)\mathcal{X}(3, 1)}_{\alpha_{231}} \lambda_1 + \underbrace{\mathcal{X}(2, 2)\mathcal{X}(3, 2)}_{\alpha_{232}} \lambda_2 + \underbrace{\mathcal{X}(2, 3)\mathcal{X}(3, 3)}_{\alpha_{233}} \lambda_3$$

$$\mathbf{W} = \begin{pmatrix} 0.417 & 0.302 & 0.186 \\ 0.302 & 0.147 & 0.346 \\ 0.186 & 0.346 & 0.397 \end{pmatrix}$$



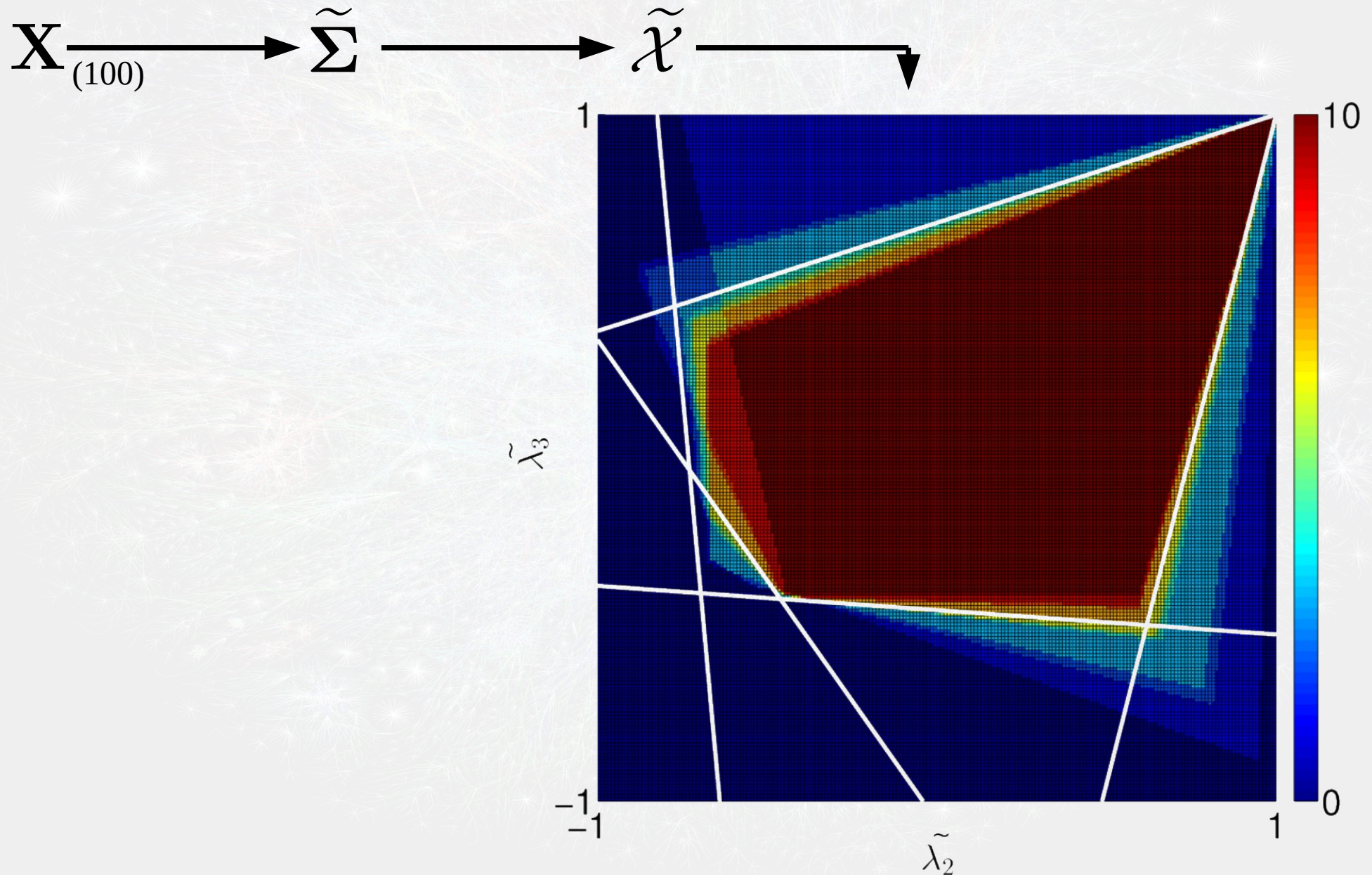
Graph inference from signals

The case of stationary signals – Using the sample covariance matrix



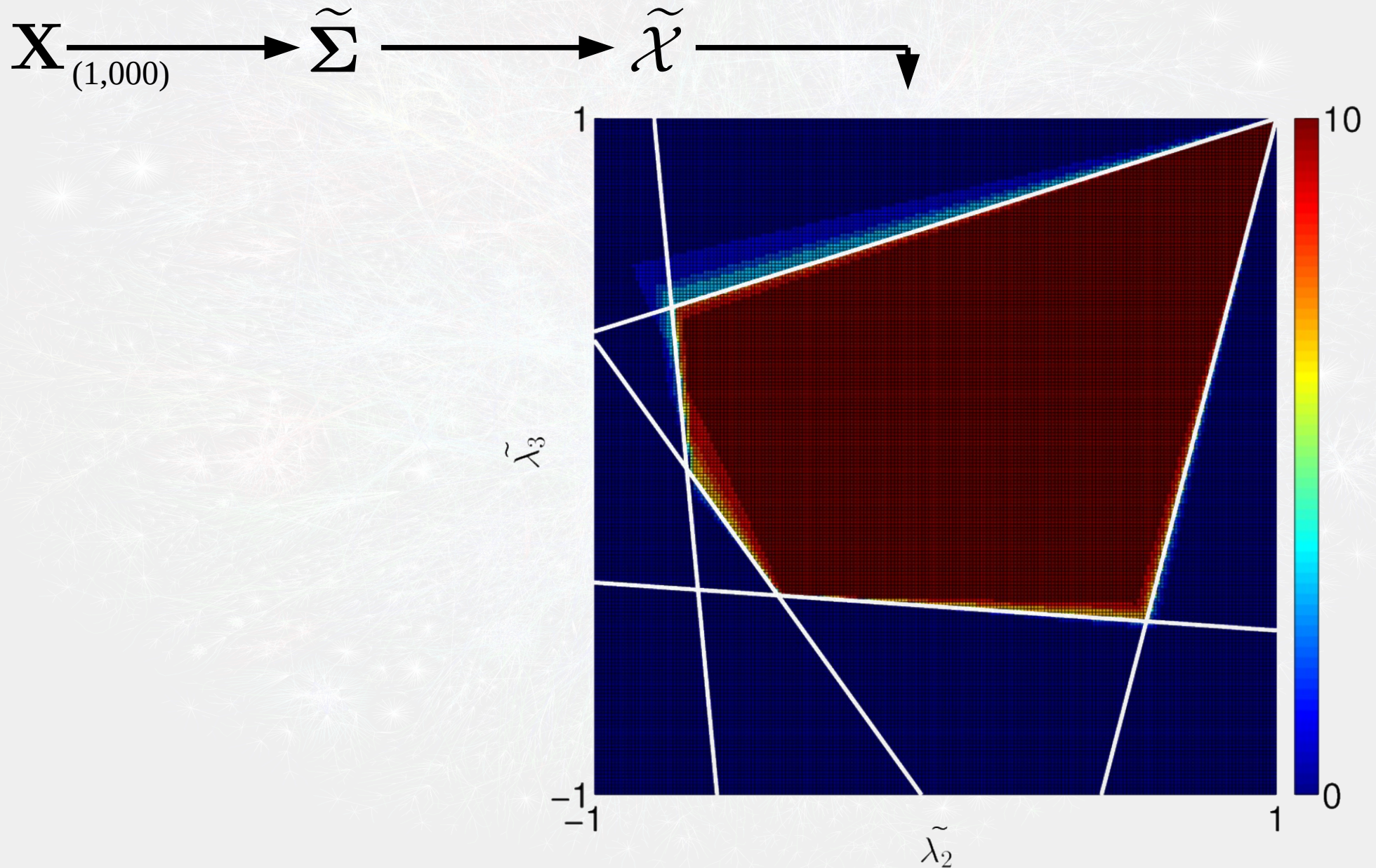
Graph inference from signals

The case of stationary signals – Using the sample covariance matrix



Graph inference from signals

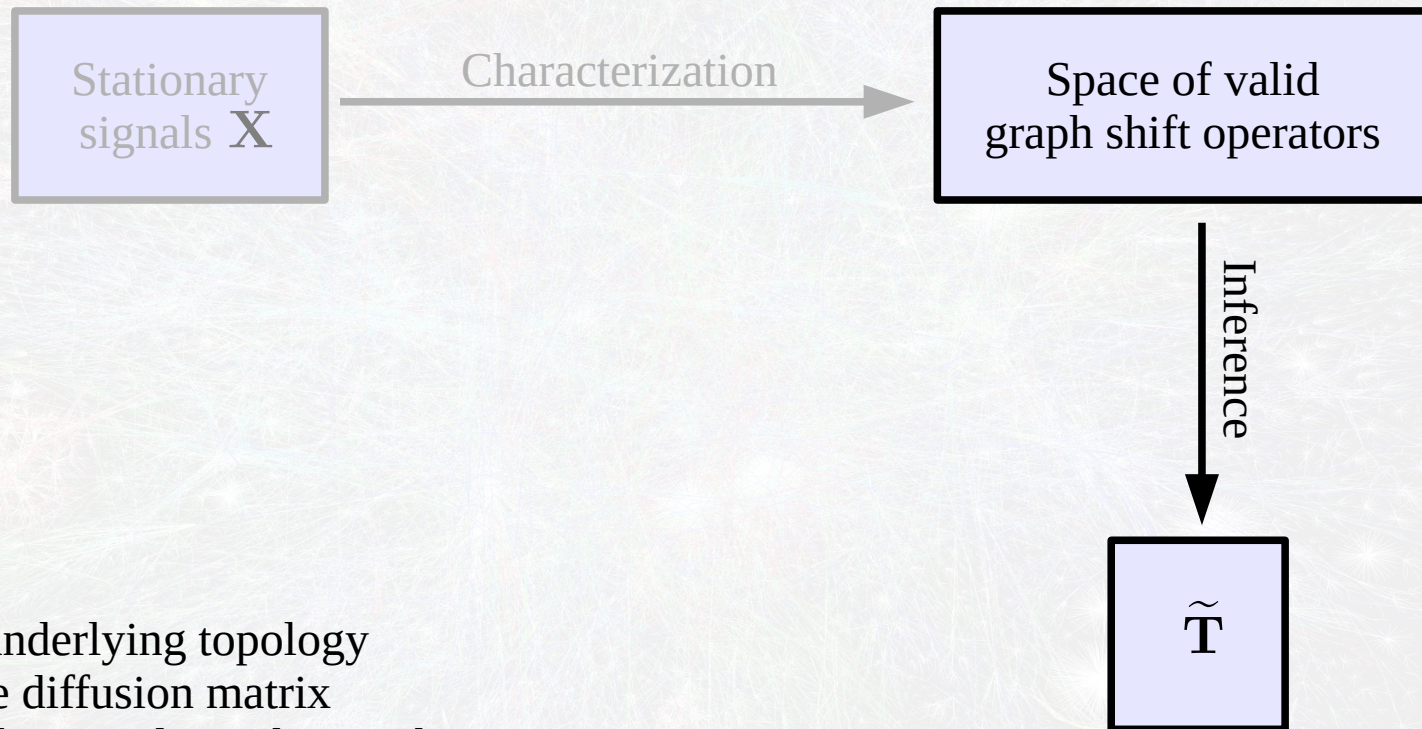
The case of stationary signals – Using the sample covariance matrix



Graph inference from signals

The case of stationary signals – Selection of a graph

Inference requires selection criteria

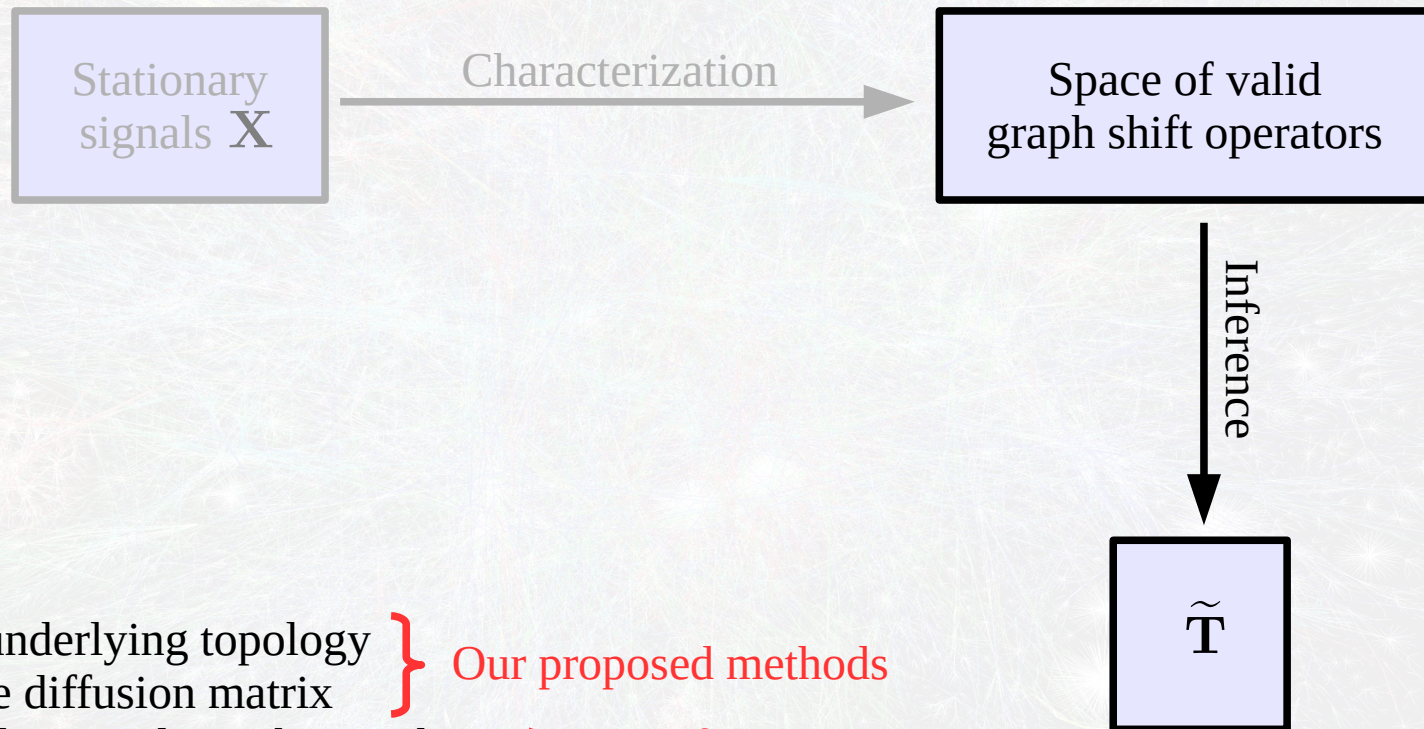


- Sparsity of the underlying topology
- Simplicity of the diffusion matrix
- Smoothness of the signals on the graph
- Inference of a (close to) binary matrix
- ...

Graph inference from signals

The case of stationary signals – Selection of a graph

Inference requires selection criteria



- Sparsity of the underlying topology
 - Simplicity of the diffusion matrix
 - Smoothness of the signals on the graph
 - Inference of a (close to) binary matrix
 - ...
- Our proposed methods
- Kalofolias?

Graph inference from signals

The case of stationary signals – Inference of a simple matrix

$$\widetilde{\lambda}_1, \dots, \widetilde{\lambda}_N = \arg \min_{\lambda_1, \dots, \lambda_N} \sum_{i=1}^N \lambda_i$$
$$\text{s. t. } \begin{cases} \text{Polytope constraints} \\ \forall i \in \{1, \dots, N\} : \lambda_i \in [-1, 1] \\ \lambda_1 = 1 \end{cases}$$

Graph inference from signals

The case of stationary signals – Inference of a simple matrix

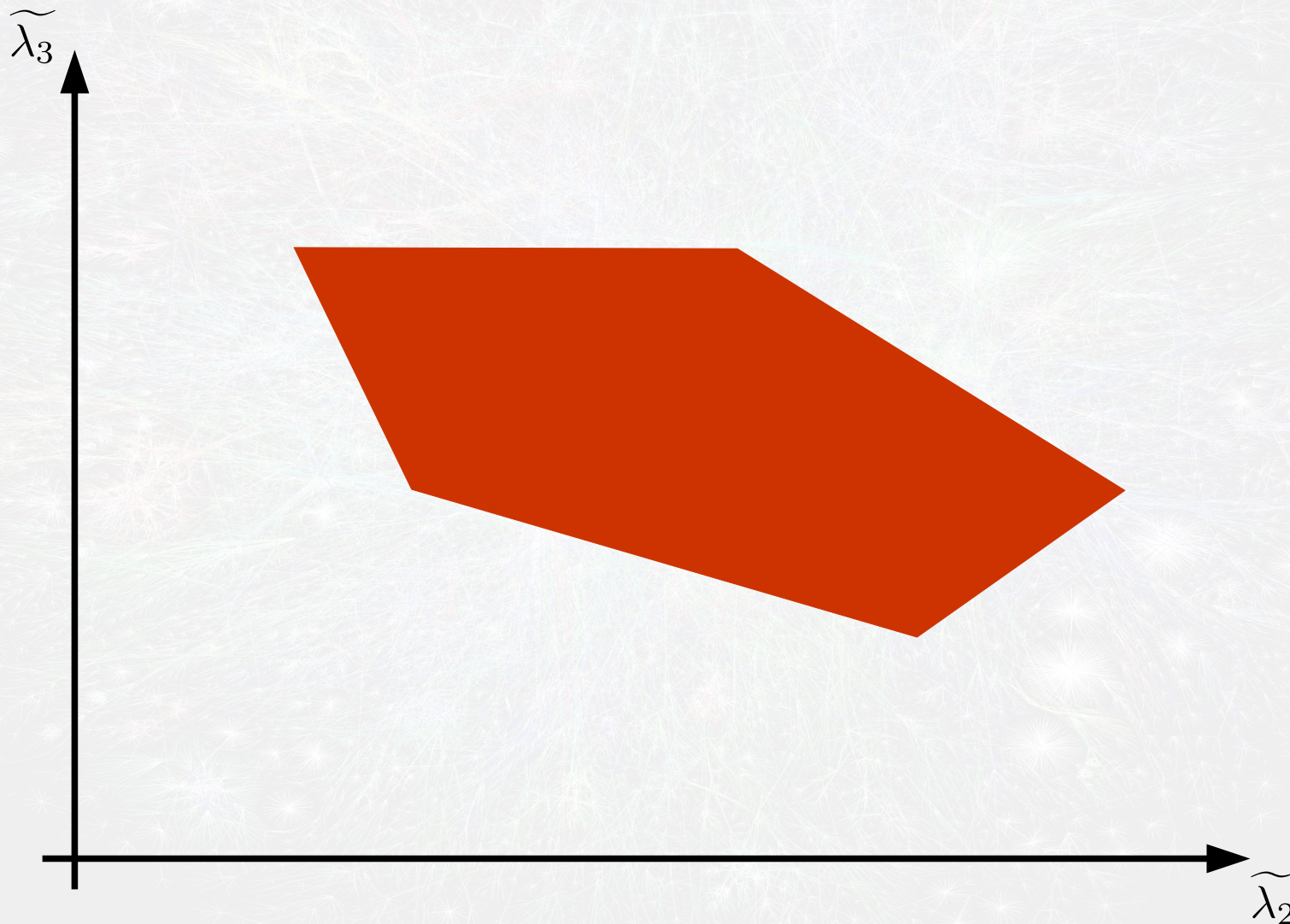
$$\widetilde{\lambda}_1, \dots, \widetilde{\lambda}_N = \arg \min_{\lambda_1, \dots, \lambda_N} \sum_{i=1}^N \lambda_i$$

Trace of the matrix

$$\text{s. t. } \begin{cases} \text{Polytope constraints} \\ \forall i \in \{1, \dots, N\} : \lambda_i \in [-1, 1] \\ \lambda_1 = 1 \end{cases}$$

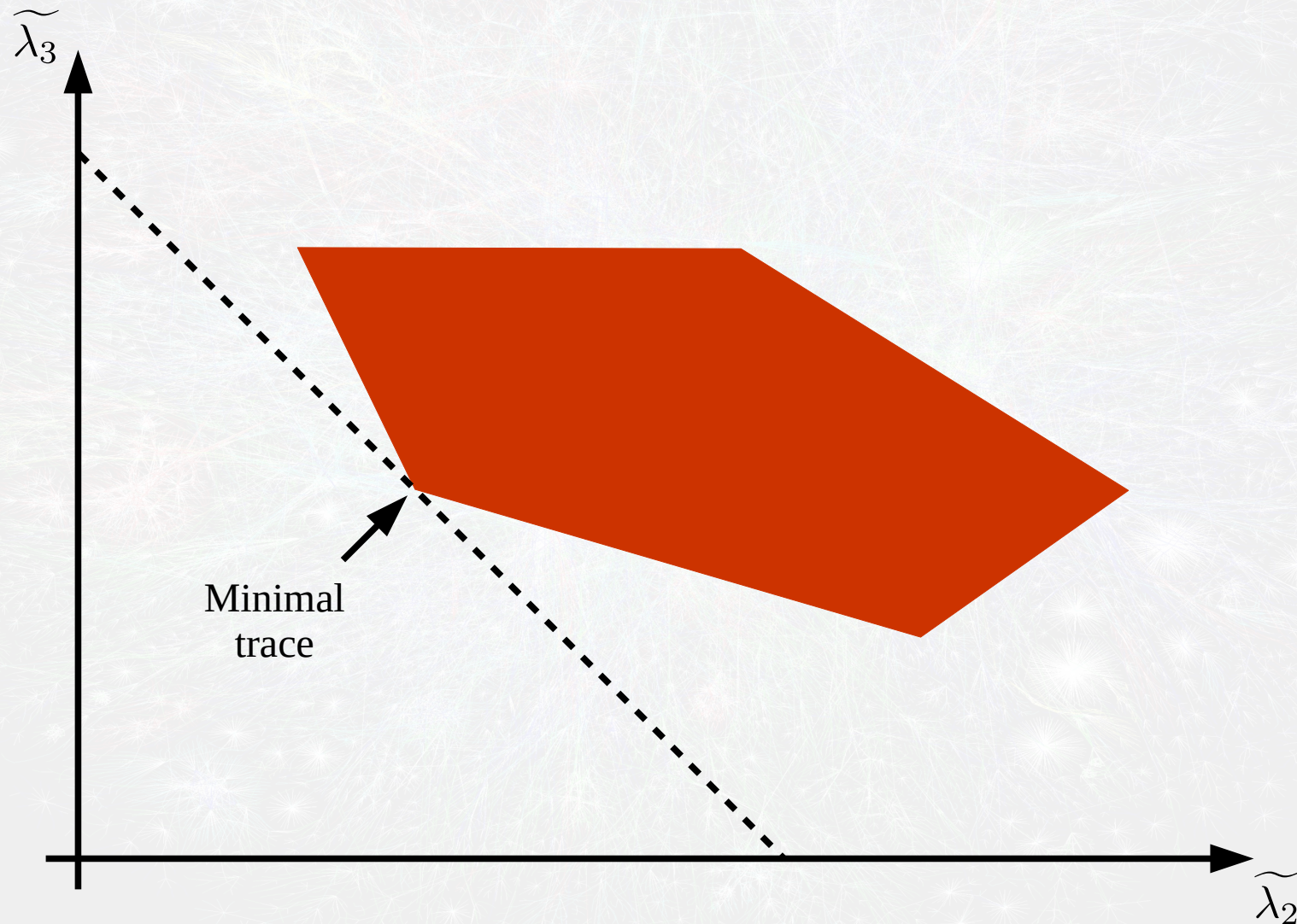
Graph inference from signals

The case of stationary signals – Graphically



Graph inference from signals

The case of stationary signals – Graphically



Graph inference from signals

The case of stationary signals – Inference of a sparse matrix

$$\widetilde{\lambda}_1, \dots, \widetilde{\lambda}_N = \arg \min_{\lambda_1, \dots, \lambda_N} \mathbf{1}_N^\top \mathcal{X} \begin{pmatrix} \lambda_1 & 0 & 0 \\ 0 & \dots & 0 \\ 0 & 0 & \lambda_N \end{pmatrix} \mathcal{X}^\top \mathbf{1}_N$$

$$\text{s. t. } \begin{cases} \text{Polytope constraints} \\ \forall i \in \{1, \dots, N\} : \lambda_i \in [-1, 1] \\ \lambda_1 = 1 \end{cases}$$

Graph inference from signals

The case of stationary signals – Inference of a sparse matrix

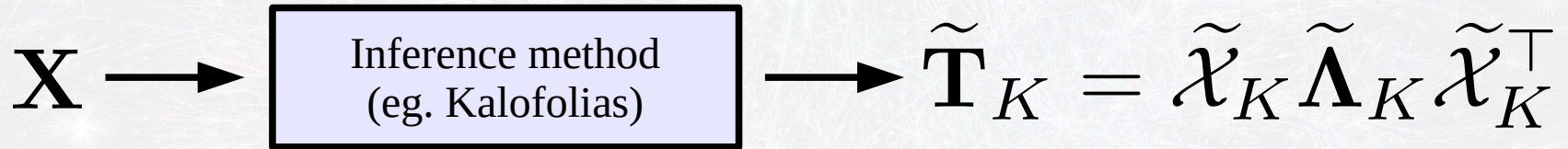
$$\widetilde{\lambda}_1, \dots, \widetilde{\lambda}_N = \arg \min_{\lambda_1, \dots, \lambda_N} \mathbf{1}_N^\top \mathcal{X} \begin{pmatrix} \lambda_1 & 0 & 0 \\ 0 & \dots & 0 \\ 0 & 0 & \lambda_N \end{pmatrix} \mathcal{X}^\top \mathbf{1}_N$$

ℓ_1 norm

$$\text{s. t. } \begin{cases} \text{Polytope constraints} \\ \forall i \in \{1, \dots, N\} : \lambda_i \in [-1, 1] \\ \lambda_1 = 1 \end{cases}$$

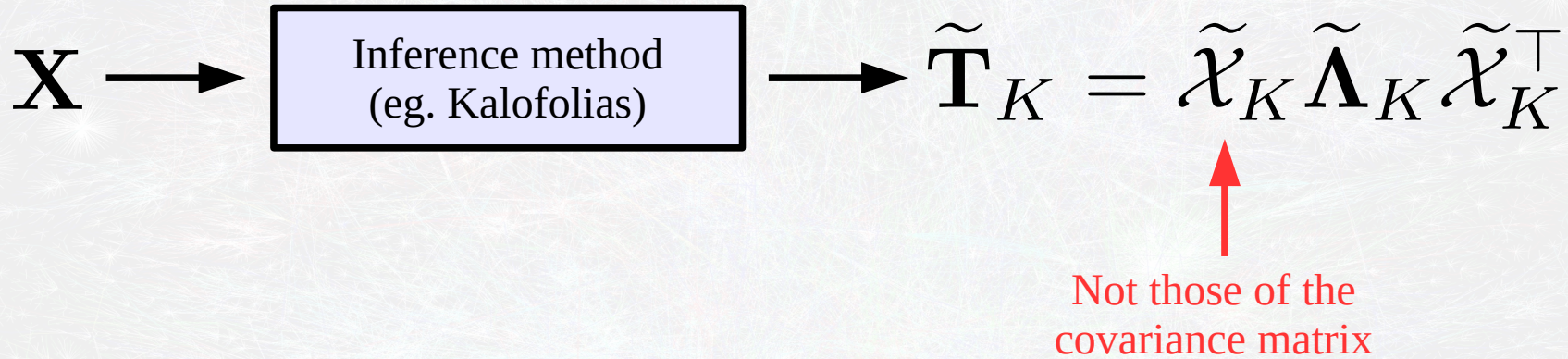
Graph inference from signals

The case of stationary signals – Regularization of other methods



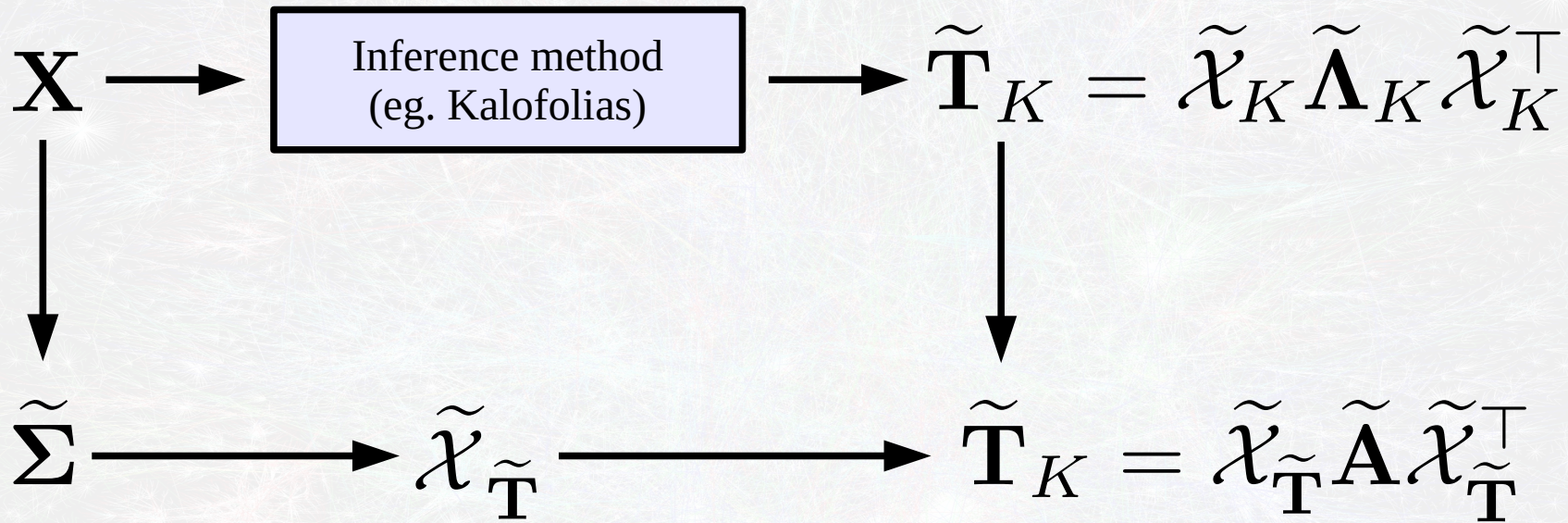
Graph inference from signals

The case of stationary signals – Regularization of other methods



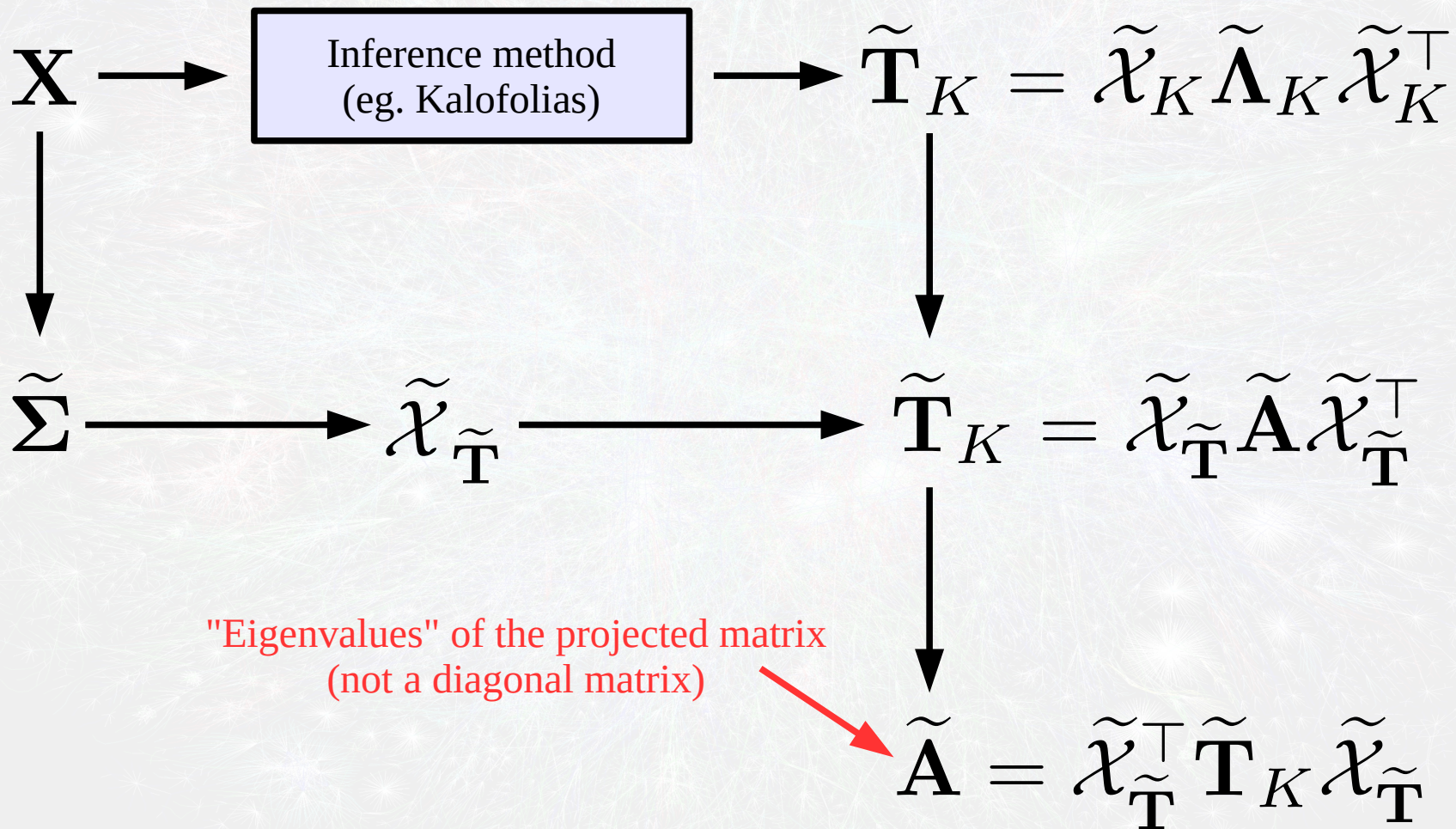
Graph inference from signals

The case of stationary signals – Regularization of other methods



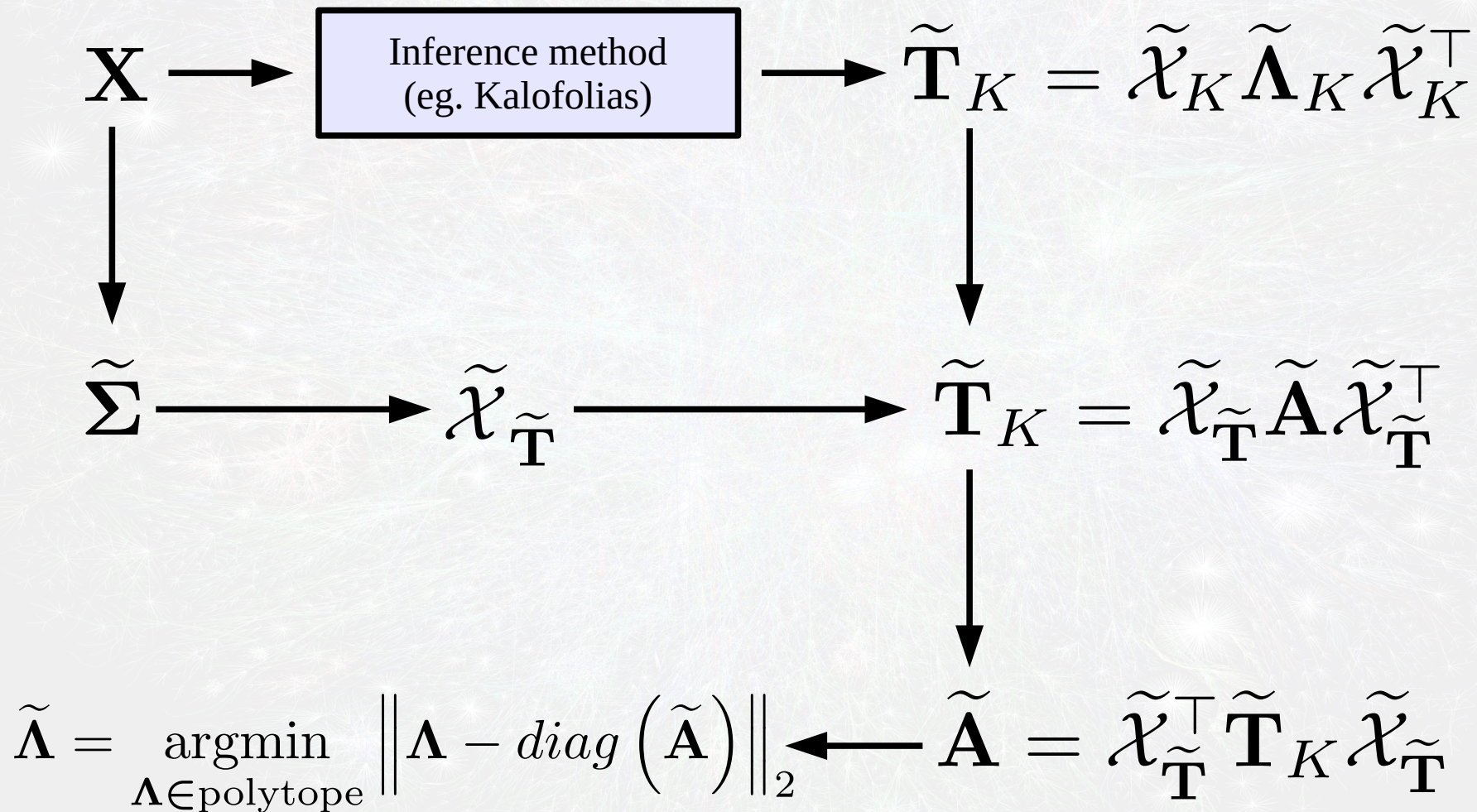
Graph inference from signals

The case of stationary signals – Regularization of other methods



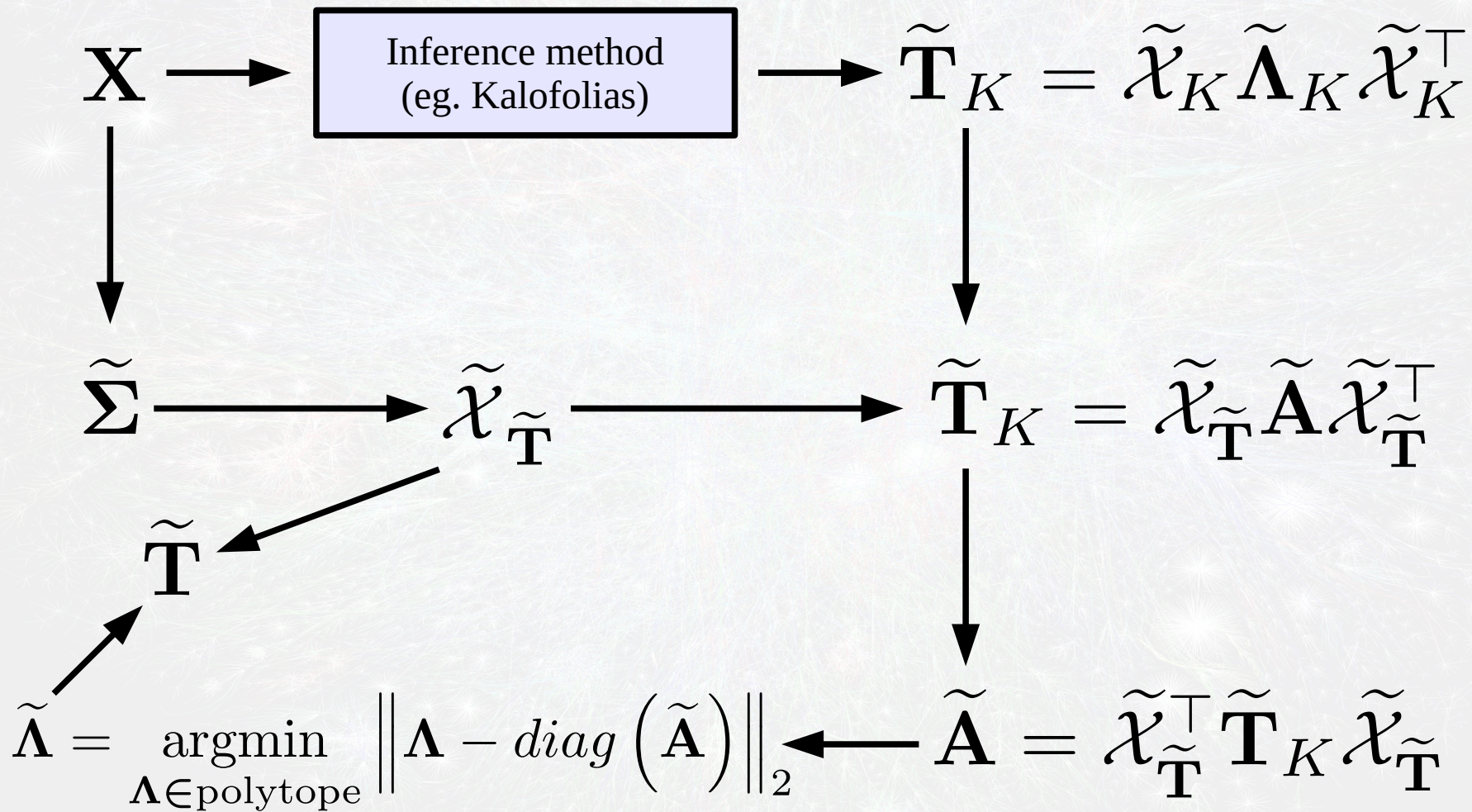
Graph inference from signals

The case of stationary signals – Regularization of other methods



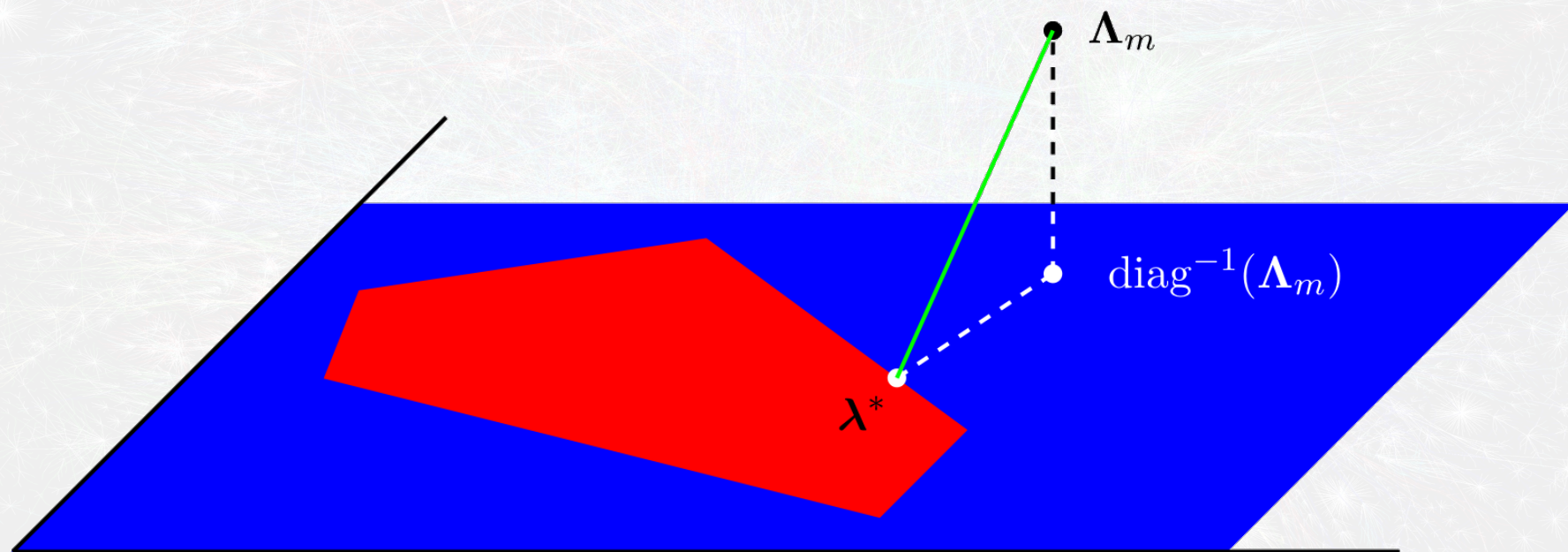
Graph inference from signals

The case of stationary signals – Regularization of other methods



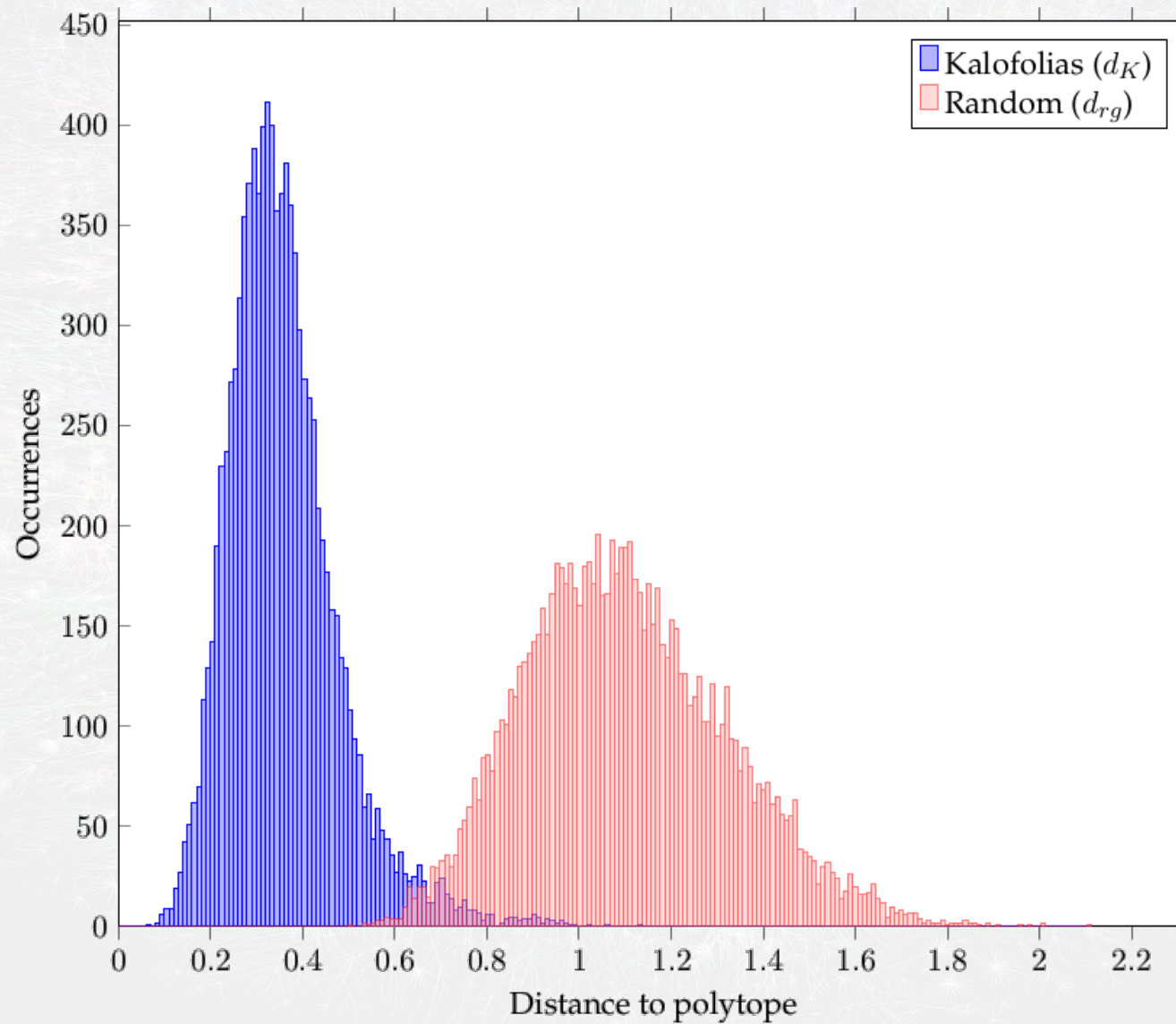
Graph inference from signals

The case of stationary signals – Graphically



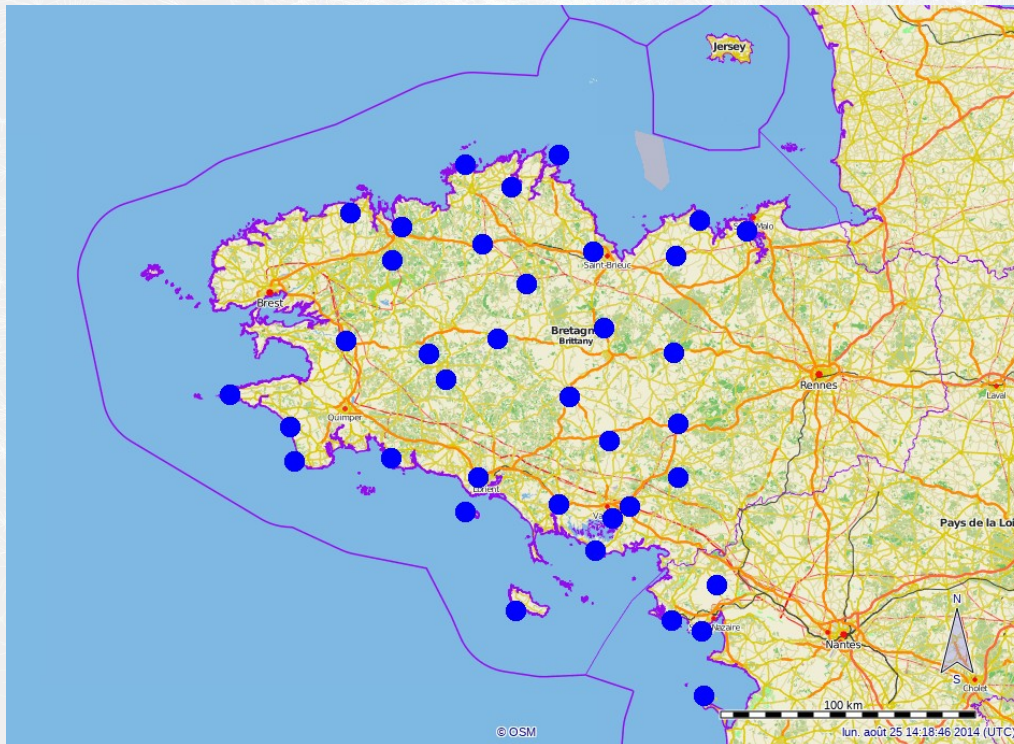
Graph inference from signals

The case of stationary signals – Does it seem right?



Graph inference from signals

The case of stationary signals – Kalofolias regularization on a dataset

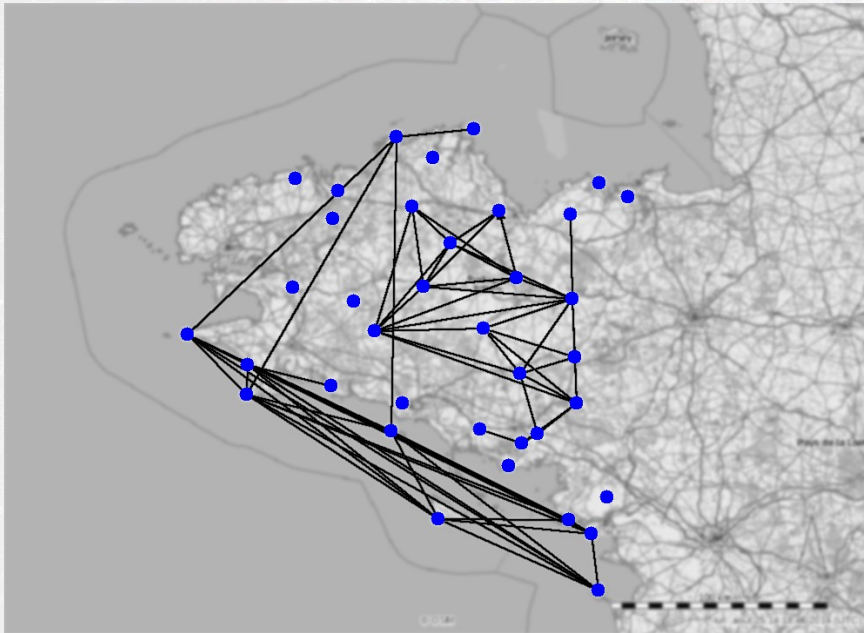


37 stations after preprocessing

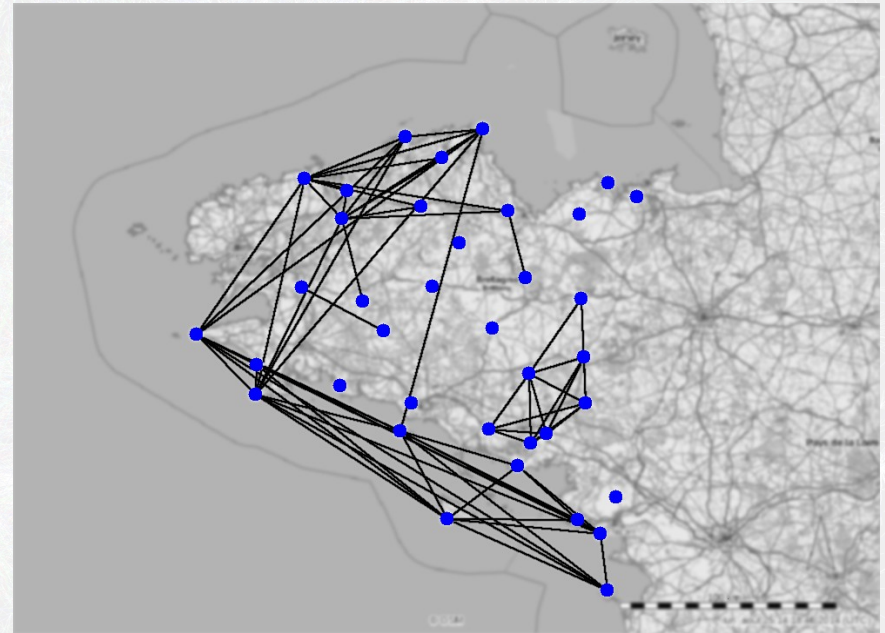
744 observations per station

Graph inference from signals

The case of stationary signals – Kalofolias regularization on a dataset



Kalofolias



Regularized Kalofolias

Graph inference from signals

The case of stationary signals – Performance evaluation on synthetic data

	polytope	$L_{1,1}$	Tr	S(X)
Simple	✓	36.9974	0.0013	0.0551
Sparse	✓	36.9971	0.9093	0.0585
Kalofolias	0.0313	36.9979	0	0.0751
Kalofolias closest	✓	36.9974	0.0298	0.0548
Segarra <i>et al.</i>	0.0062	36.9993	1.97×10^{-5}	0.0245
Segarra <i>et al.</i> closest	✓	36.9974	0.0046	0.0551
Graphical lasso	1.3730	35.3539	13.4977	32.8421
Graphical lasso closest	✓	36.9984	13.3584	0.0335

► A few words on graph signal processing

- * What is it?
- * The main ideas
- * Some problems and applications

► Graph inference from signals

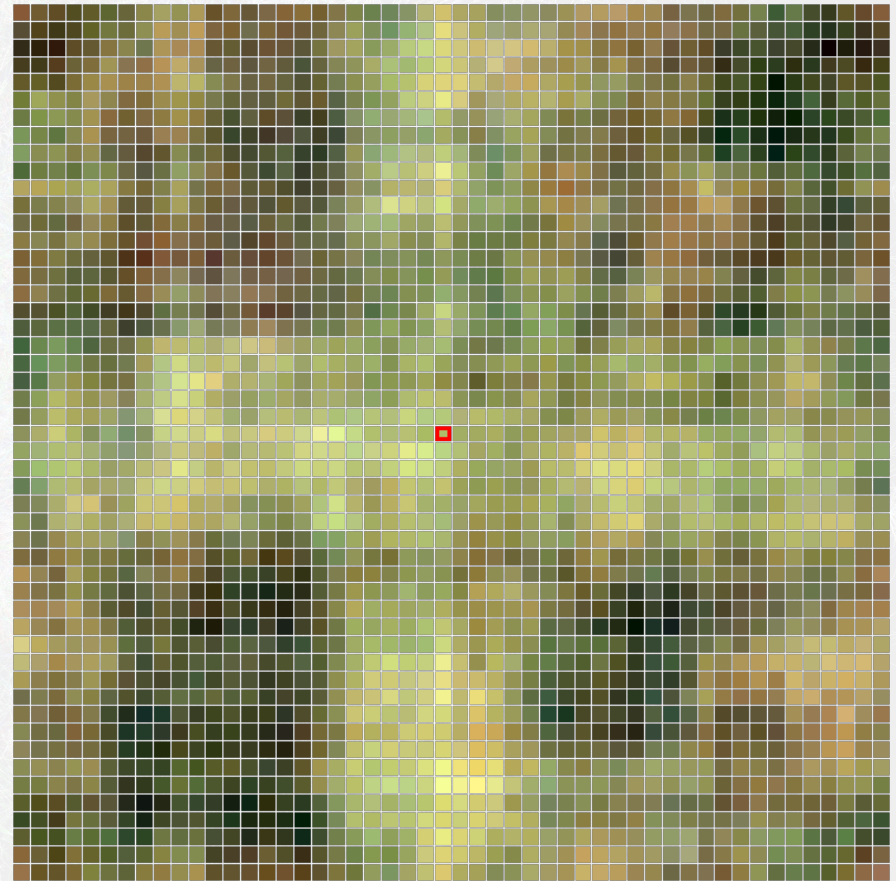
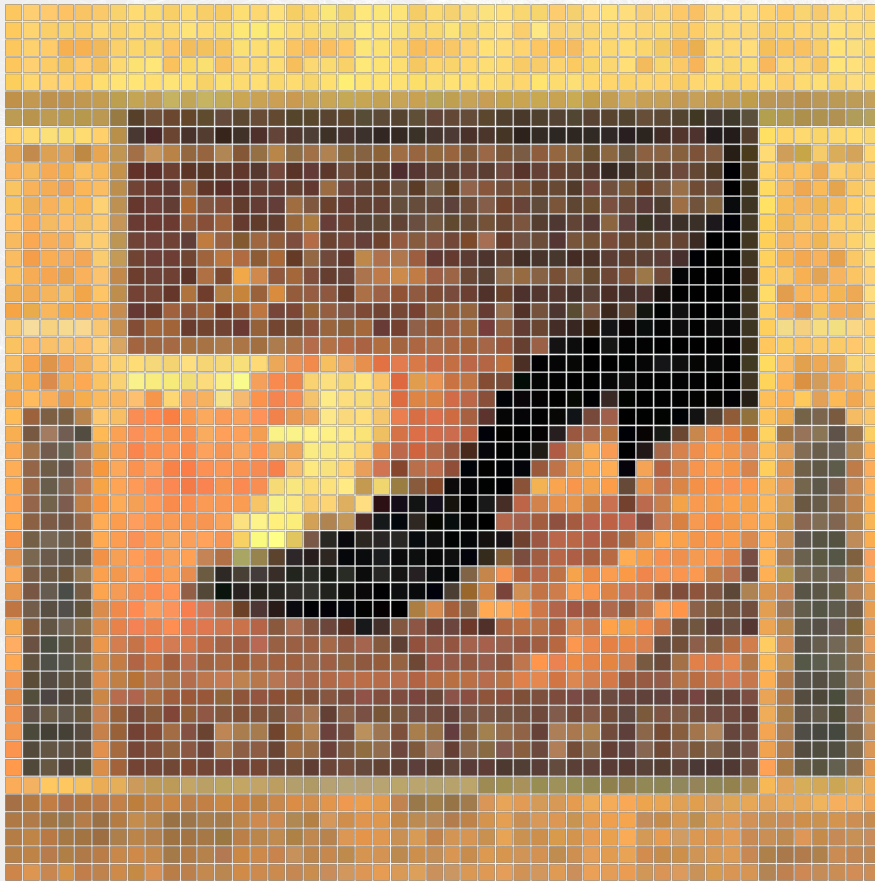
- * Why it may be necessary
- * The case of stationary signals

► Translations on irregular graphs

- * How we define them
- * How we identify them
- * Examples and applications

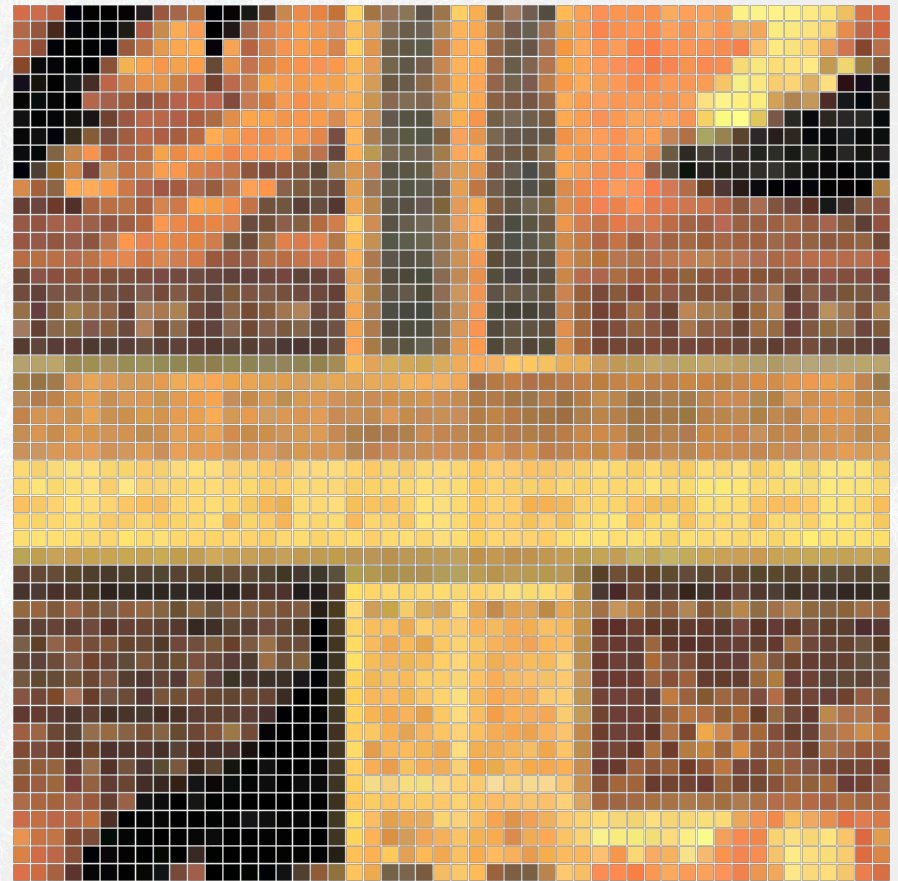
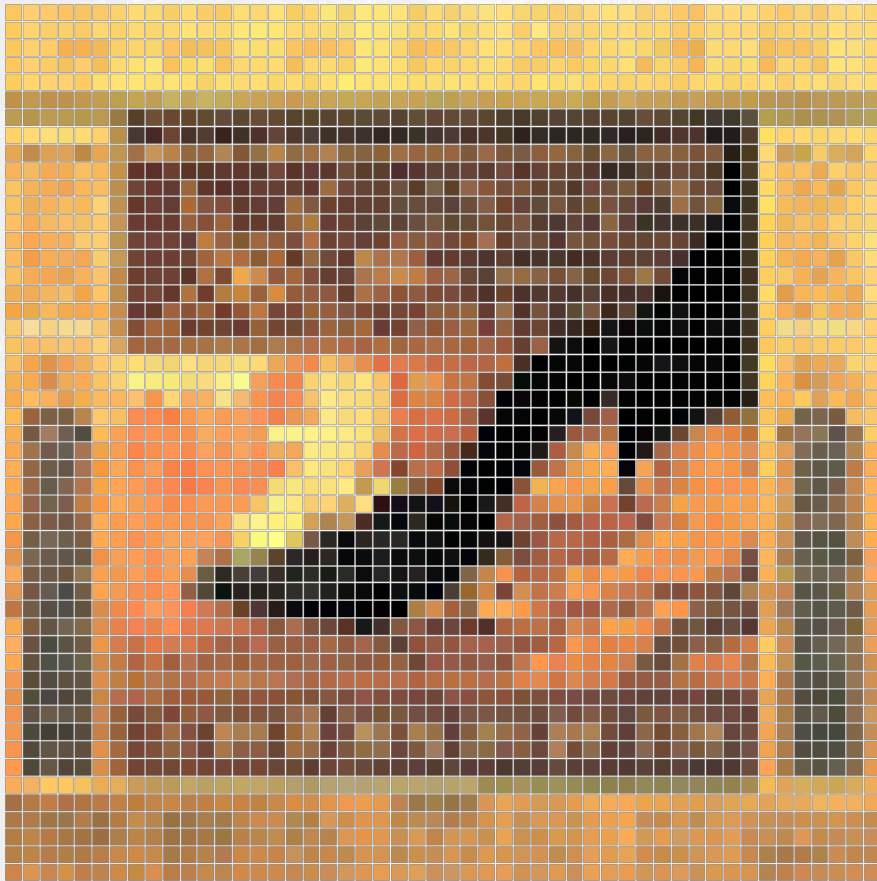
Translations on irregular graphs

How to define them – Translation in the spectral domain



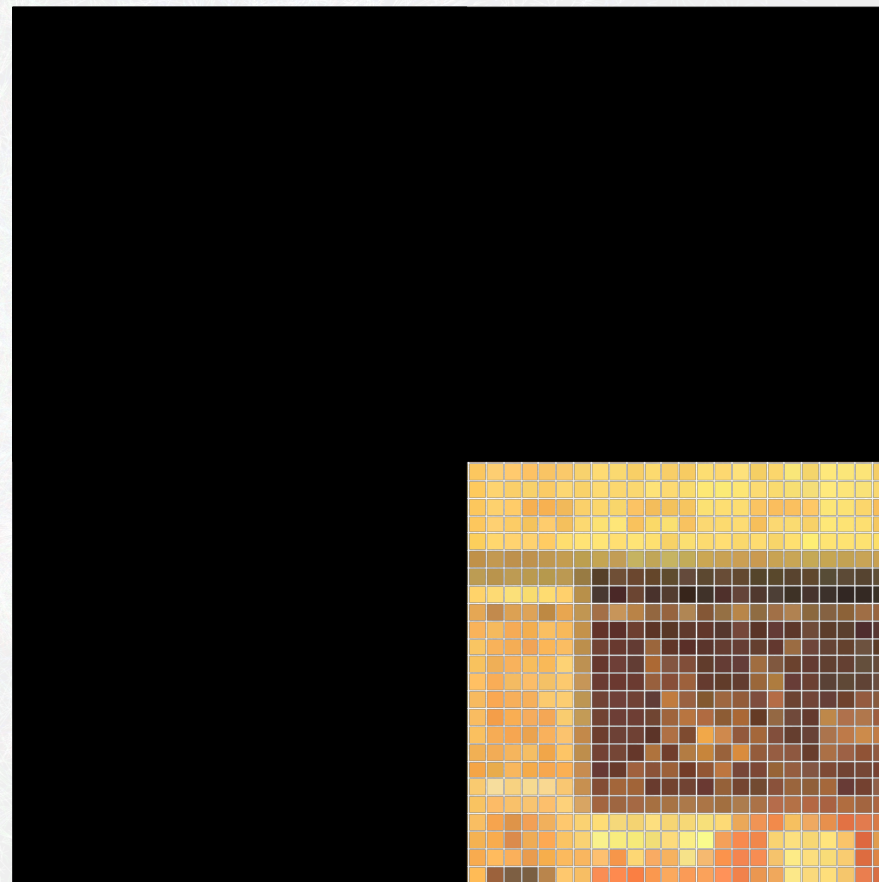
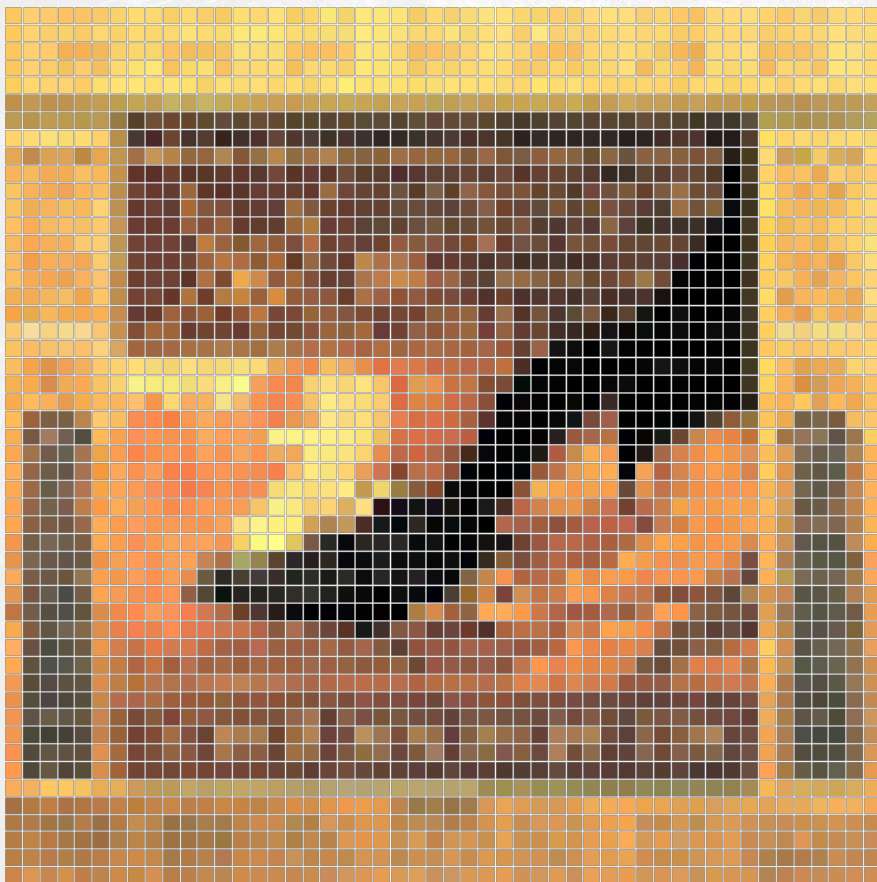
Translations on irregular graphs

How to define them – Euclidean translation on a torus



Translations on irregular graphs

How to define them – Euclidean translation on a grid



Translations on irregular graphs

How to define them – Three simple properties

► Injectivity

$$\forall v_1, v_2 \in \mathcal{V} : (\psi(v_1) = \psi(v_2) \neq \perp) \Rightarrow (v_1 = v_2)$$

► Restriction to the edges

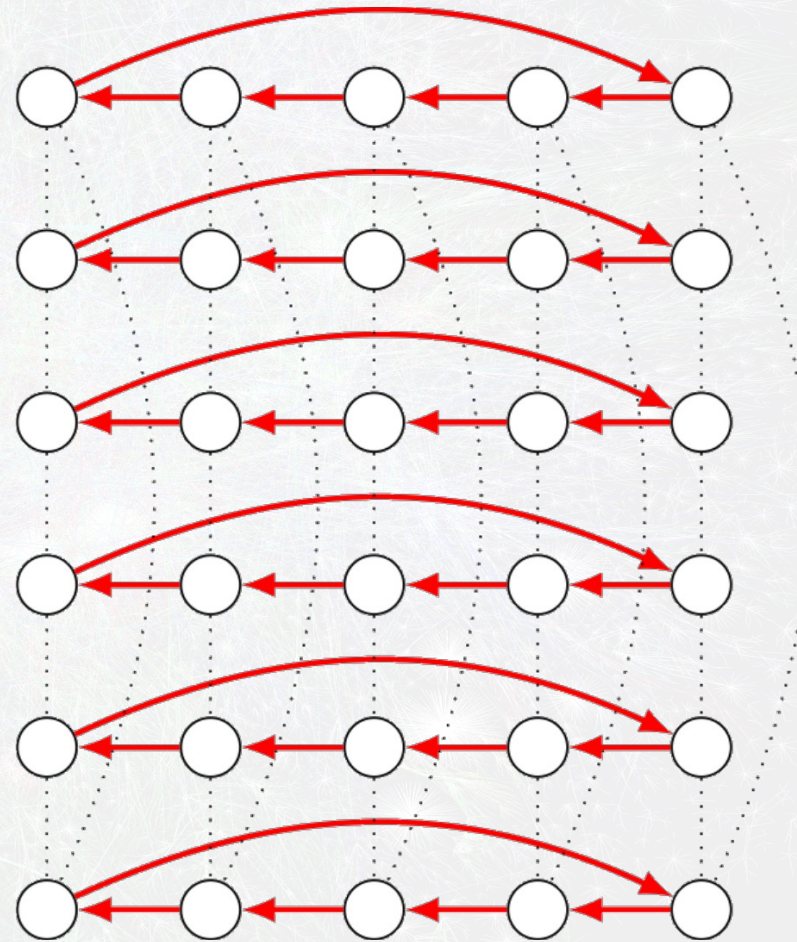
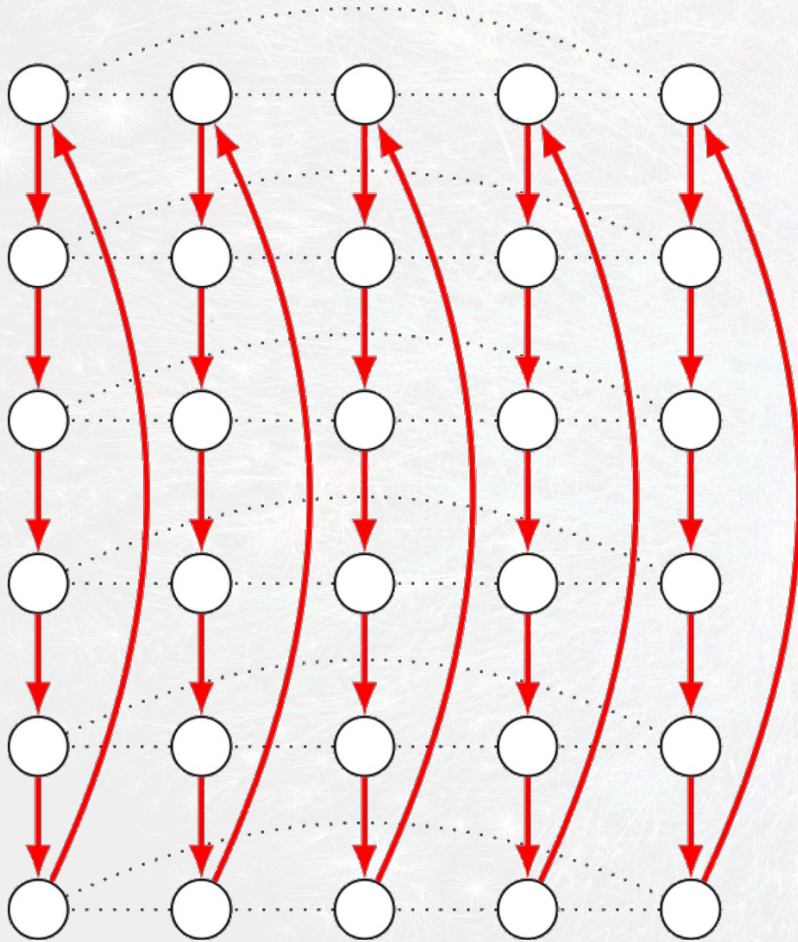
$$\forall v \in \mathcal{V} : (\{v, \psi(v)\} \in \mathcal{E}) \vee (\psi(v) = \perp)$$

► Preservation of the neighborhood

$$\begin{aligned} \forall v_1, v_2 \in \mathcal{V} : (\{v_1, v_2\} \in \mathcal{E} \Leftrightarrow \{\psi(v_1), \psi(v_2)\} \in \mathcal{E}) \\ \vee (\psi(v_1) = \perp) \vee (\psi(v_2) = \perp) \end{aligned}$$

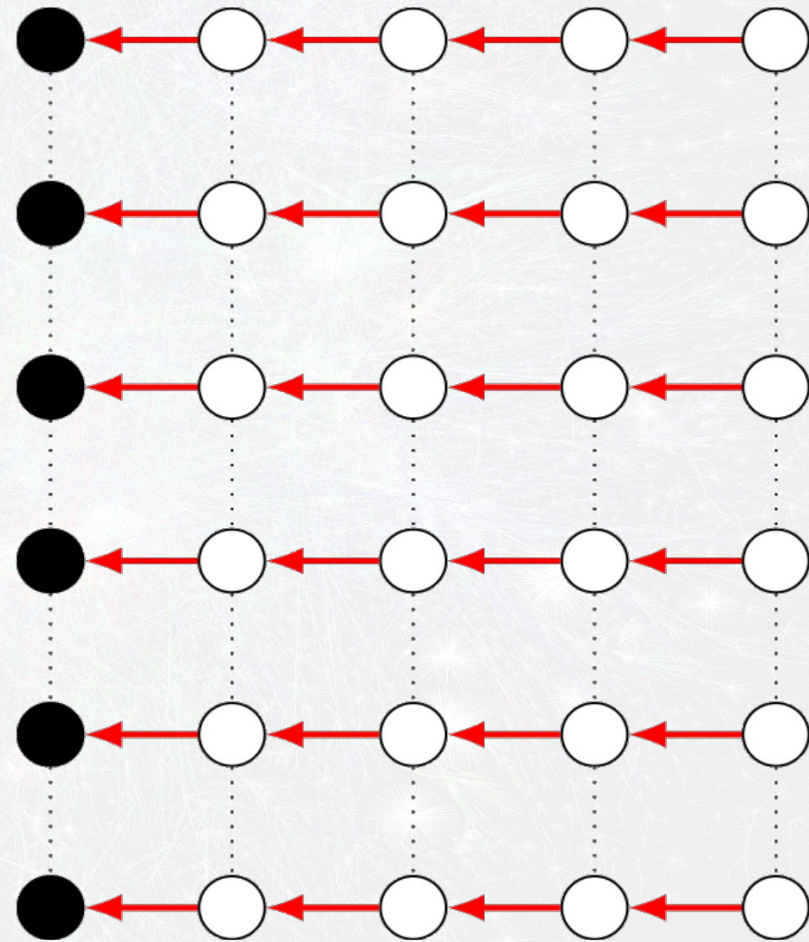
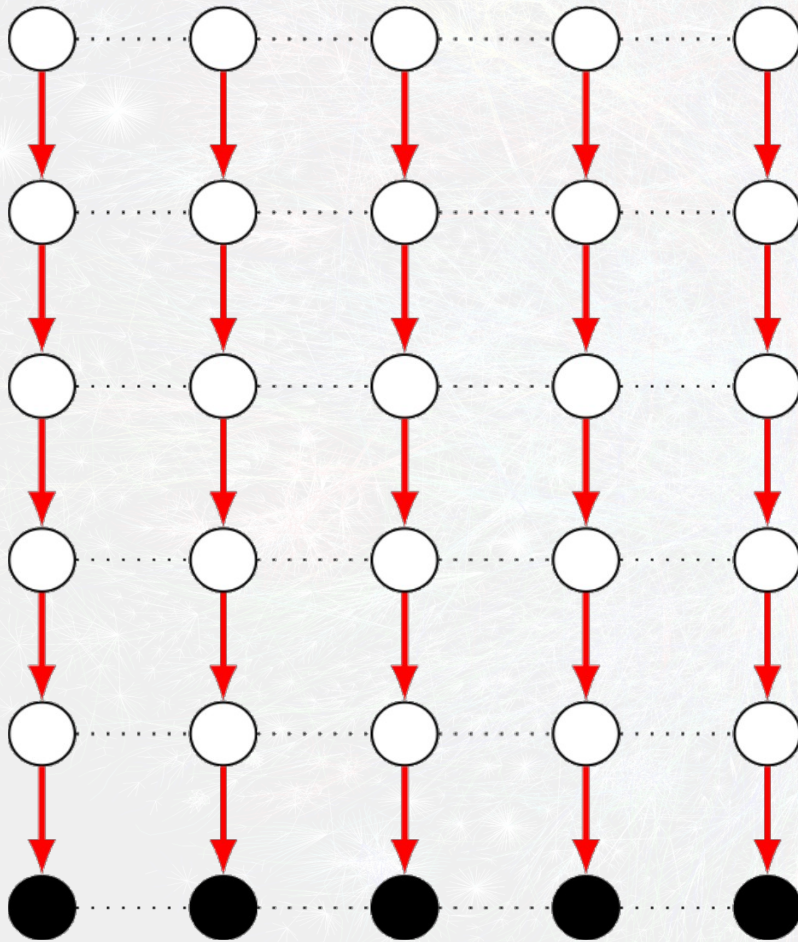
Translations on irregular graphs

How to identify them – Only lossless translations on a torus graph



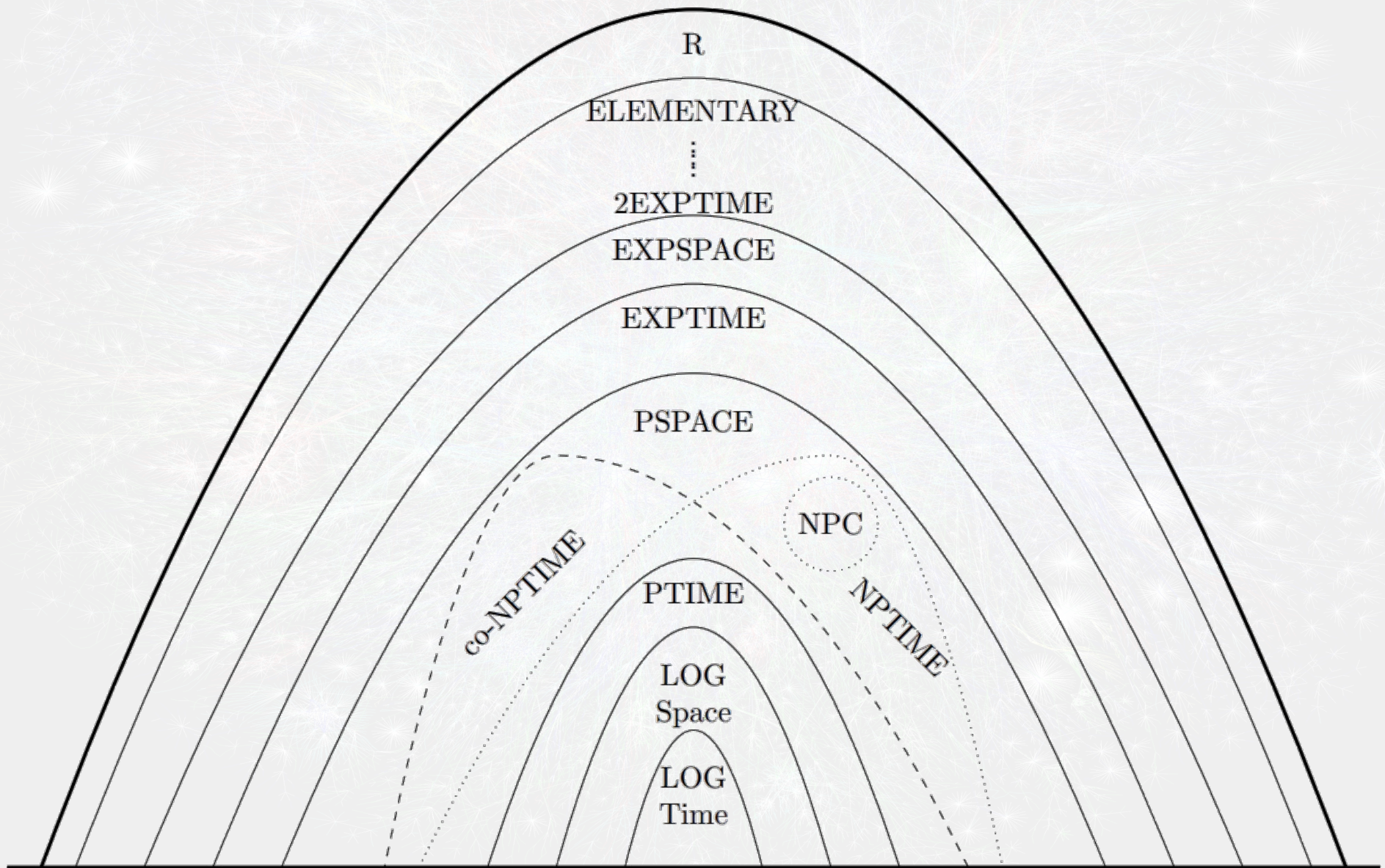
Translations on irregular graphs

How to identify them – Translations with minimal loss on a grid graph



Translations on irregular graphs

How to identify them – Identifying translations is an NP-complete problem



Translations on irregular graphs

How to identify them – Necessity to relax the problem

$$\mathbf{A}_\psi^* = \arg \min_{\mathbf{A}_\psi \in \{0,1\}^{N \times N}} \|\mathbf{A} \cdot \mathbf{A}_\psi - \mathbf{A}_\psi \cdot \mathbf{A}\|_1$$

Minimize deformation

$$s.t. \left\{ \begin{array}{l} \forall i, j \in [1, N] : \mathbf{A}_\psi[i, j] \leq \mathbf{A}[i, j] \\ \forall j \in [1, N] : \sum_{i=1}^N \mathbf{A}_\psi[i, j] \leq 1 \\ \forall i \in [1, N] : \sum_{j=1}^N \mathbf{A}_\psi[i, j] \leq 1 \\ \|\mathbf{A}_\psi\|_1 = C \end{array} \right.$$

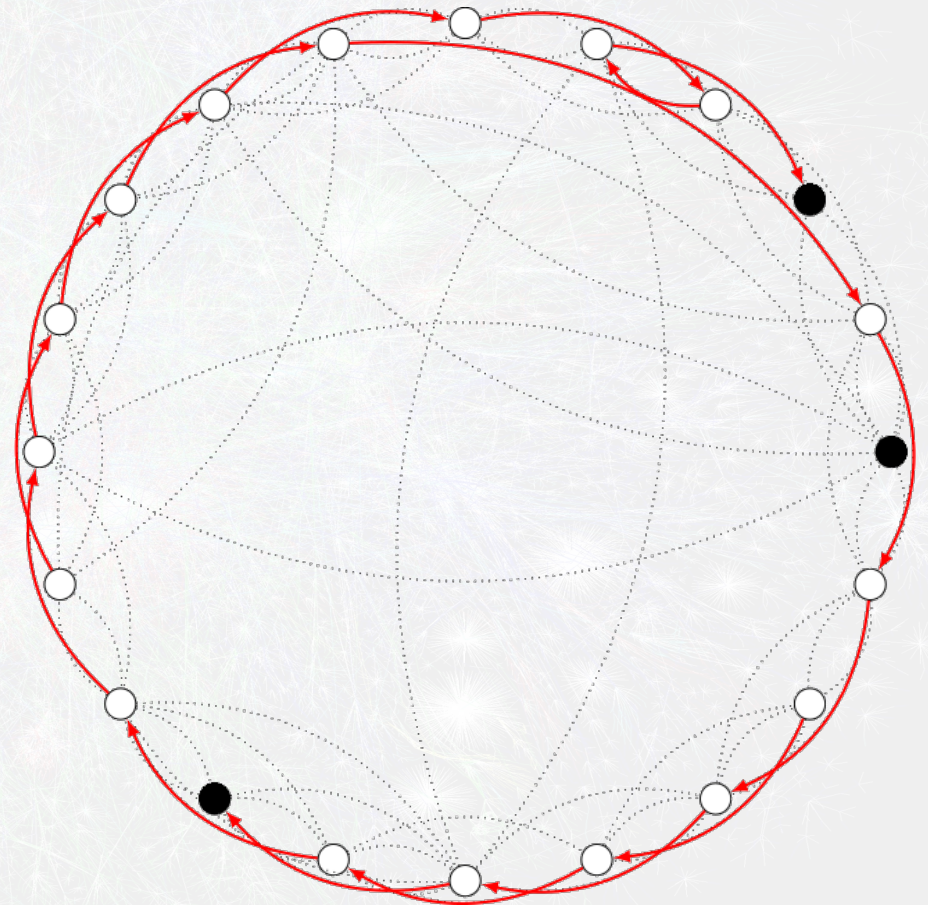
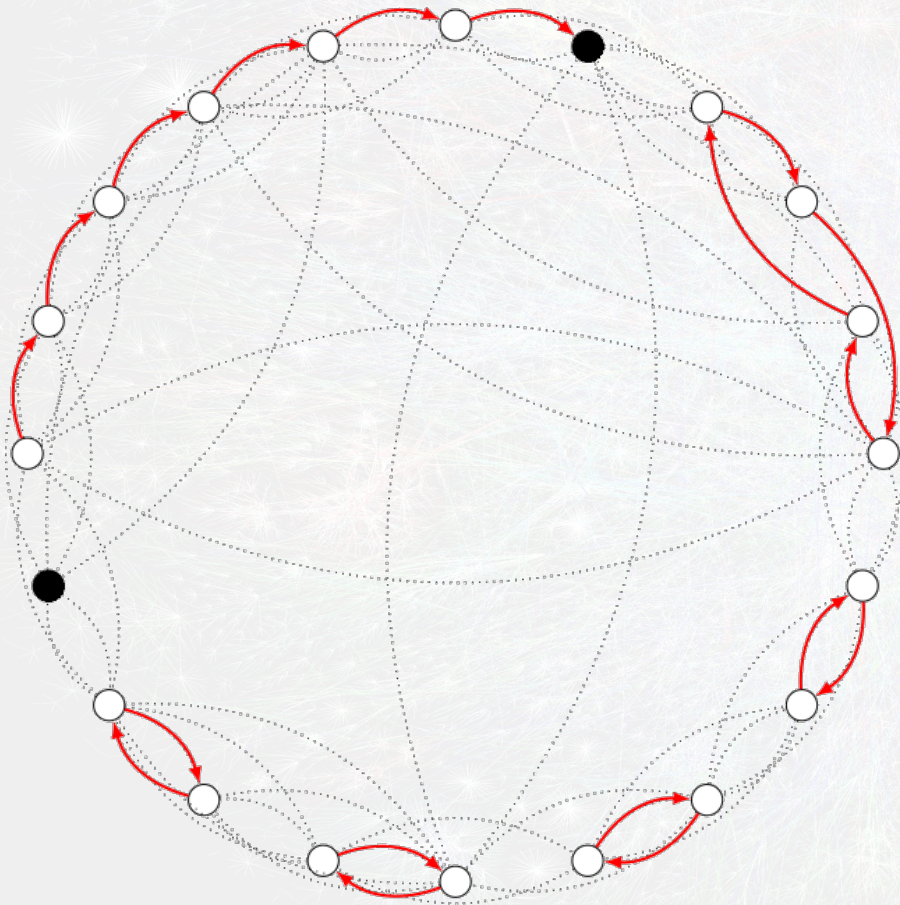
Restriction to edges

Injectivity

Avoid empty solution

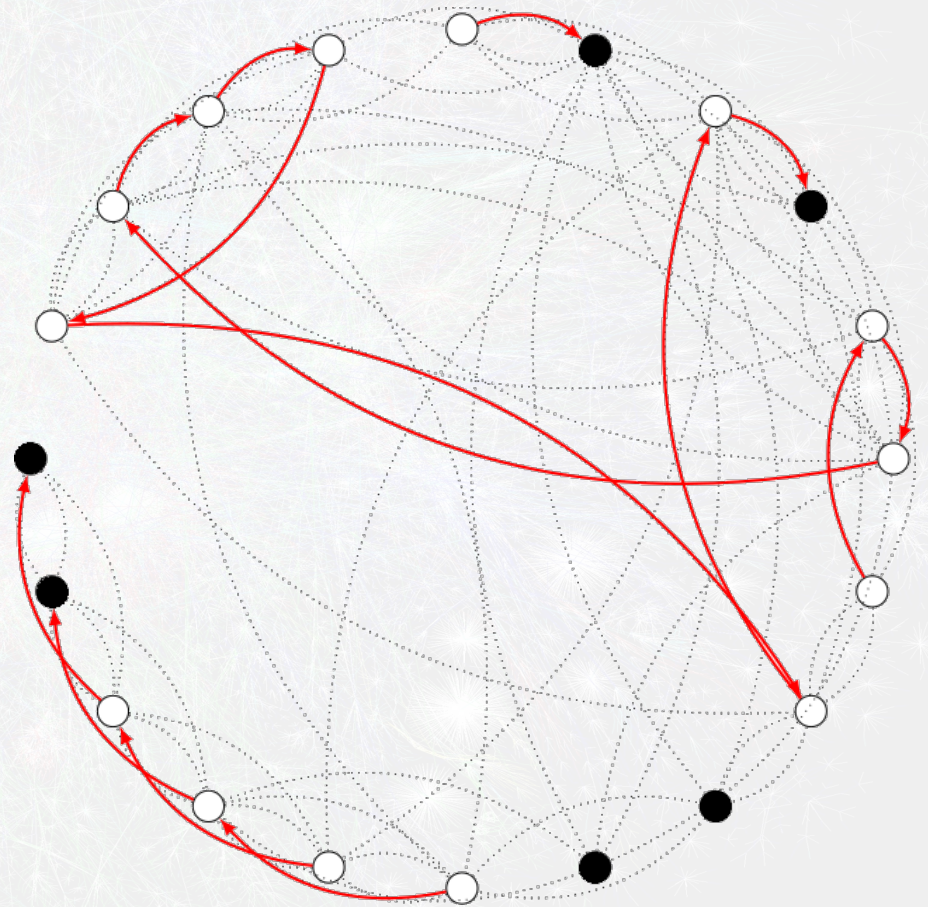
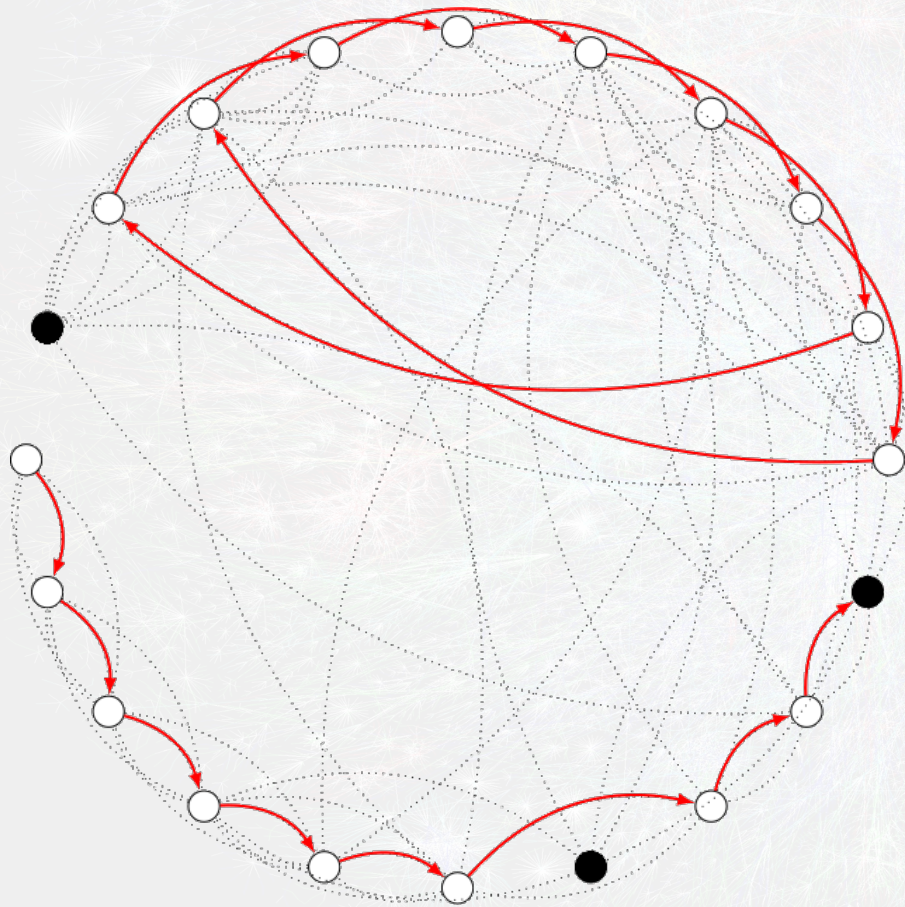
Translations on irregular graphs

Examples and applications – Results on a Watts-Strogatz graph ($P=0.1$)



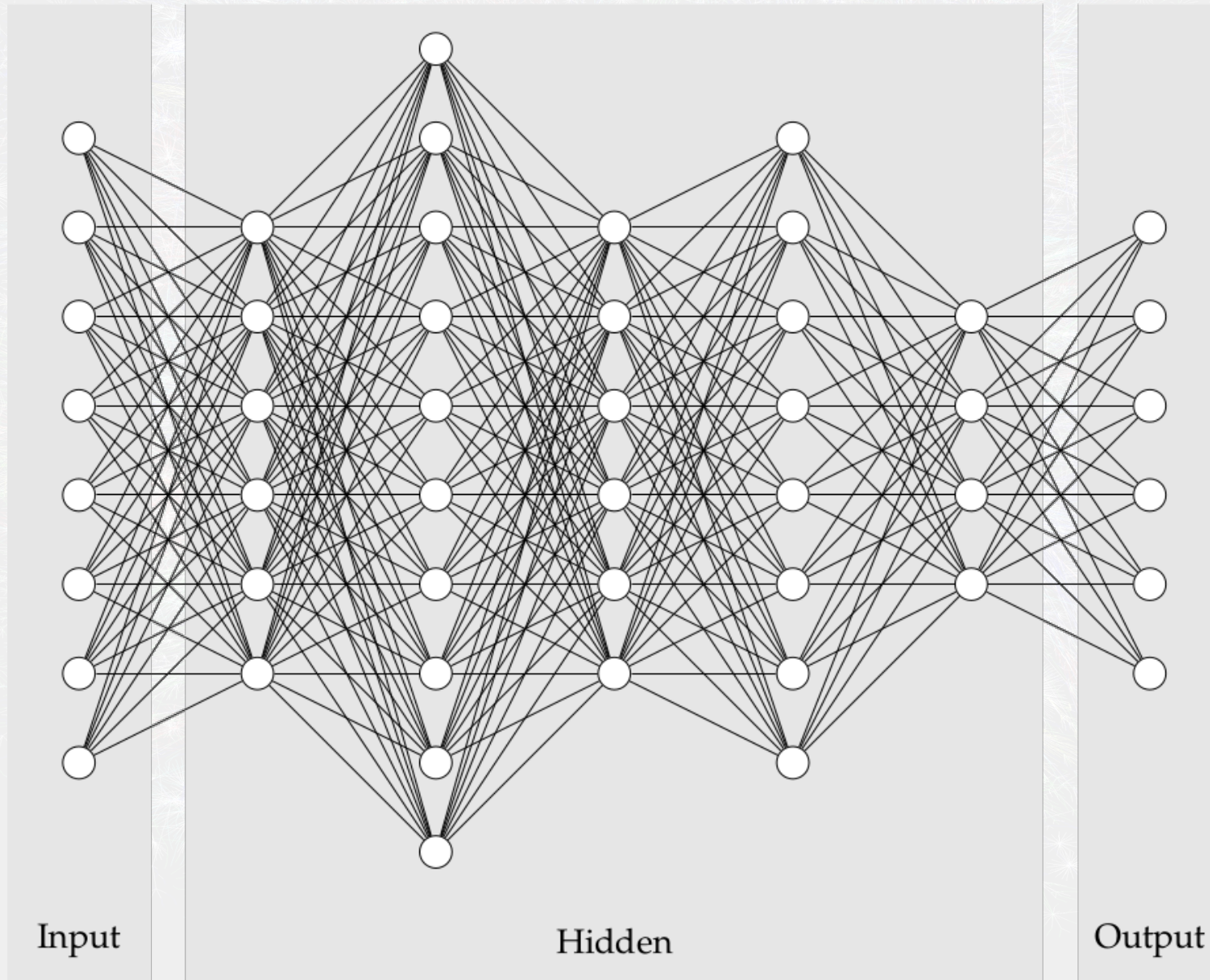
Translations on irregular graphs

Examples and applications – Results on a Watts-Strogatz graph ($P=0.3$)



Translations on irregular graphs

Examples and applications – Neural networks



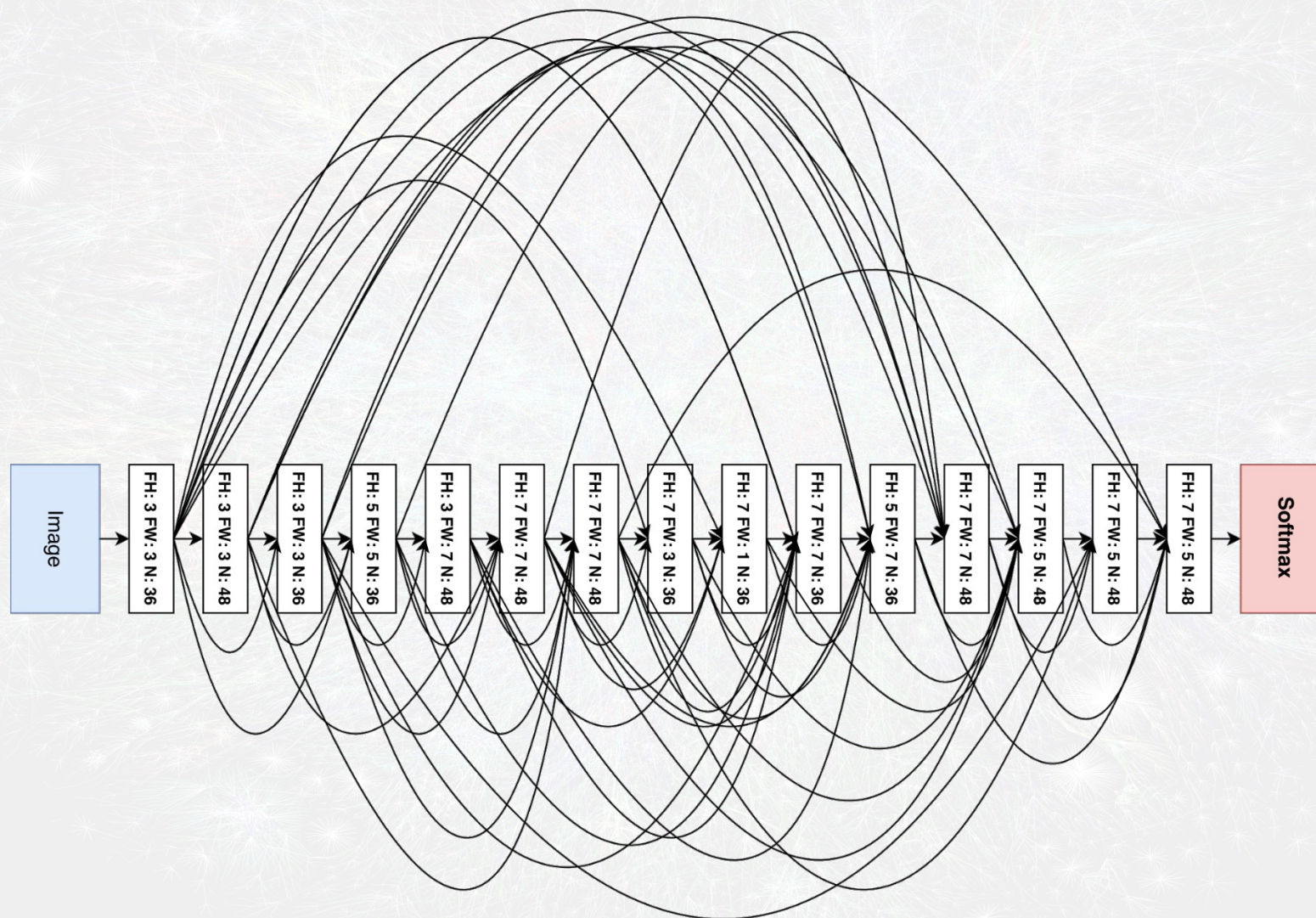
Translations on irregular graphs

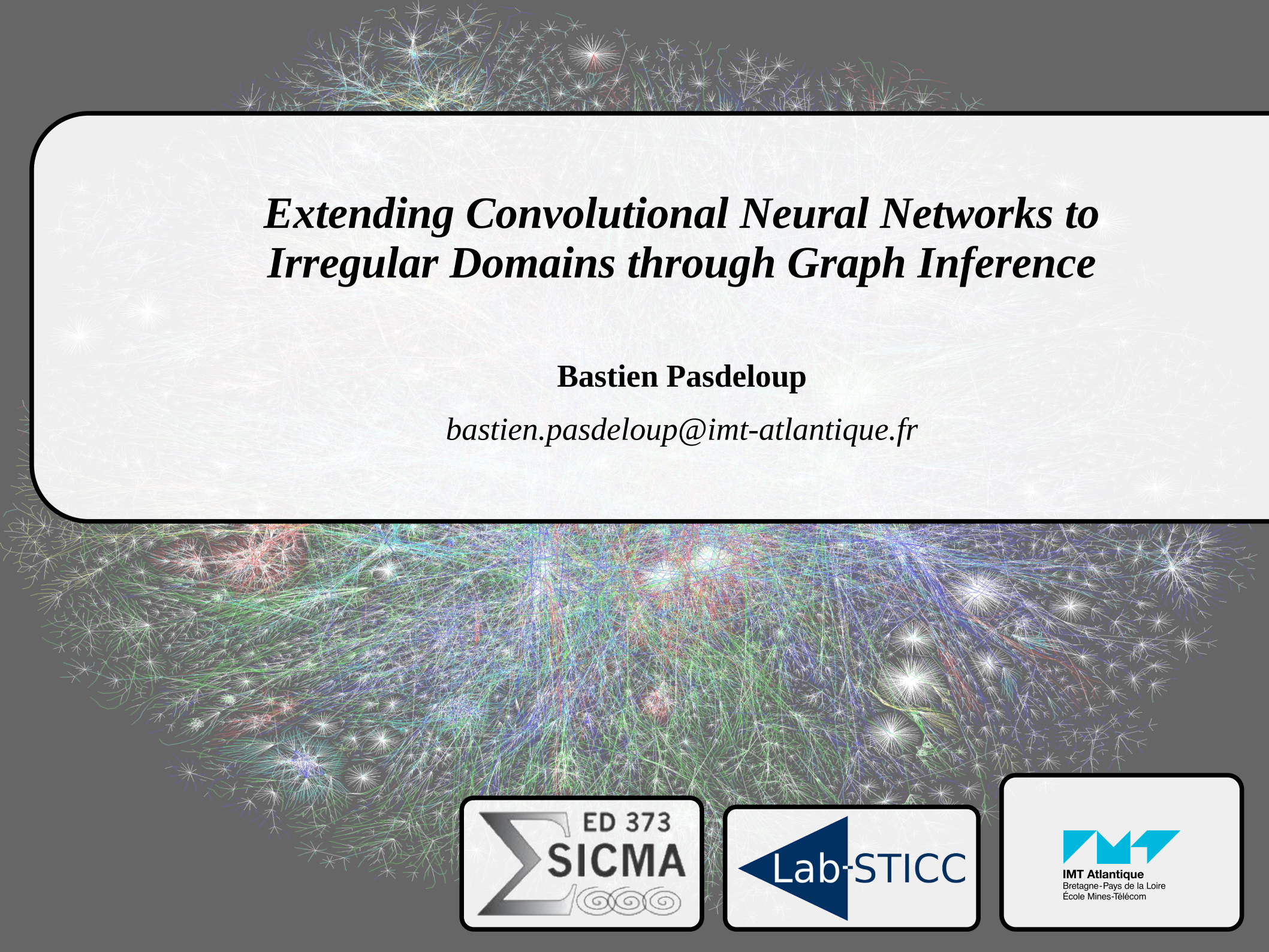
Examples and applications – Convolutional neural networks



Translations on irregular graphs

Examples and applications – Convolutional neural networks





Extending Convolutional Neural Networks to Irregular Domains through Graph Inference

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